Screening deployed arms by pseudo-outcome risk share

Gradual shifts, large batches

1 What was run

Four fixed linear scorecards compete on successive batches. The reward is negative batch mean squared error. Excess on a round is the gap between the served scorecard and the best scorecard on that batch. The FSDS weight of scorecard k is the held-out residual variance of its doubly robust pseudo-outcome, divided by the sum of those residuals across the four scorecards. The pseudo-outcome compares a fixed reference window (the first six batches) with the latest six batches. The share is refreshed every eight rounds from round 16. It is computed from every scorecard's prediction on the logged batch, so it does not require that the scorecard was served.

The index is the usual UCB, discounted UCB, or Thompson draw, minus a tax $\eta\sigma w_k$. Here σ is the standard deviation of the four arm rewards on the reference window, and $\eta \in \{0.5, 1, 2\}$. An absolute penalty $\lambda = 5$, not scaled by σ, is reported as a check. No rule uses $\lambda = 20$ or $\lambda = 40$. Epsilon-greedy at 5%, 10%, and 15% is the uniform-exploration comparator, and one extra rule puts the $\eta = 1$ tax on the 10% rule.

Each batch has 200 rows. There are 80 rounds. The mixing weight stays at 0 through round 20 and then ramps linearly to 1 at the last round. The next batch is not announced. The share is the adaptation: it is recomputed from the latest six batches, on a fixed calendar, with no change-point indicator. Post-shift totals use rounds whose mixing weight is at least 1/2. Early totals use rounds whose mixing weight is exactly 0. The ramp in between is reported separately. Four seeds, 20261018 through 20261021, are averaged. Parentheses in the tables are the seed standard deviation.

The covariate pools all have at least 10,000 rows, or are synthetic of that size. Columns are standardized.

Pool	Rows used	Columns
Gaussian	20,000	30
Gaussian	20,000	12
Student-t_5	20,000	30
AR(1) Gaussian, correlation 0.6	20,000	30
California housing	20,640	8
Covertype, draw from 581,012	30,000	54

Real matrices supply covariates only. The label moves gradually: a linear block relocates, a sign flips along the ramp, a block dies, a nonlinear term fades in, or the noise scale grows. The claim assumes there is no jump. A change-point that replaces the mechanism between two rounds is a different problem, and it is not used below. The headline number is the median, across the six pools, of the four-seed mean. California's nonlinear add-on produces a very large squared term, so a raw mean would follow that one pool.

A second horse-race replaces scorecards by four linear policies on a horizon of four steps, with 200 episodes in each round. The same residual share is computed from each policy's scalar self-score. Three monitors are compared: an oracle that sees every policy's return, an on-policy log of the served policy only, and a least-squares reward fit that imagines the unserved policies.

2 What the share is doing

On a gradual relocation the share does not chase the scorecard whose predictions move the most. It chases the scorecard that has stopped explaining the outcome.

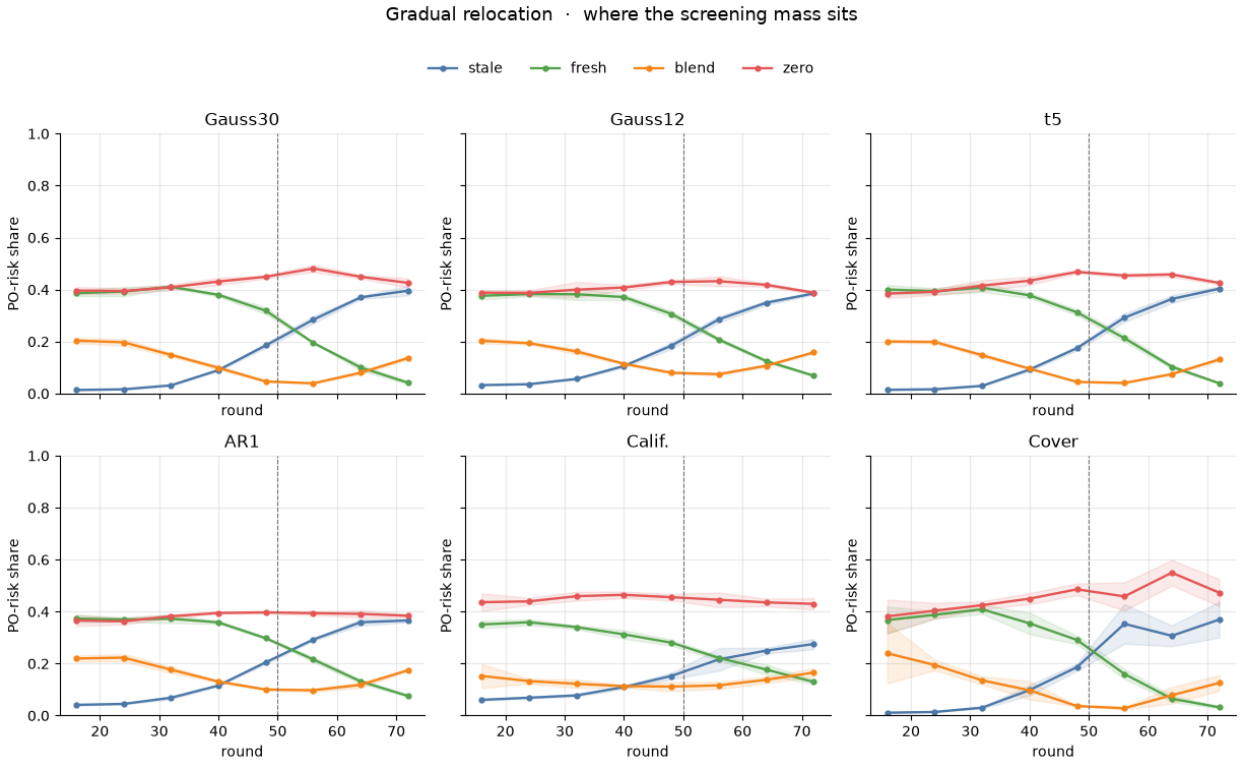


Figure 1: PO-risk shares on a gradual relocation. Before the ramp, the fresh block and the zero score hold the mass, and the stale block, which still carries the signal, sits near zero. After the mixing weight passes 1/2 (dashed line), the fresh share falls and the stale share rises to meet the zero score. The same crossing appears on all six pools. California, with only eight columns, separates more slowly.

Before the move, y is the stale block. A regression of the pseudo-outcome on that score absorbs the contrast, so its residual share is small. The fresh block is orthogonal to y, and the zero score has nothing to project onto, so those two residuals take the share. After the move, y lives on the fresh block. The fresh residual collapses and the stale residual inflates, because the current-window leftover is now orthogonal to the stale score. The zero score never explains y, so it keeps a large share throughout. Post-shift means on the Gaussian $d = 30$ pool are about 0.35 on the stale block, 0.11 on the fresh block, 0.09 on the blend, and 0.45 on zero. Covertype is sharper: about 0.34, 0.09, 0.08, and 0.49.

The tax $-\eta\sigma w_k$ is therefore a screen on currently unexplained scores. An arm that moves with the world sheds the tax. An arm that moves against the world, or that never carried the signal, keeps it. That is why a penalty on the raw change in the arm's own prediction would be the wrong

instrument: the fresh card's fitted mean moves the most, and that movement is the signal.

While the mixing weight is still zero, UCB and UCB with $\eta \in \{1, 2\}$ have the same early excess (median 16.1 on gradual relocation). The relevant arm already wins the reward index, and the tax on the irrelevant arms does not change the argmax. Epsilon-greedy pays immediately: its early median is 19.9 at 10%.

3 Where the tax earns its keep

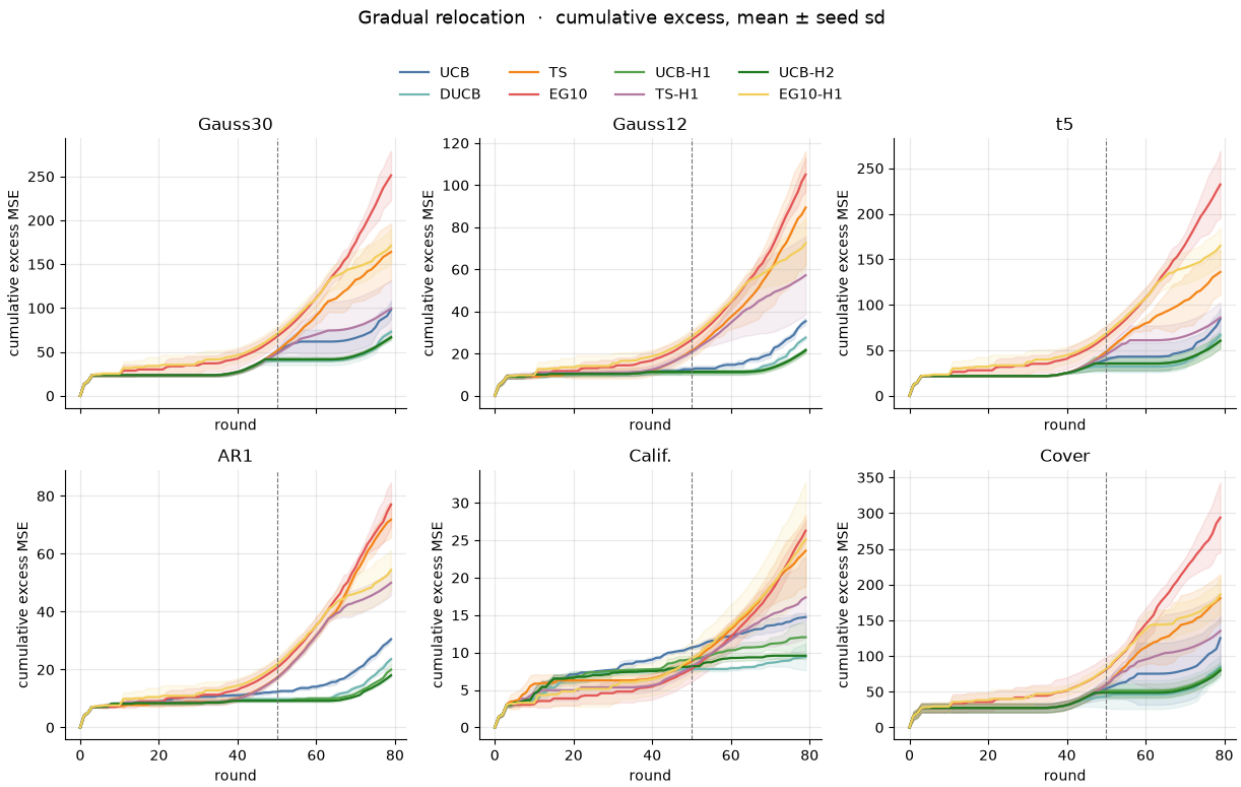


Figure 2: Cumulative excess on a gradual relocation. UCB with $\eta = 1$ or 2 stays flat after the dashed line. Epsilon-greedy at 10%, with or without the tax, keeps accumulating. Thompson sampling without the tax tracks the exploring rules; with $\eta = 1$ it moves toward UCB.

Under a ramp, three regimes show up.

The signal changes subspace. Gradual relocation, the nonlinear add-on, and a dying block are this case. The median post-shift excess on gradual relocation is 34.6 for UCB, 24.6 for discounted UCB, 80.0 for Thompson sampling, and 125.0 for epsilon-greedy at 10%. UCB with $\eta = 2$ and discounted UCB with $\eta = 2$ both sit at 17.6. That removes 49% of the UCB total and 86% of the epsilon-greedy total, and it is lower than discounting alone on all six pools. The same ordering holds at $\eta = 1$ (17.8). Epsilon at 5% is still 100.6, and epsilon at 15% is 82.6: a larger uniform share spends more whole batches on a wrong scorecard and does not locate the fresh block any faster. On the nonlinear add-on the best median is discounted UCB at $\eta = 2$, 2.80, against 8.48 for UCB and 21.9 for epsilon at 10%. On a gradual death the same rule is 27.1, against 41.6 for discounting alone and 52.2 for UCB.

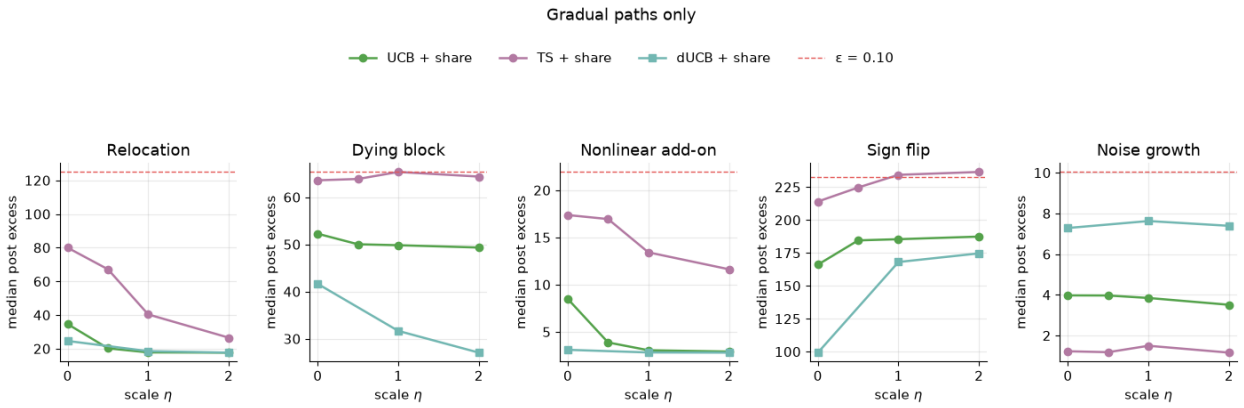


Figure 3: Median post-shift excess against the scale η, gradual paths only. The dashed line is epsilon-greedy at 10%. Relocation, a dying block, and a nonlinear add-on slope down. A gradual sign flip slopes up. Noise growth is already small for Thompson sampling, and discounting sits above UCB because the best card never changes.

The signal flips sign along the ramp. A within-window least squares absorbs the sign into the two window-specific slopes. The residual then cannot prefer the flipped card. On the gradual flip the two signed scores stay close to each other and the zero score takes the largest share, about 0.38 to 0.58 depending on the pool. The tax therefore does not choose the current sign. Discounted UCB has median post-shift excess 99.2, plain UCB 166, and UCB at $\eta = 1$ is 185. The shut-off is visible in the share, before any regret is computed. If the live scores are tied and the null score holds the mass, set $\eta = 0$ and discount the reward. Discounting wins by following the recent reward, which moves toward zero while the sign is in between. It does not discover the flipped card: that card's early mean is poor and it is rarely pulled.

Only the noise scale changes. The linear score stays correct, and its share stays small, about 0.10 to 0.18. The tax lands on zero and on the irrelevant block. Discounting discards a history that is still valid, so discounted UCB is worse than UCB (7.28 against 3.97). Thompson sampling already concentrates on the unchanged arm. A small tax on top of it is the best rule on this ramp: absolute $\lambda = 5$ has median 0.90, and Thompson at $\eta = 2$ is 1.16.

4 How large the effect is

The median is a summary. The claim that the tax works has to survive two stricter checks: it has to appear on each pool, and the gap has to be larger than the seed standard deviation of either rule. Table **??** and Table **??** are those checks. Every raw mean, for every rule and every pool, is in the appendix.

Gradual relocation is the clean effect. UCB at $\eta = 2$ cuts the post-shift excess on all six pools, by 46% to 65%: Gaussian $d = 30$ from 50.6 to 25.1, Gaussian $d = 12$ from 22.9 to 10.3, Student-t_5 from 46.3 to 25.0, AR(1) from 18.2 to 8.8, California housing from 4.2 to 1.5, Covertype from 73.1 to 31.2. On each pool the gap exceeds the larger seed standard deviation, and the taxed seed standard deviation is itself smaller (Covertype falls from 20.7 to 1.2). Against discounted UCB the same gap clears the seed noise on five pools. California is the exception: discounting alone is already 1.65 with seed sd 1.50, and the tax moves the point estimate to 1.49. That pool has only

eight columns, so the two blocks are short and the share separates more slowly. Against $\varepsilon = 0.10$ the removed fraction is 85% to 92% on every pool, and the gap is several times the seed noise.

The hit rate explains what kind of improvement this is. On the four large synthetic pools, UCB's post-shift hit rate is about 0.38 to 0.44, and the taxed rule is about 0.47 to 0.51. Excess falls by half while the fraction of rounds on the unique best card rises by only about ten points. The tax is not finding a hidden optimum on every round. It is refusing the stale card and the zero card. The blend, which is close to the fresh block during the ramp, absorbs many of the remaining rounds, so the excess of those rounds is small. Epsilon-greedy's hit rate on the same pools is about 0.03. It almost never serves the best card after the shift, which is why its excess is 57 to 217.

The early window, mixing weight exactly zero, does not pay for this. UCB and UCB at $\eta = 2$ have the same early total on Gaussian $d = 30$ (23.5), Gaussian $d = 12$ (10.4), Student-t_5 (21.8), and Covertype (27.3). AR(1) moves from 8.95 to 8.43, and California from 7.06 to 6.62. Epsilon at 10% is already higher on five of the six pools before anything moves.

Where the same bar is cleared, and where it is not. Table **??** counts a pool only when the candidate beats the baseline by more than the larger seed standard deviation. Every row is a ramp. Growing noise, with Thompson at $\eta = 2$, clears UCB on all six pools and discounted UCB on all six. The best card never changed, the share stays on the irrelevant scores, and the tax is a small push on top of a posterior that was already in the right place. A gradual sign flip clears UCB on none of the six pools and discounted UCB on none. The negative direction is the same one the share predicts: the tax pushes away from zero while zero is the hedge during the ramp. It still beats $\varepsilon = 0.10$ on all six pools, because uniform exploration remains the more expensive mistake.

A dying block and a nonlinear add-on sit between these. Discounted UCB with a modest tax has the best median, but the per-pool gap over discounting alone often sits inside the seed noise. The reliable piece is the comparison with UCB and with epsilon-greedy on the pools where the linear signal is large. Gaussian $d = 12$ is the instructive small case for a dying block: plain UCB's post-shift excess is already about 2, and the tax cannot improve a problem that was almost solved. California's nonlinear add-on is not evidence either way. One squared housing coordinate makes the excess and the seed sd both enormous, which is why the headline number is a median.

Table 1: Gradual relocation, pool by pool. Post-shift excess MSE, mean over four seeds, seed sd in parentheses. The last two columns are the fraction removed by UCB at $\eta = 2$.

	UCB	dUCB	$\text{UCB}_{\eta=2}$	$\text{dUCB}_{\eta=2}$	$\varepsilon.10$	TS	$\text{TS}_{\eta=2}$	cut UCB	cut ε
Gaussian $d = 30$	50.61 (8.51)	32.71 (6.18)	25.12 (1.51)	25.12 (1.51)	187.11 (32.16)	116.40 (30.74)	37.17 (7.16)	50%	87%
Gaussian $d = 12$	22.87 (0.43)	16.43 (1.31)	10.31 (0.13)	10.31 (0.13)	79.78 (9.26)	69.17 (25.32)	20.13 (7.30)	55%	87%
Student-t_5	46.29 (4.72)	34.85 (3.66)	24.98 (1.09)	24.98 (1.09)	170.17 (33.87)	90.73 (25.06)	32.91 (6.79)	46%	85%
AR(1)	18.22 (1.10)	13.86 (0.66)	8.80 (0.52)	8.34 (0.59)	57.42 (9.38)	55.57 (6.56)	19.01 (7.32)	52%	85%
California	4.23 (0.63)	1.65 (1.50)	1.49 (0.58)	0.22 (0.35)	18.75 (1.63)	15.08 (4.61)	7.76 (2.66)	65%	92%
Covertype	73.10 (20.68)	42.22 (0.64)	31.17 (1.16)	31.17 (1.16)	216.59 (51.67)	125.18 (31.74)	35.15 (3.04)	57%	86%

A cut is (baseline − taxed)/baseline. On every pool the taxed rule also has the smaller seed sd. The path is a ramp. There is no change-point.

Table 2: Gradual paths only. A pool is counted when the candidate beats the baseline by more than the larger of the two seed standard deviations.

	candidate	beats UCB	beats dUCB	beats ε.10
Gradual relocation	$\text{UCB}_{\eta=2}$	6/6	5/6	6/6
Gradual dying block	$\text{dUCB}_{\eta=2}$	3/6	2/6	5/6
Gradual nonlinear	$\text{dUCB}_{\eta=2}$	4/6	0/6	5/6
Gradual sign flip	$\text{UCB}_{\eta=1}$	0/6	0/6	6/6
Gradual noise	$\text{TS}_{\eta=2}$	6/6	6/6	4/6

Jump paths are outside the claim. California's nonlinear add-on has seed standard deviations in the thousands, so that pool rarely clears the bar.

Table 3: Post-shift hit rate, Gradual relocation. Fraction of rounds on the single best scorecard.

	UCB	dUCB	UCB$_{\eta=2}$	dUCB$_{\eta=2}$	TS	TS$_{\eta=2}$	ε.10
Gaussian $d = 30$	0.38 (0.09)	0.51 (0.02)	0.51 (0.02)	0.51 (0.02)	0.07 (0.07)	0.40 (0.05)	0.03 (0.03)
Gaussian $d = 12$	0.44 (0.02)	0.49 (0.02)	0.50 (0.03)	0.50 (0.03)	0.07 (0.13)	0.30 (0.14)	0.02 (0.03)
Student-t_5	0.42 (0.06)	0.47 (0.00)	0.47 (0.00)	0.47 (0.00)	0.13 (0.09)	0.39 (0.07)	0.03 (0.03)
AR(1)	0.42 (0.02)	0.48 (0.02)	0.50 (0.03)	0.50 (0.03)	0.01 (0.02)	0.23 (0.15)	0.03 (0.03)
California	0.52 (0.10)	0.75 (0.17)	0.73 (0.03)	0.92 (0.08)	0.03 (0.05)	0.14 (0.17)	0.03 (0.03)
Covertype	0.38 (0.13)	0.47 (0.06)	0.50 (0.03)	0.50 (0.03)	0.14 (0.13)	0.47 (0.05)	0.03 (0.03)

Hit rate counts only the unique best card. Excess can fall while the hit rate stays near one half, when the screen stops serving the stale card and a blend is close.

Table 4: Post-shift hit rate, Gradual dying block. Fraction of rounds on the single best scorecard.

	UCB	dUCB	$\text{UCB}_{\eta=2}$	$\text{dUCB}_{\eta=2}$	TS	$\text{TS}_{\eta=2}$	$\varepsilon.10$
Gaussian $d = 30$	0.01 (0.02)	0.12 (0.08)	0.01 (0.02)	0.27 (0.02)	0.01 (0.02)	0.01 (0.02)	0.05 (0.02)
Gaussian $d = 12$	0.82 (0.10)	0.71 (0.07)	0.73 (0.08)	0.67 (0.07)	0.17 (0.17)	0.14 (0.18)	0.06 (0.02)
Student-t_5	0.00 (0.00)	0.12 (0.11)	0.01 (0.02)	0.29 (0.07)	0.00 (0.00)	0.00 (0.00)	0.04 (0.02)
AR(1)	0.62 (0.31)	0.90 (0.09)	0.78 (0.17)	0.86 (0.03)	0.01 (0.02)	0.10 (0.08)	0.05 (0.02)
California	0.76 (0.02)	0.89 (0.05)	0.84 (0.06)	0.91 (0.05)	0.03 (0.03)	0.08 (0.08)	0.07 (0.03)
Covertype	0.05 (0.10)	0.18 (0.28)	0.12 (0.23)	0.27 (0.31)	0.01 (0.02)	0.03 (0.07)	0.04 (0.02)

Hit rate counts only the unique best card. Excess can fall while the hit rate stays near one half, when the screen stops serving the stale card and a blend is close.

Table 5: Post-shift hit rate, Gradual sign flip. Fraction of rounds on the single best scorecard.

	UCB	dUCB	$\text{UCB}_{\eta=2}$	$\text{dUCB}_{\eta=2}$	TS	$\text{TS}_{\eta=2}$	$\varepsilon.10$
Gaussian $d = 30$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.03 (0.03)
Gaussian $d = 12$	0.18 (0.18)	0.17 (0.12)	0.07 (0.11)	0.06 (0.08)	0.06 (0.06)	0.03 (0.05)	0.03 (0.03)
Student-t_5	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.03 (0.03)
AR(1)	0.31 (0.18)	0.26 (0.11)	0.31 (0.09)	0.25 (0.13)	0.03 (0.05)	0.03 (0.05)	0.03 (0.03)
California	0.35 (0.06)	0.37 (0.09)	0.42 (0.06)	0.32 (0.12)	0.03 (0.07)	0.08 (0.10)	0.03 (0.03)
Covertype	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.03 (0.03)

Hit rate counts only the unique best card. Excess can fall while the hit rate stays near one half, when the screen stops serving the stale card and a blend is close.

Table 6: Post-shift hit rate, Gradual noise growth. Fraction of rounds on the single best scorecard.

	UCB	dUCB	$\text{UCB}_{\eta=2}$	$\text{dUCB}_{\eta=2}$	TS	$\text{TS}_{\eta=2}$	$\varepsilon.10$
Gaussian $d = 30$	0.93 (0.00)	0.88 (0.02)	0.93 (0.00)	0.88 (0.03)	0.99 (0.02)	1.00 (0.00)	0.94 (0.03)
Gaussian $d = 12$	0.82 (0.02)	0.67 (0.00)	0.83 (0.03)	0.68 (0.02)	0.88 (0.02)	0.90 (0.03)	0.89 (0.04)
Student-t_5	0.93 (0.03)	0.88 (0.02)	0.96 (0.02)	0.87 (0.00)	0.99 (0.02)	0.99 (0.02)	0.94 (0.03)
AR(1)	0.76 (0.02)	0.59 (0.02)	0.77 (0.03)	0.61 (0.02)	0.87 (0.00)	0.87 (0.03)	0.89 (0.04)
California	0.64 (0.02)	0.50 (0.00)	0.66 (0.02)	0.52 (0.02)	0.82 (0.06)	0.84 (0.06)	0.88 (0.06)
Covertype	0.96 (0.02)	0.90 (0.00)	0.96 (0.02)	0.92 (0.02)	0.99 (0.02)	1.00 (0.00)	0.94 (0.03)

Hit rate counts only the unique best card. Excess can fall while the hit rate stays near one half, when the screen stops serving the stale card and a blend is close.

5 Why epsilon-greedy loses at this batch size

Epsilon-greedy buys robustness by spending a fixed fraction of every batch on a uniform draw. That price is proportional to the batch size times the gap between a random scorecard and the best one. At 200 rows the gap is paid in full on each exploratory round, before the shift and after it. On gradual relocation the post-shift median rises from 101 at 5% to 125 at 10%, and the 15% rule is 83 only because a few seeds stumbled onto the fresh block, not because the explore rate is well tuned. None of the three is close to UCB, and putting the $\eta = 1$ tax on the 10% rule only brings the median from 125 to 72.5. The uniform draws still fire after the share has already identified the stale block.

This does not contradict the small-batch finding that a few percent of exploration is useful when nothing else is watching the environment. Here something else is watching. The share is computed from logged predictions of every deployed score, on a window of 1,200 rows, and it does not consume a served batch. Uniform traffic is then the expensive sensor, and the residual share is the cheap one. The stationary period already shows the toll: epsilon at 10% has higher early excess on every family, while the tax does not.

The scale of the tax matters as much as its presence. $\eta\sigma w_k$ is in reward units. A full share at $\eta = 1$ is one reference-window standard deviation, which nudges the index without erasing it. Absolute $\lambda = 5$ matches that nudge on the Gaussian relocation, where σ is a few units, and it is too small on the California nonlinear add-on, where the same $\lambda = 5$ stays next to untaxed UCB while the scaled tax drops the excess. One η travels across the six pools. One raw λ does not. Values $\eta > 2$ are not needed for the subspace moves, and they are actively harmful on a sign flip, which is the reason the grid stops at 2.

6 The same share on policies

Four linear policies, one per feature block plus a zero policy, run for four steps. The environment coefficient ramps from the first block to the second, and in the double path onward to the third. There is no jump between rounds. The same share is refreshed on a fixed calendar, because the next batch is not announced. The scalar score of a policy is the return it would earn if the world matched its own coefficient. The residual of that score is again the share.

Phase 0 is unaffected by the tax, as in the scorecard experiment. After the move, UCB at $\eta = 2$ is the lowest median on all six pools in the single-shift path, and it is below discounted UCB. Thompson sampling needs the larger tax: at $\eta = 2$ its phase-1 median falls from 49.6 to 22.6, still above discounted UCB. Epsilon-greedy remains the most expensive rule in both phases.

The monitor is the practical question. The oracle share uses every policy's realized return. The fitted monitor does not. It fits the reward coefficient by least squares on the transitions of the served policy, imagines every policy's return under that fit, and only then computes the residual share. On the single move, UCB at $\eta = 1$ with the fitted monitor has median phase-1 excess 16.2, the same as the oracle tax at $\eta = 1$ and the same as discounted UCB. On the second move of the double path the fitted monitor stays with the oracle: phase-2 medians are 11.0 for oracle UCB at $\eta = 2$, 14.6 for the fitted monitor at $\eta = 1$, 16.4 for discounted UCB, and 42.6 for plain UCB. The on-policy monitor, which assigns residual 1 to any policy missing from a window, does not survive the second move. Its phase-2 median is 96, worse than plain UCB. A policy that was not served in the last four rounds is not a high-residual policy. Treating missingness as risk taxes the arm one most needs to inspect.

So the deployable version of this horse-race is not "log only what you served." It is "fit the

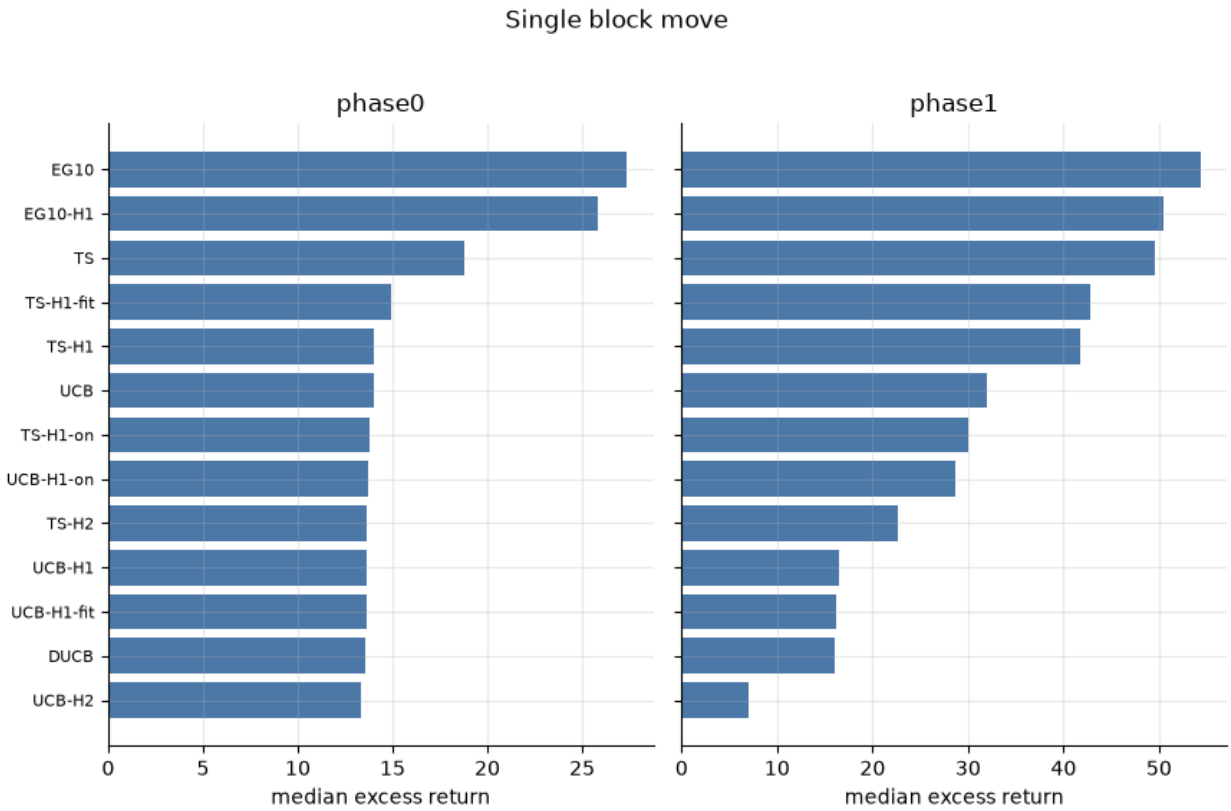


Figure 4: Single block move. Phase 0, before the new block has majority weight, is essentially tied for every rule except epsilon-greedy. Phase 1 median excess return is 7.1 for UCB at $\eta = 2$, 16.0 for discounted UCB, 16.2 for the fitted monitor at $\eta = 1$, 32.0 for plain UCB, and 54.4 for epsilon-greedy at 10%.

reward on what you served, imagine the other policies, and put the residual share on a discounted or undiscounted UCB index, with η at 1 or 2." Turn the tax off when the share collapses onto a single null score and the live scores are tied. That pattern is the sign-flip failure, and it is readable from the weights alone.

7 Operating summary

The operating assumption is a ramp. The next batch is unknown, so the share is refreshed from the latest window on a fixed calendar. No rule below is tuned for a jump.

What the share does	Screen the score that no longer explains the batch. Release the score the ramp has moved onto.
Relocation	UCB or discounted UCB at $\eta = 2$. Median post excess 17.6, against 34.6, 24.6, and 125 for UCB, discounted UCB, and $\varepsilon = 0.10$. The gap clears the seed noise on all six pools against UCB.
Dying block	Discounted UCB at $\eta = 2$. Median 27.1, against 41.6 and 52.2.
Nonlinear add-on	Discounted UCB at $\eta = 2$. Median 2.80, against 3.10 and 8.48. The gain over discounting alone is inside the seed noise; the gain over UCB is not.
Sign flip along the ramp	Tax off. Discount the reward. Median 99.2, against 166 for UCB and 185 for UCB at $\eta = 1$.
Noise growth	Thompson sampling with a small tax. Median 0.90 at $\lambda = 5$, 1.16 at $\eta = 2$. Do not discount.
Before the ramp	The tax is inert. Epsilon-greedy is not.
Policies	Same ramp. Imagine unserved policies from a reward fit. Do not impute a residual of 1 for a policy you did not serve.

Batch size 200 and pools of at least 10,000 rows are what make the comparison sharp. The pseudo-outcome is estimated on 1,200 rows in each window, and an exploratory batch is too expensive to spend uniformly once that estimate is available.

A Figures

The figures in this appendix that carry a jump in the mixing weight are not part of the claim. The claim is the ramp: the weight is zero through round 20 and then linear.

A.1 Cumulative excess

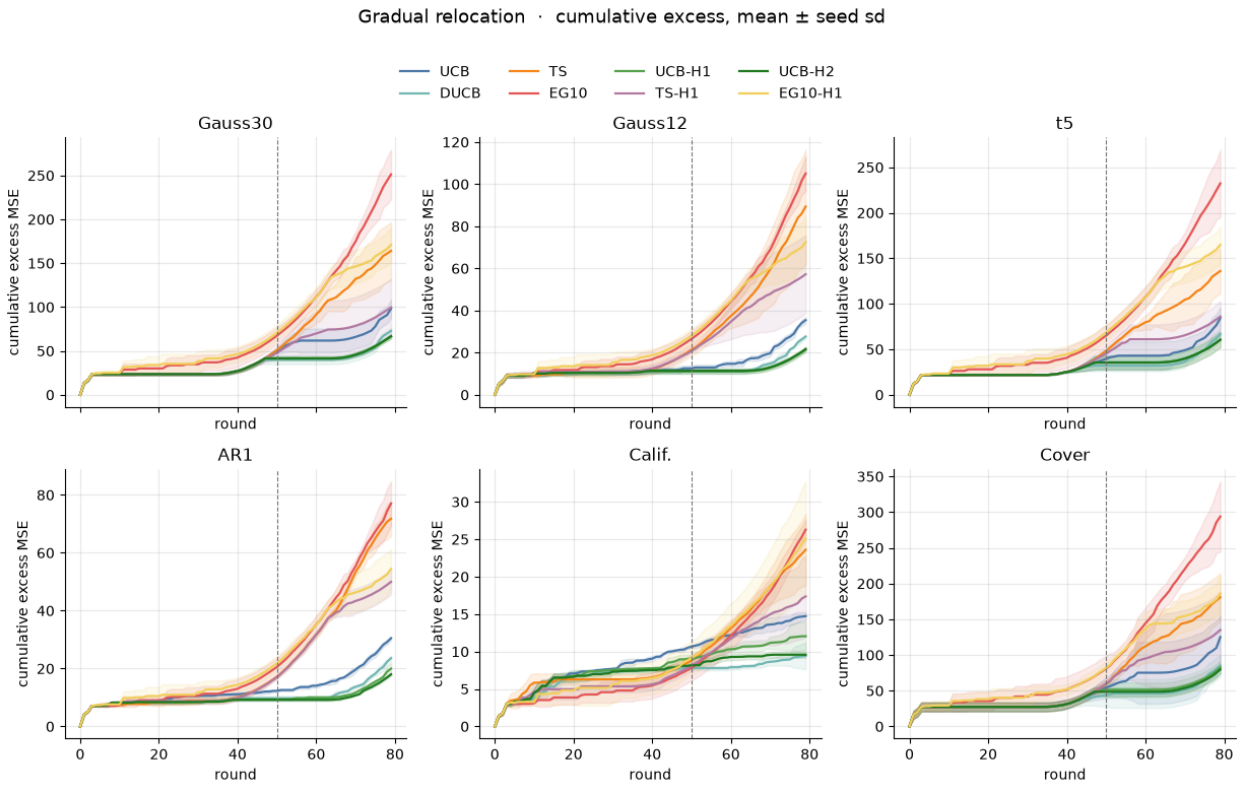


Figure 5: Cumulative excess MSE, gradual relocation. Mean across four seeds, band is the seed standard deviation. The dashed line is the first round with mixing weight at least 1/2.

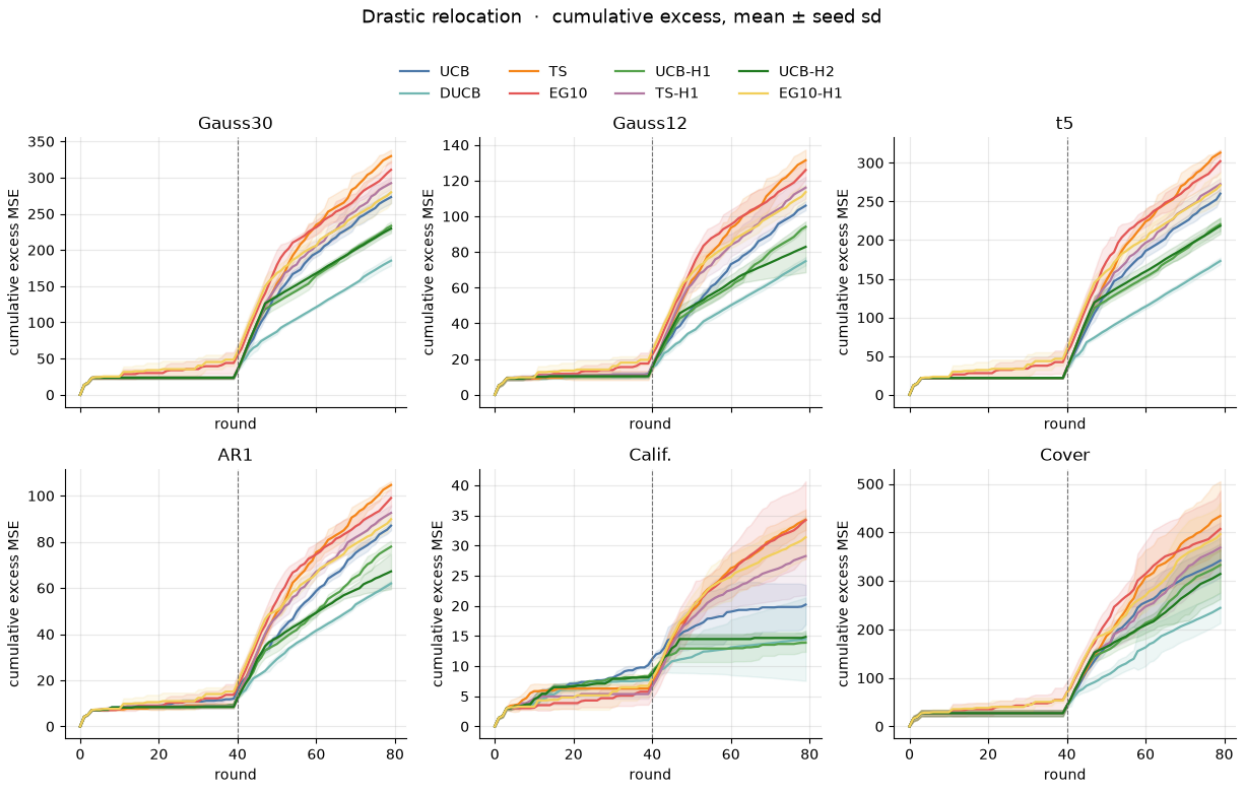


Figure 6: Cumulative excess MSE, drastic relocation. Mean across four seeds, band is the seed standard deviation. The dashed line is the first round with mixing weight at least 1/2.

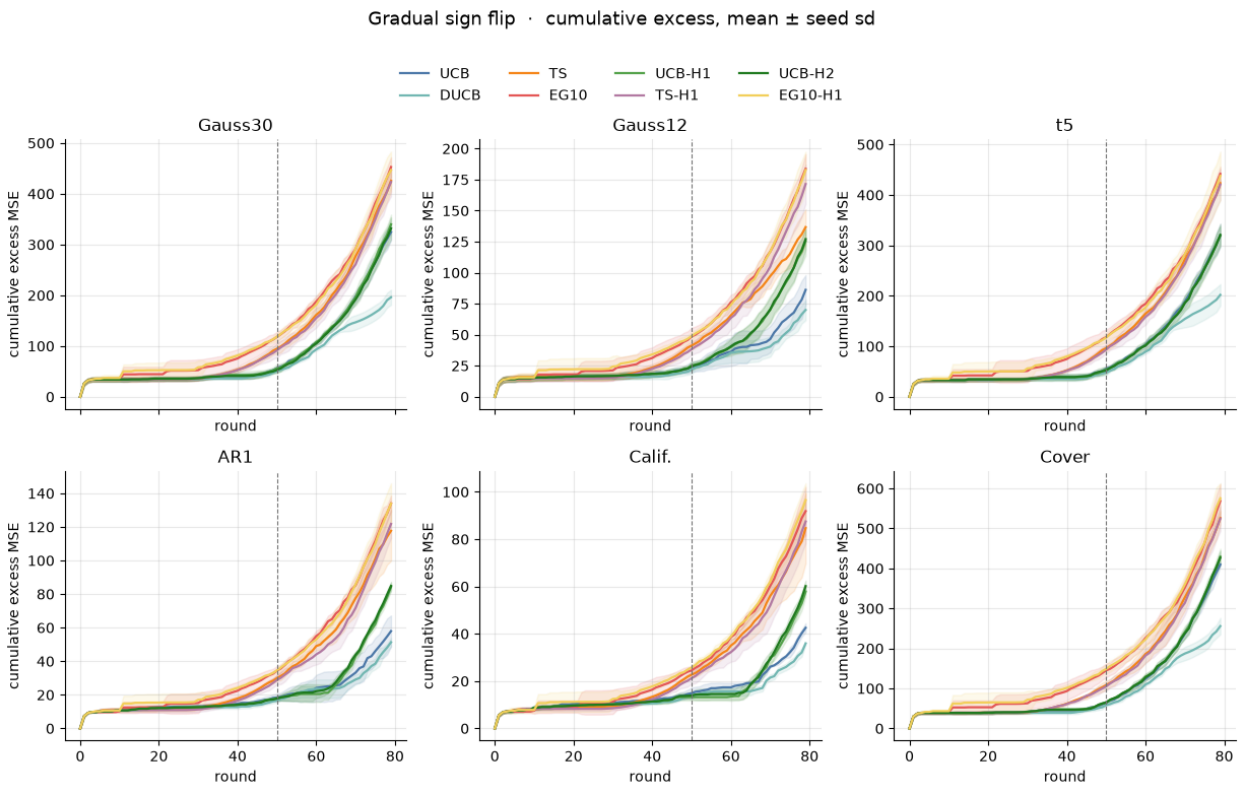


Figure 7: Cumulative excess MSE, gradual sign flip. Mean across four seeds, band is the seed standard deviation. The dashed line is the first round with mixing weight at least 1/2.

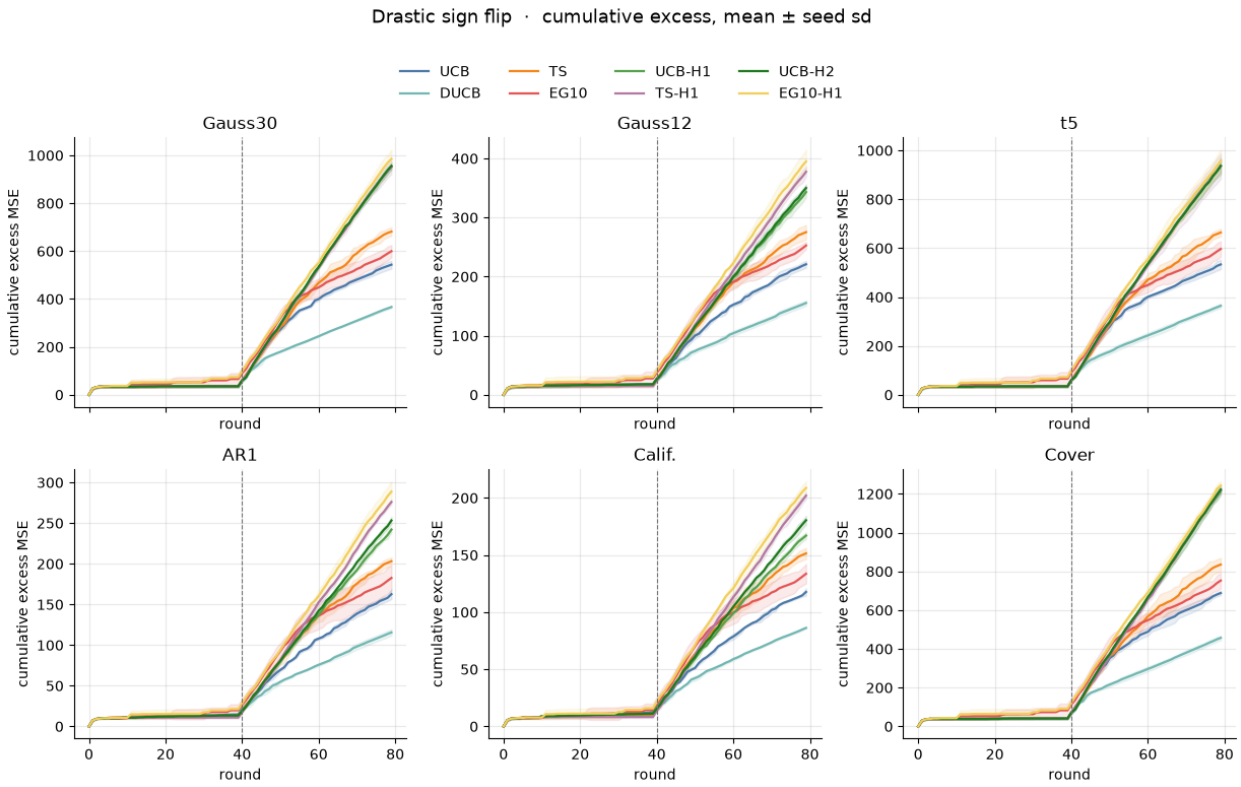


Figure 8: Cumulative excess MSE, drastic sign flip. Mean across four seeds, band is the seed standard deviation. The dashed line is the first round with mixing weight at least 1/2.

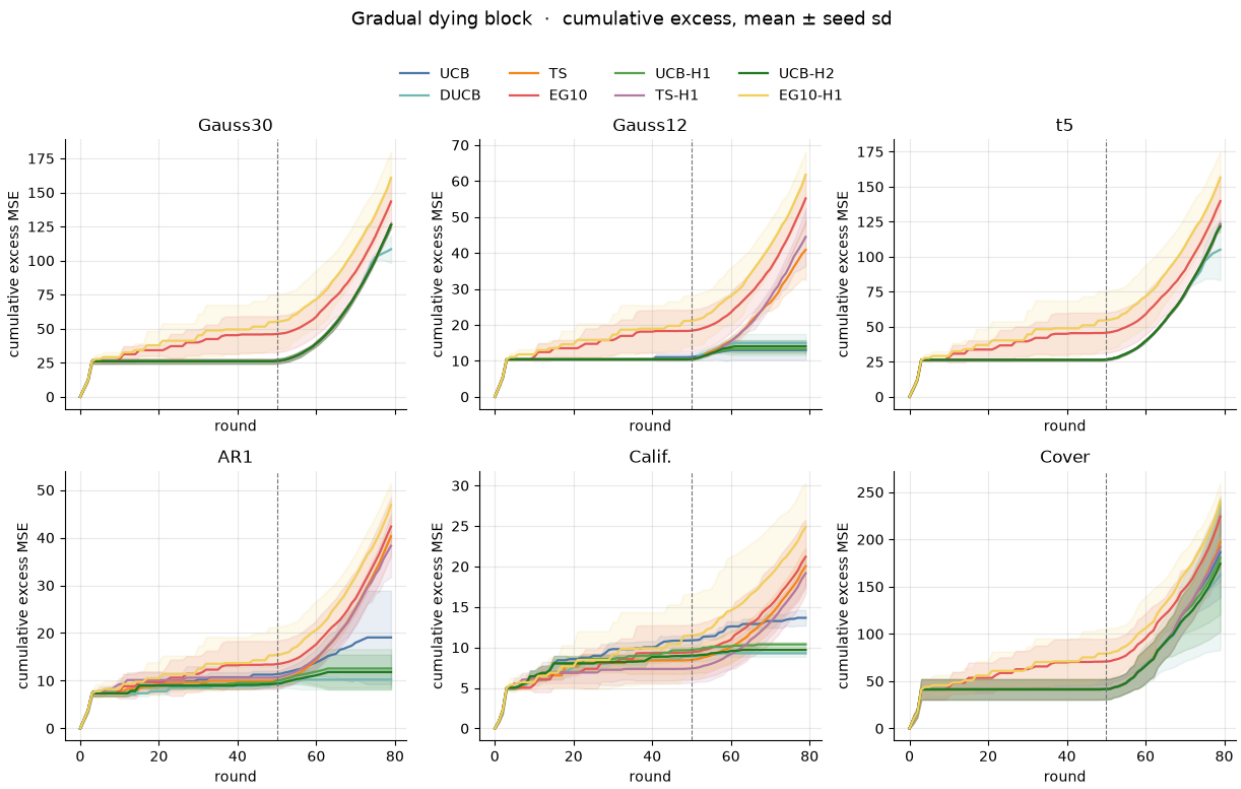


Figure 9: Cumulative excess MSE, gradual dying block. Mean across four seeds, band is the seed standard deviation. The dashed line is the first round with mixing weight at least 1/2.

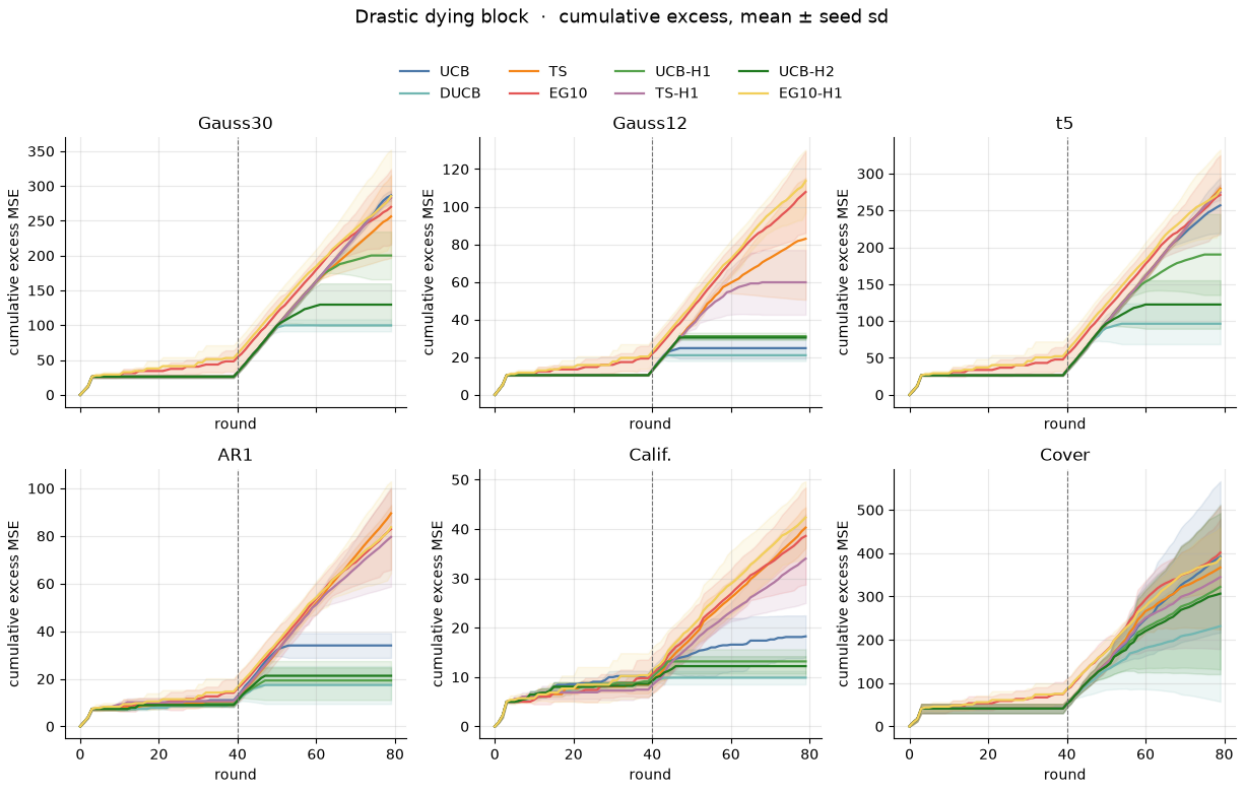


Figure 10: Cumulative excess MSE, drastic dying block. Mean across four seeds, band is the seed standard deviation. The dashed line is the first round with mixing weight at least 1/2.

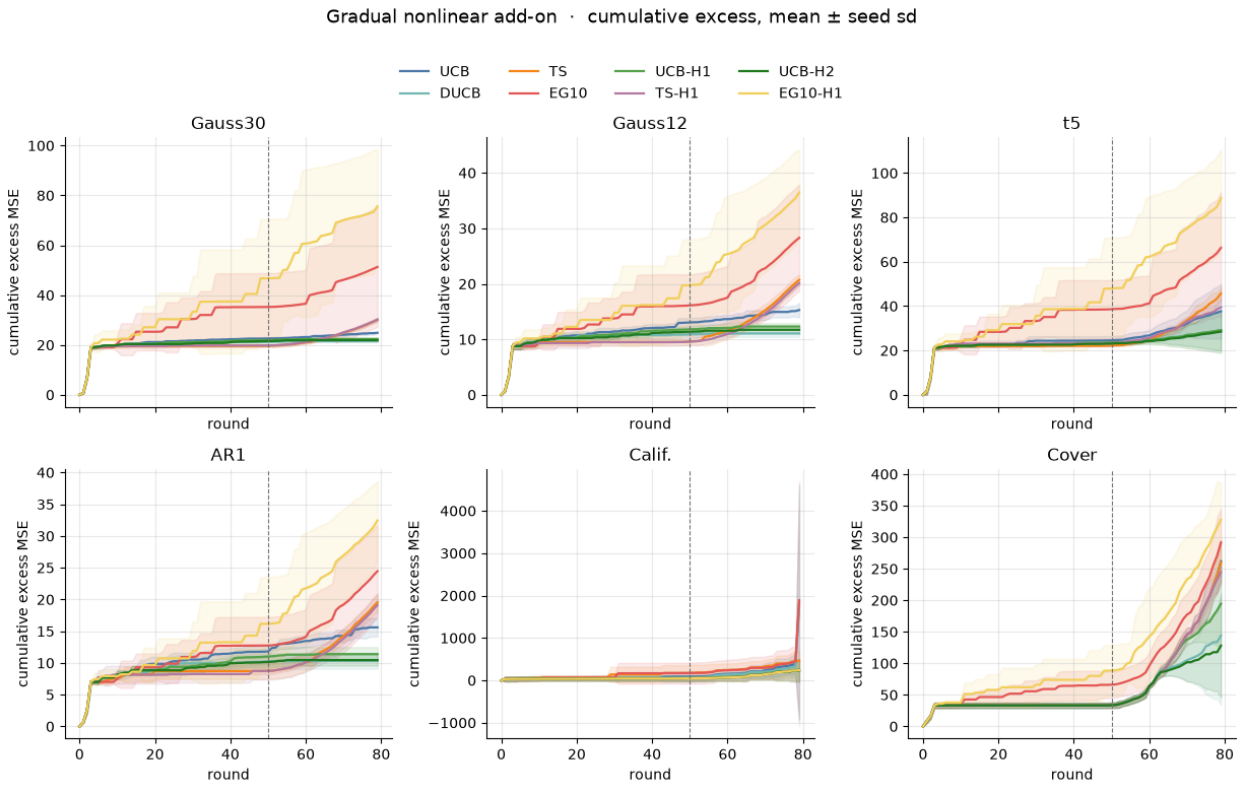


Figure 11: Cumulative excess MSE, gradual nonlinear add-on. Mean across four seeds, band is the seed standard deviation. The dashed line is the first round with mixing weight at least 1/2.

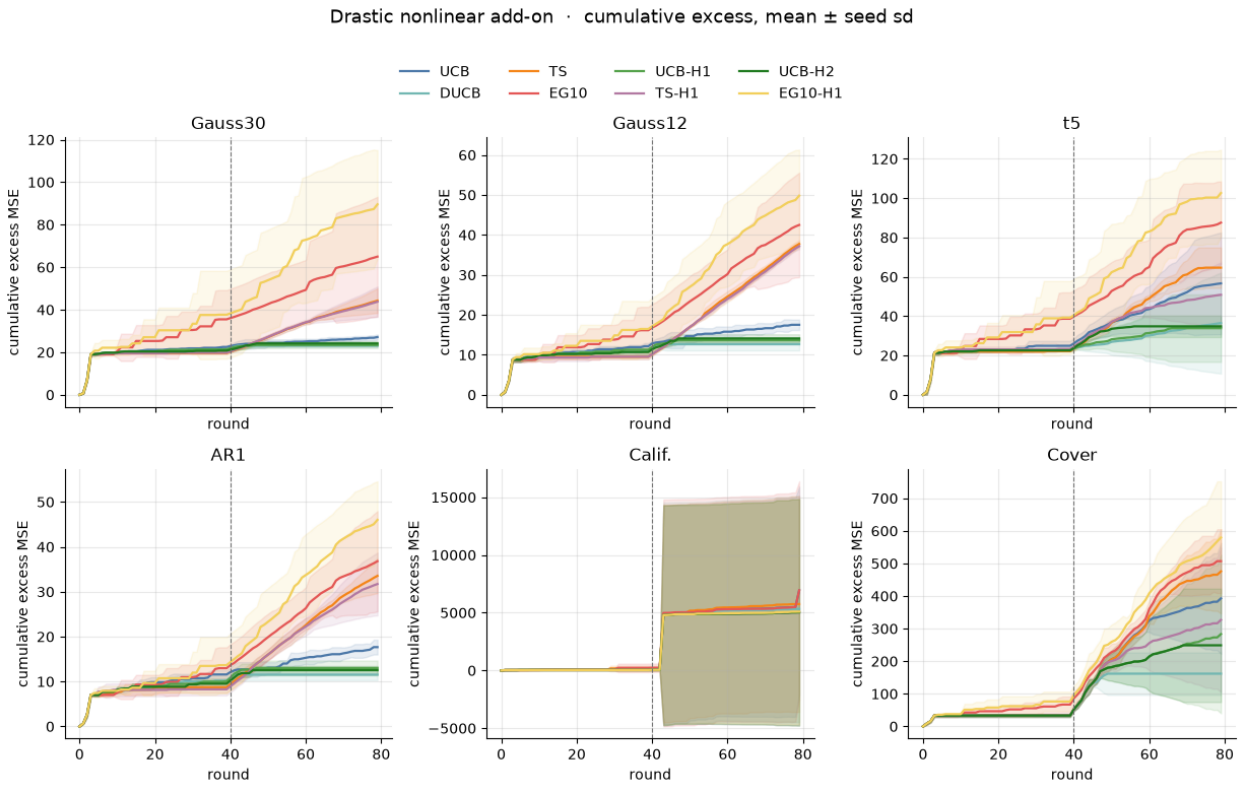


Figure 12: Cumulative excess MSE, drastic nonlinear add-on. Mean across four seeds, band is the seed standard deviation. The dashed line is the first round with mixing weight at least 1/2.

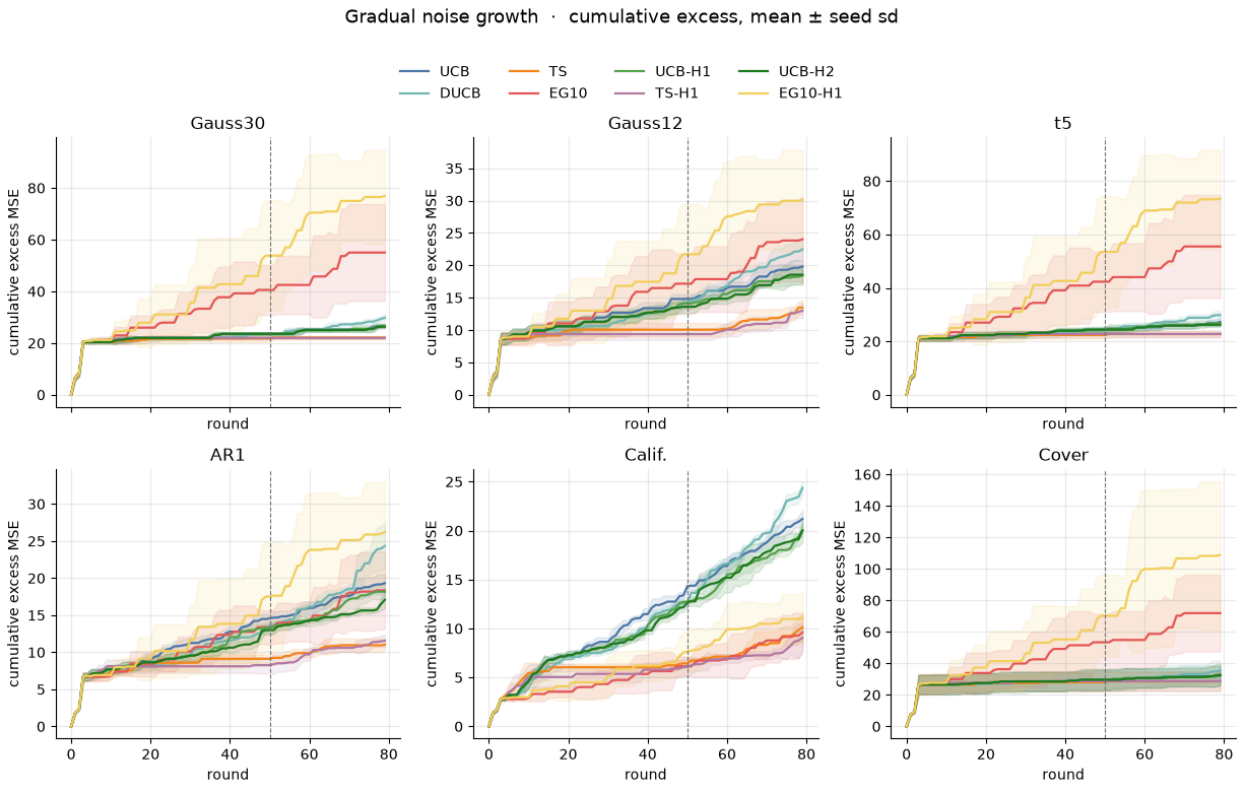


Figure 13: Cumulative excess MSE, gradual noise growth. Mean across four seeds, band is the seed standard deviation. The dashed line is the first round with mixing weight at least 1/2.

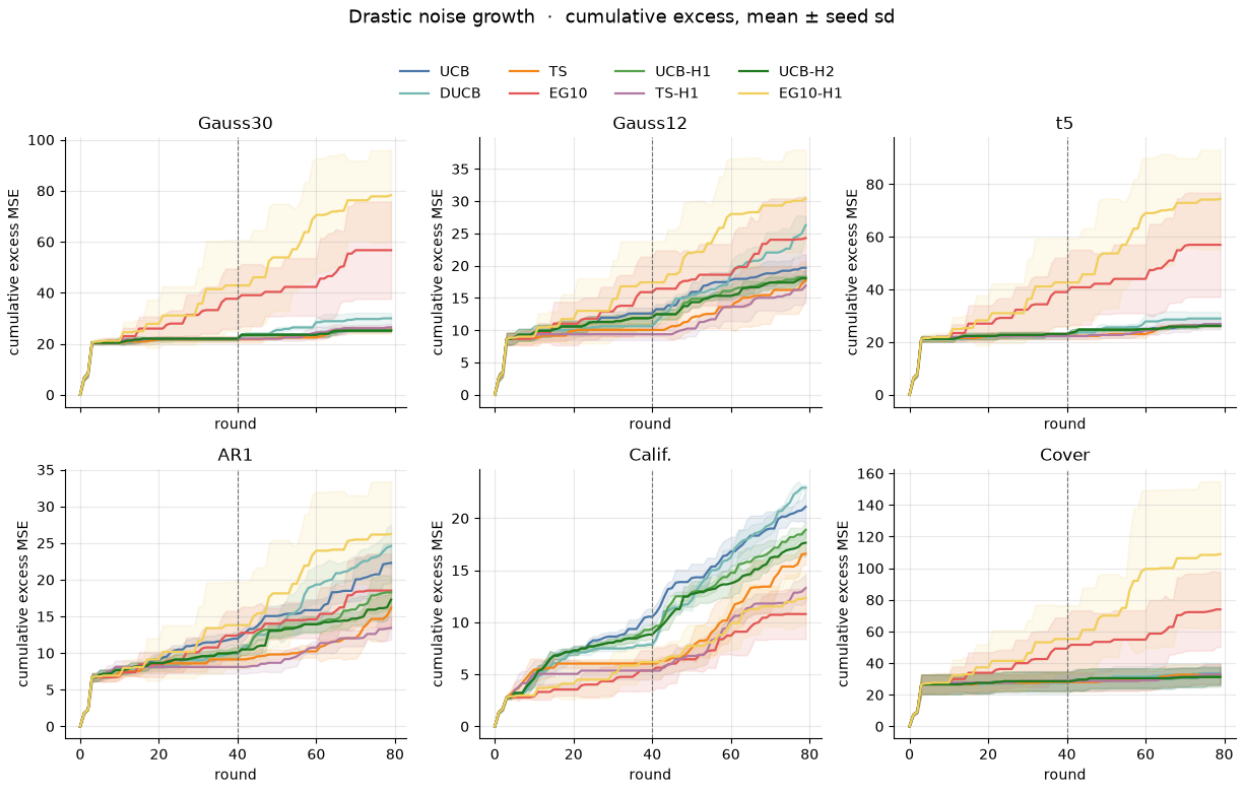


Figure 14: Cumulative excess MSE, drastic noise growth. Mean across four seeds, band is the seed standard deviation. The dashed line is the first round with mixing weight at least 1/2.

A.2 PO-risk shares

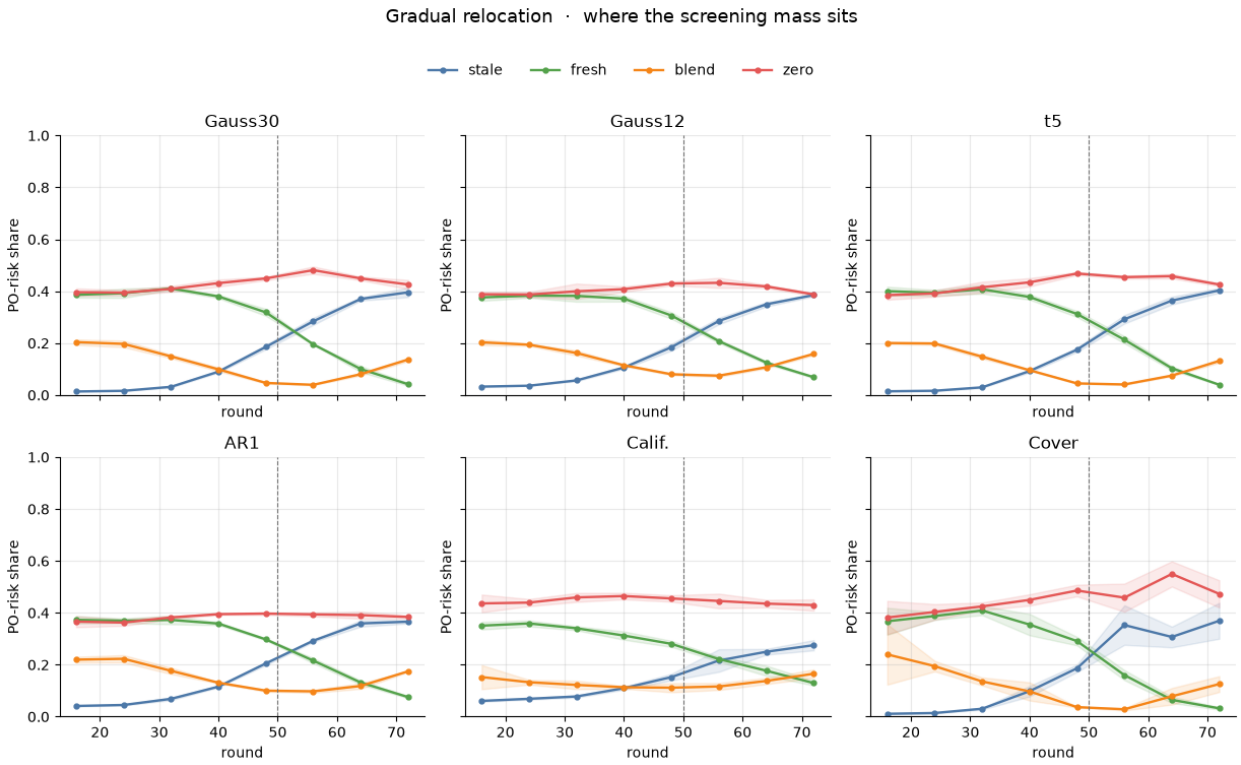


Figure 15: PO-risk shares, gradual relocation. Mean across four seeds.

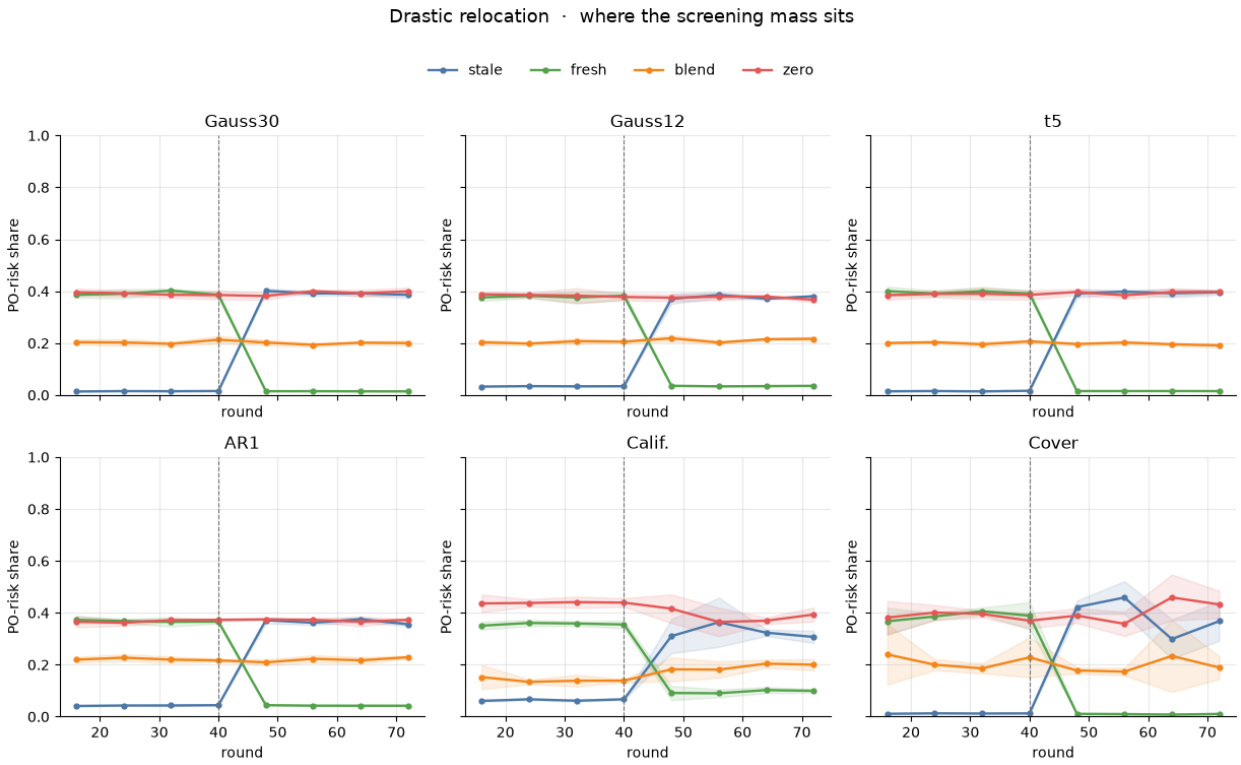


Figure 16: PO-risk shares, drastic relocation. Mean across four seeds.

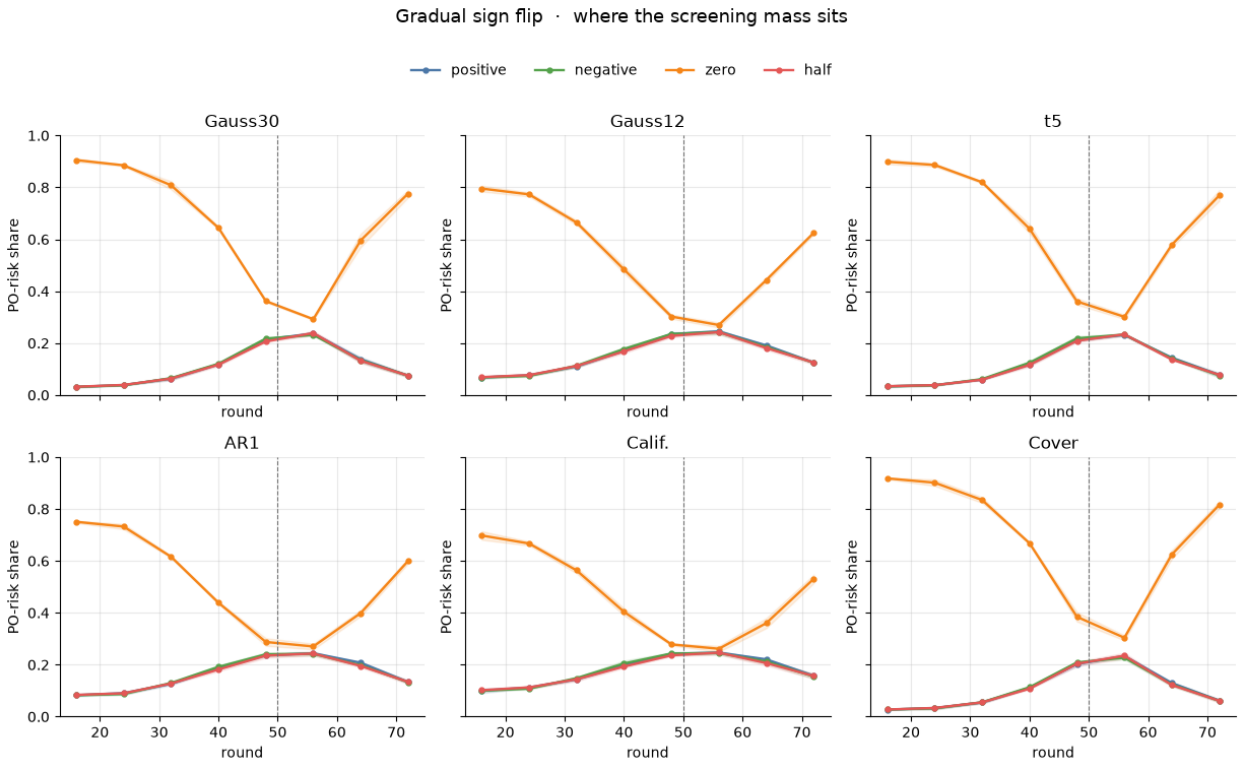


Figure 17: PO-risk shares, gradual sign flip. Mean across four seeds.

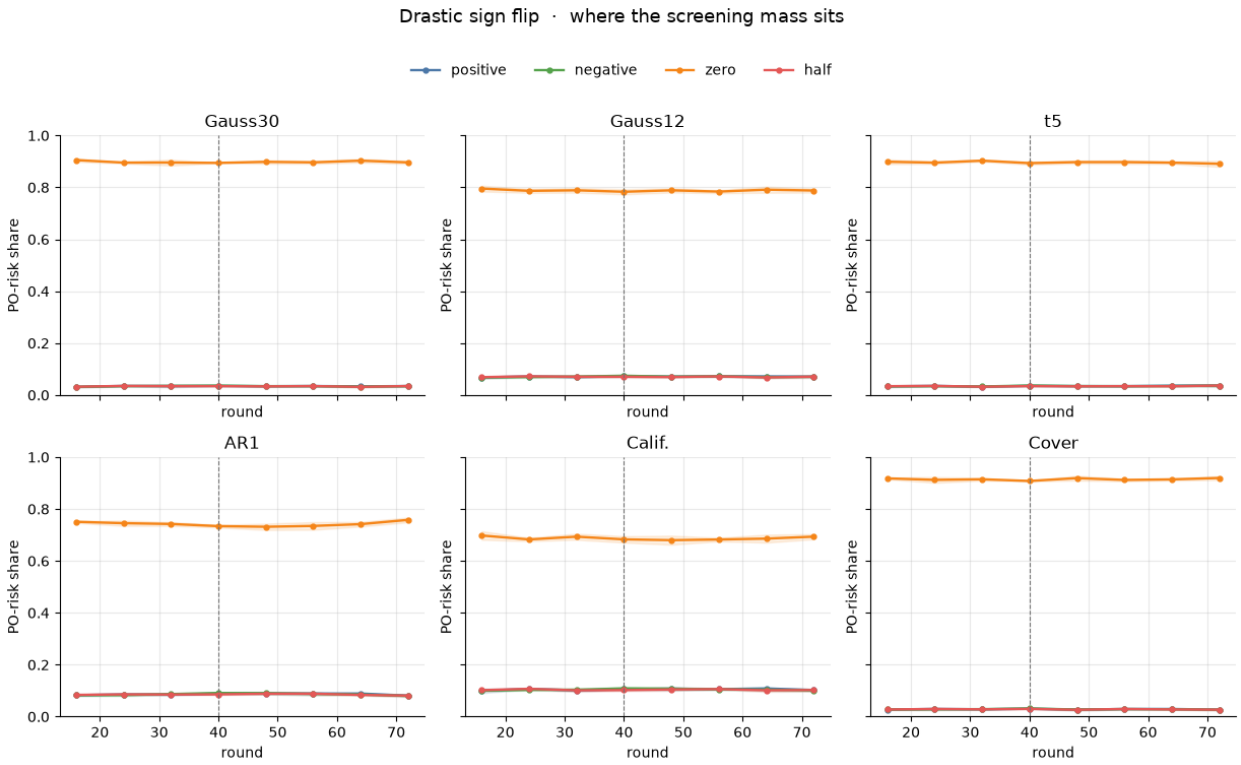


Figure 18: PO-risk shares, drastic sign flip. Mean across four seeds.

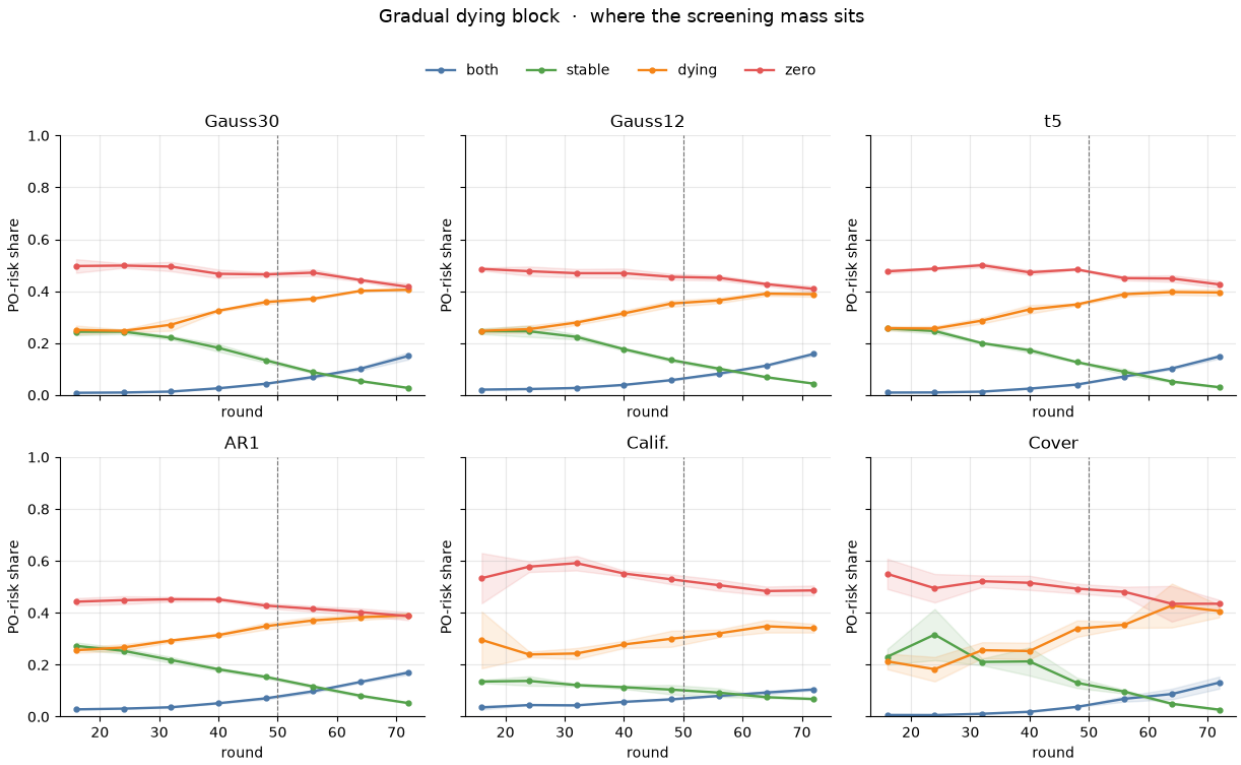


Figure 19: PO-risk shares, gradual dying block. Mean across four seeds.

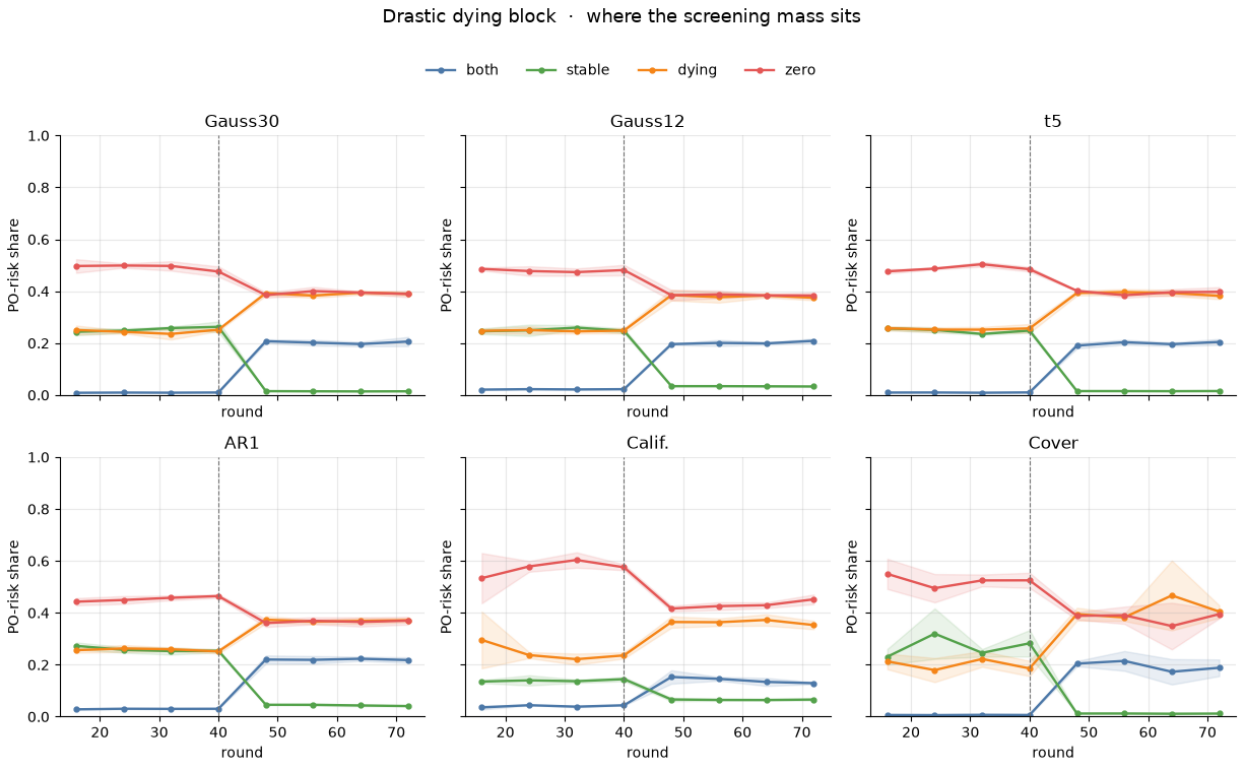


Figure 20: PO-risk shares, drastic dying block. Mean across four seeds.

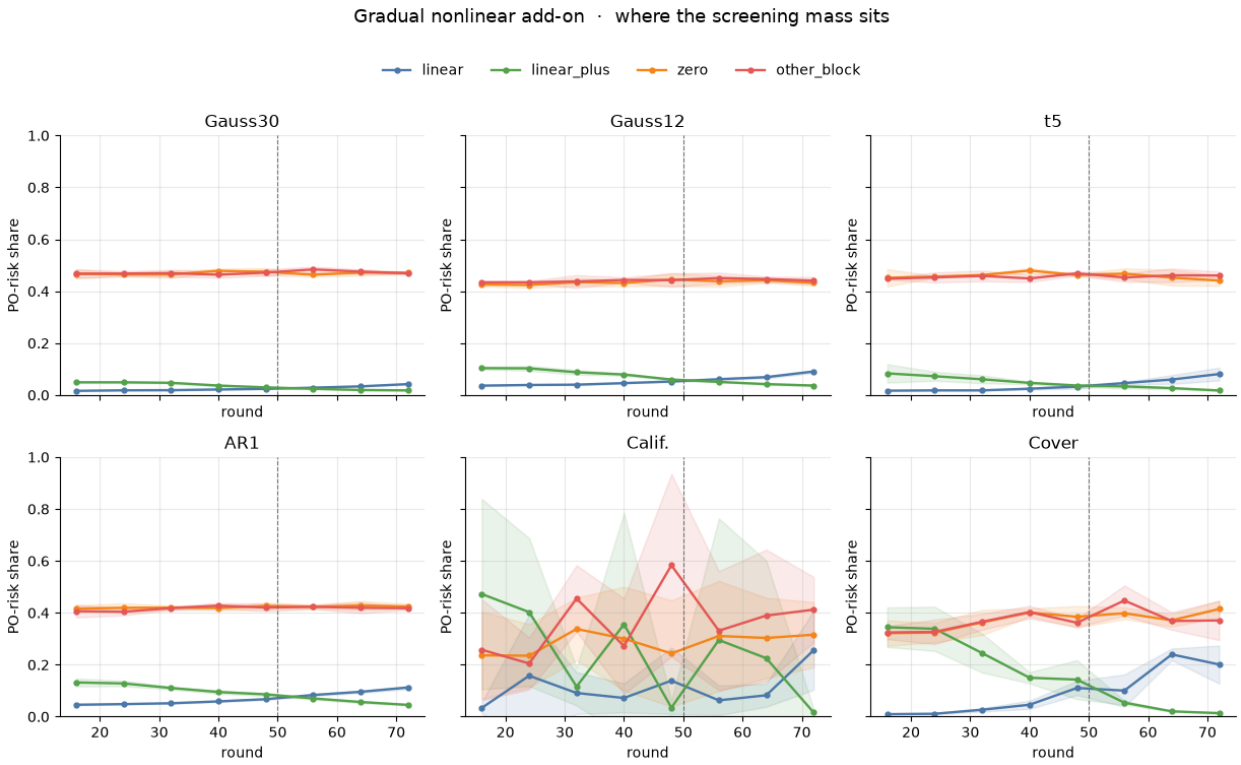


Figure 21: PO-risk shares, gradual nonlinear add-on. Mean across four seeds.

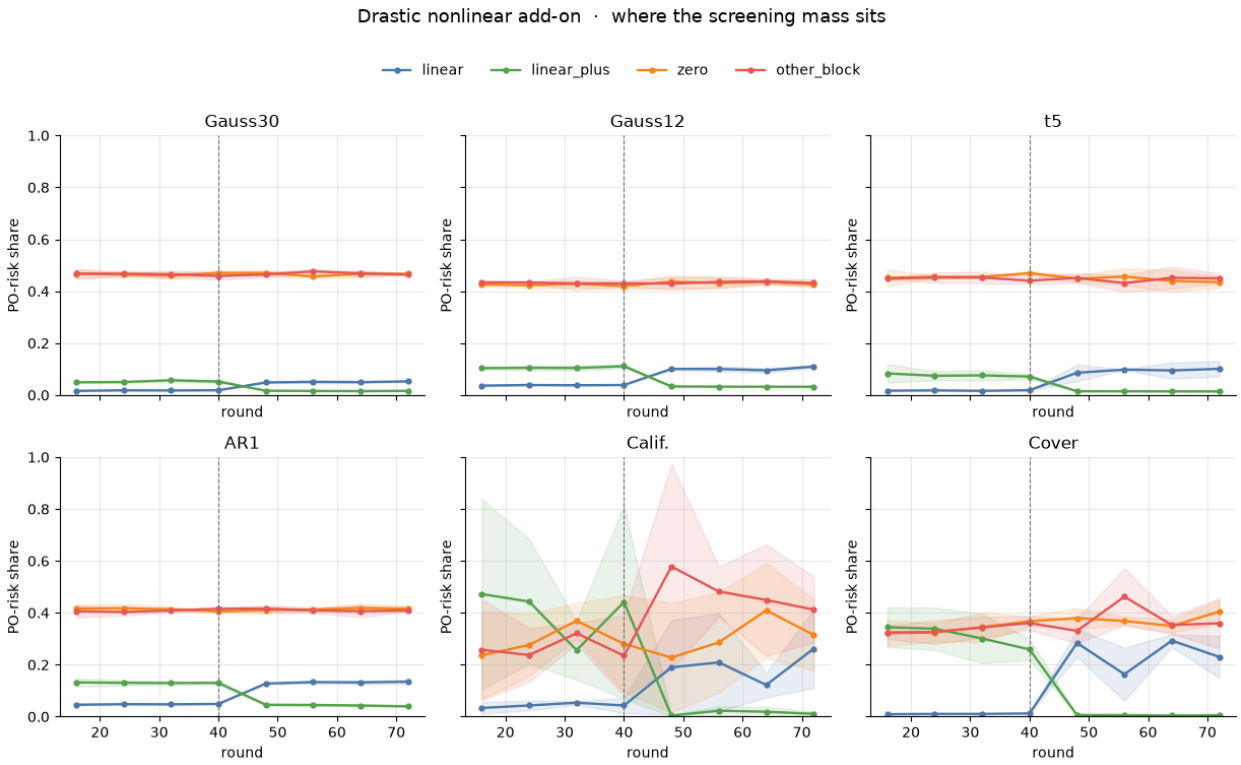


Figure 22: PO-risk shares, drastic nonlinear add-on. Mean across four seeds.

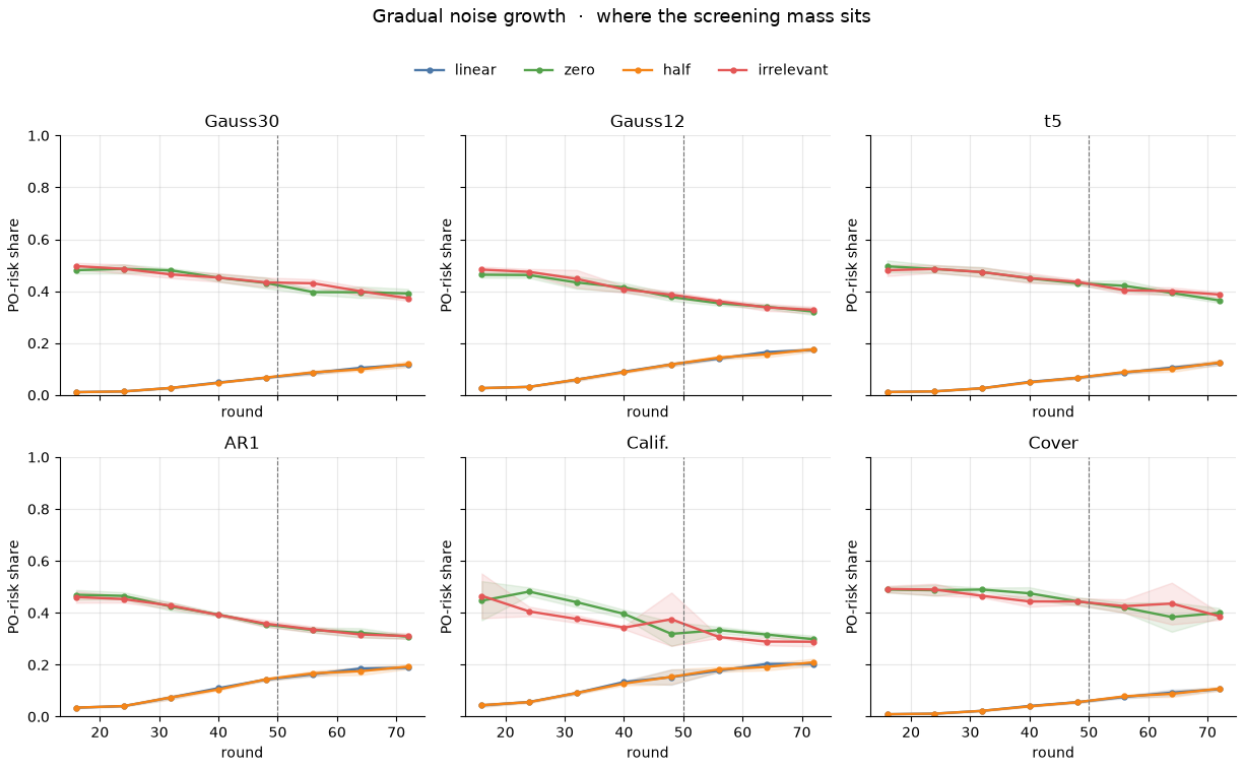


Figure 23: PO-risk shares, gradual noise growth. Mean across four seeds.

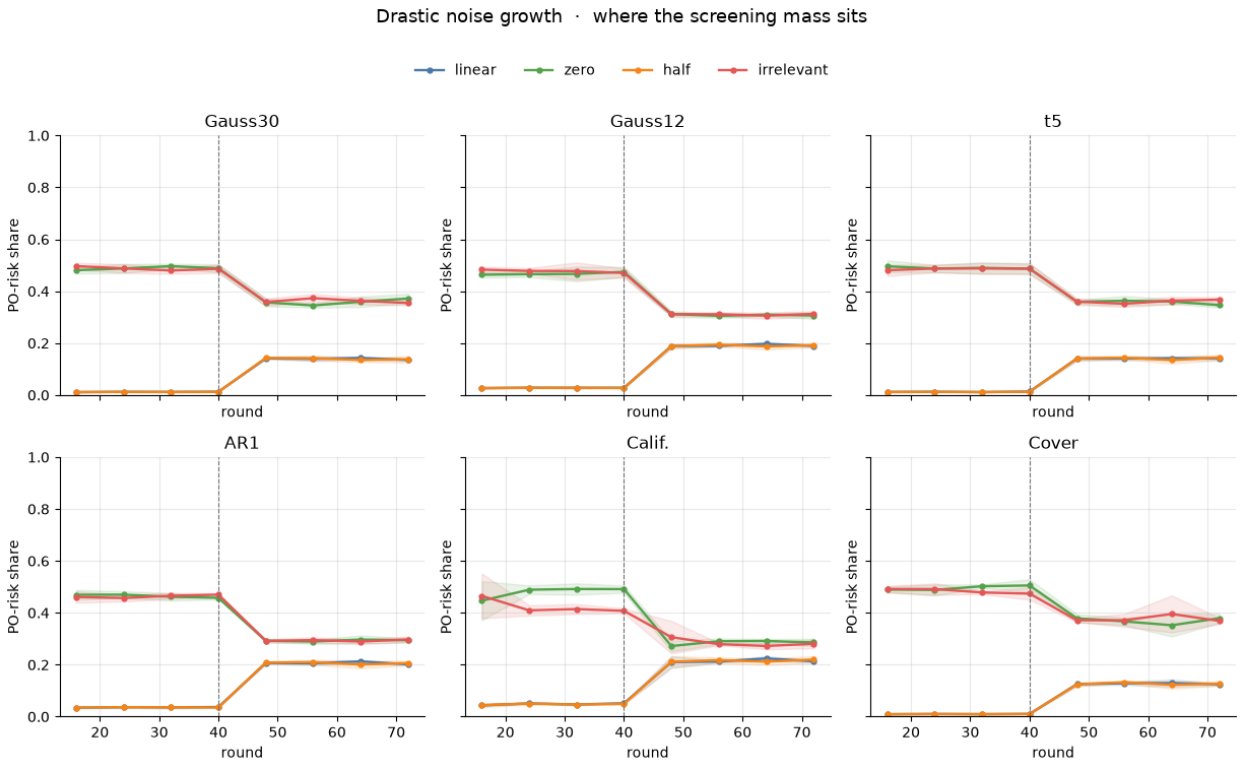


Figure 24: PO-risk shares, drastic noise growth. Mean across four seeds.

A.3 Median rankings and the scale profile

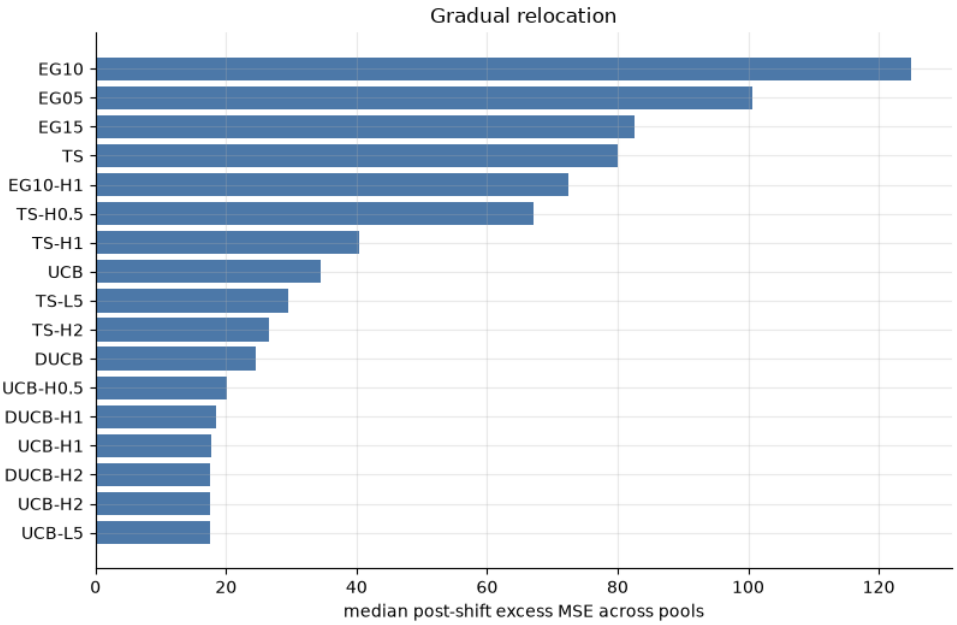


Figure 25: Median post-shift excess across the six pools, gradual relocation. Lower is better.

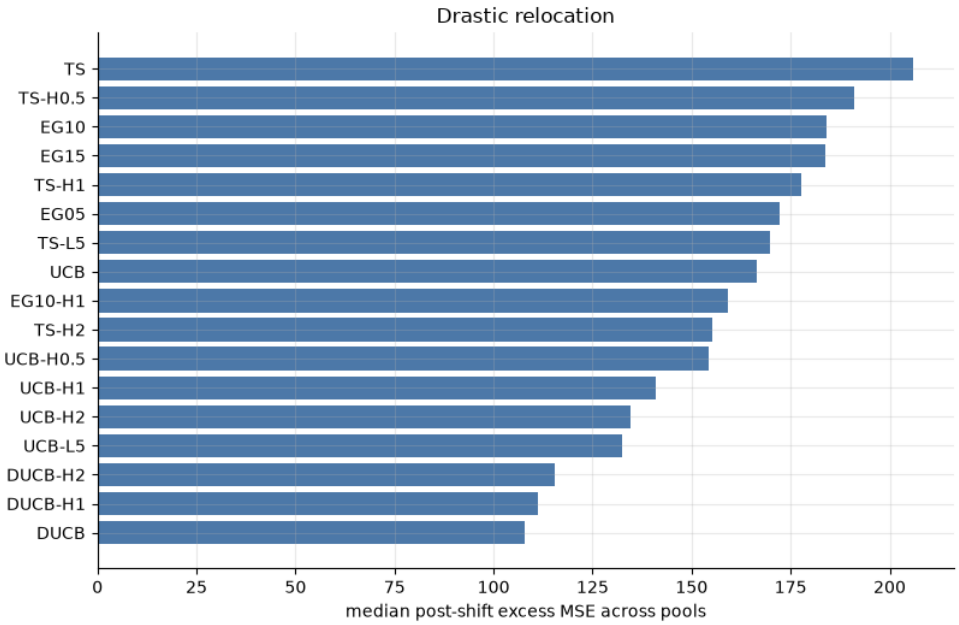


Figure 26: Median post-shift excess across the six pools, drastic relocation. Lower is better.

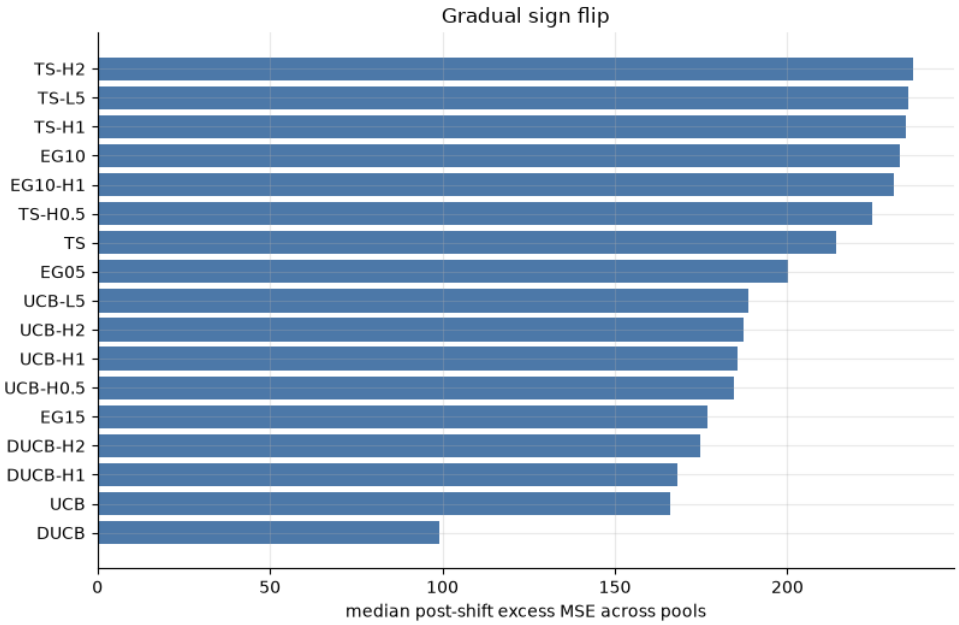


Figure 27: Median post-shift excess across the six pools, gradual sign flip. Lower is better.

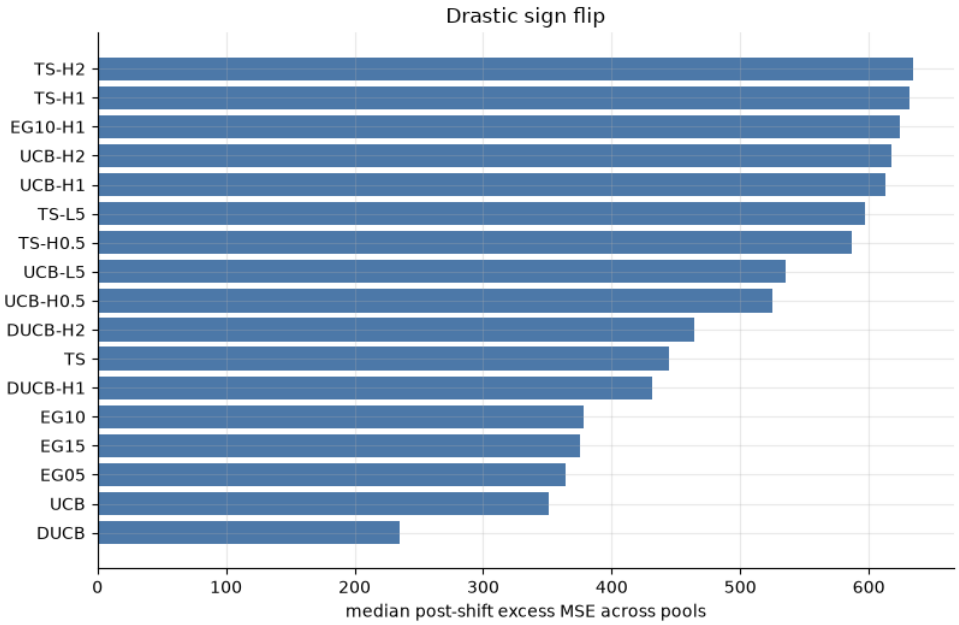


Figure 28: Median post-shift excess across the six pools, drastic sign flip. Lower is better.

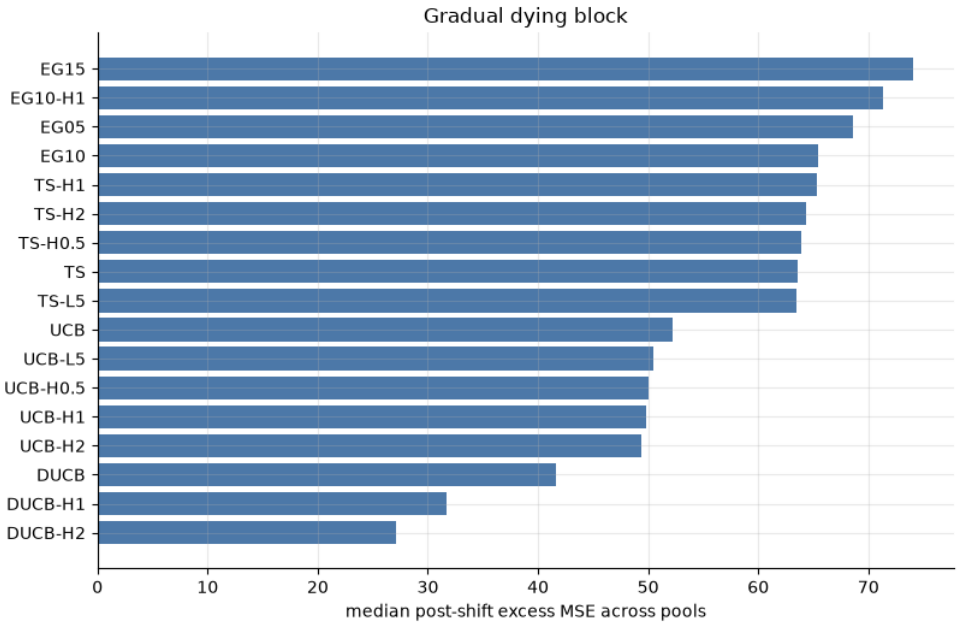


Figure 29: Median post-shift excess across the six pools, gradual dying block. Lower is better.

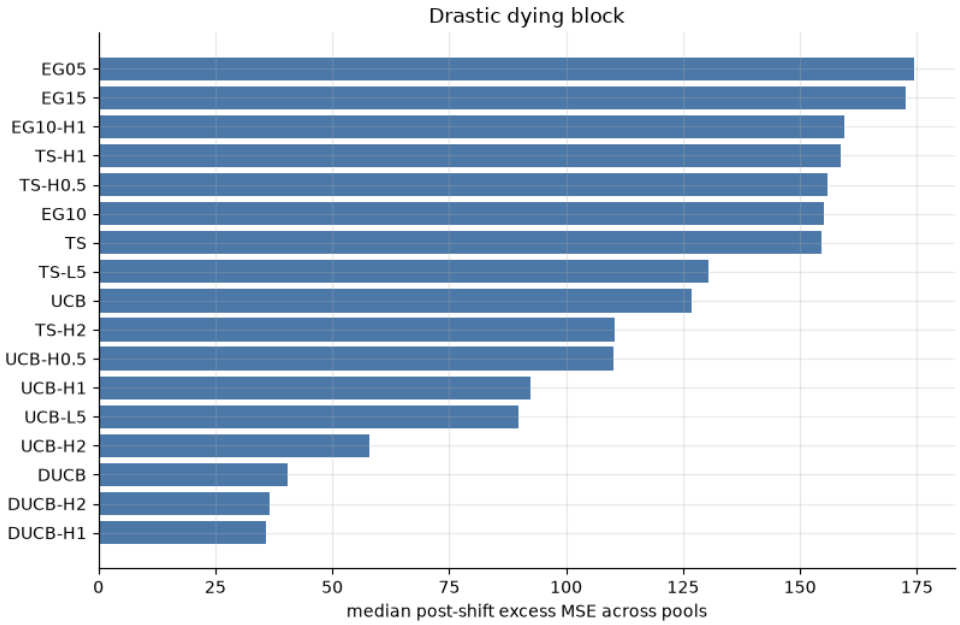


Figure 30: Median post-shift excess across the six pools, drastic dying block. Lower is better.

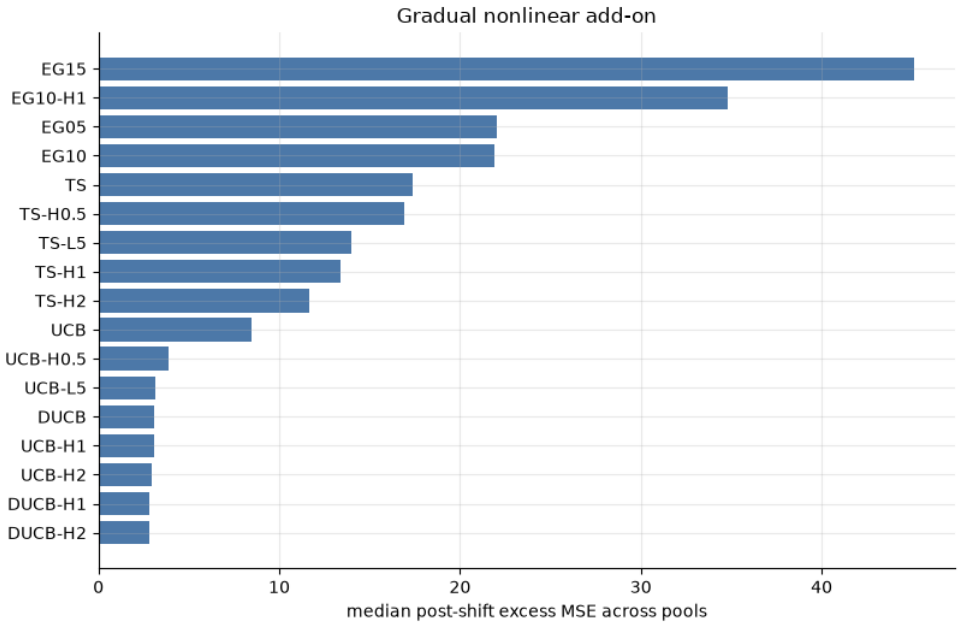


Figure 31: Median post-shift excess across the six pools, gradual nonlinear add-on. Lower is better.

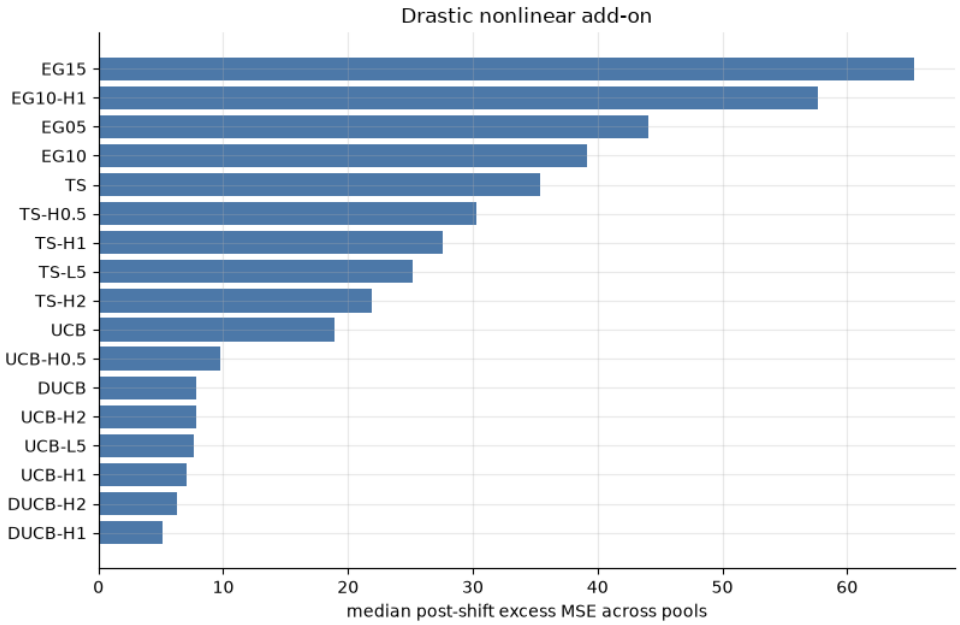


Figure 32: Median post-shift excess across the six pools, drastic nonlinear add-on. Lower is better.

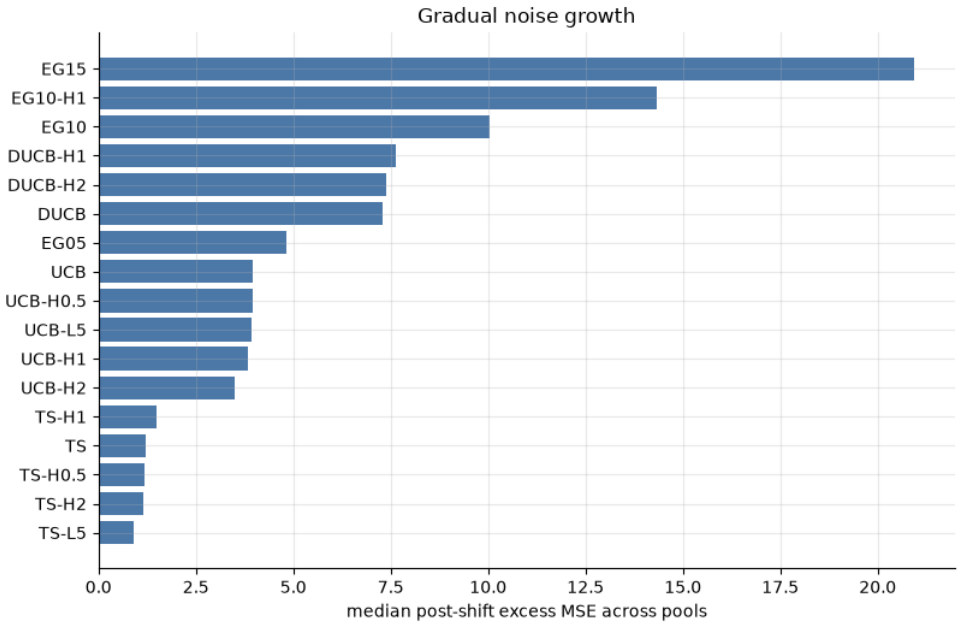


Figure 33: Median post-shift excess across the six pools, gradual noise growth. Lower is better.

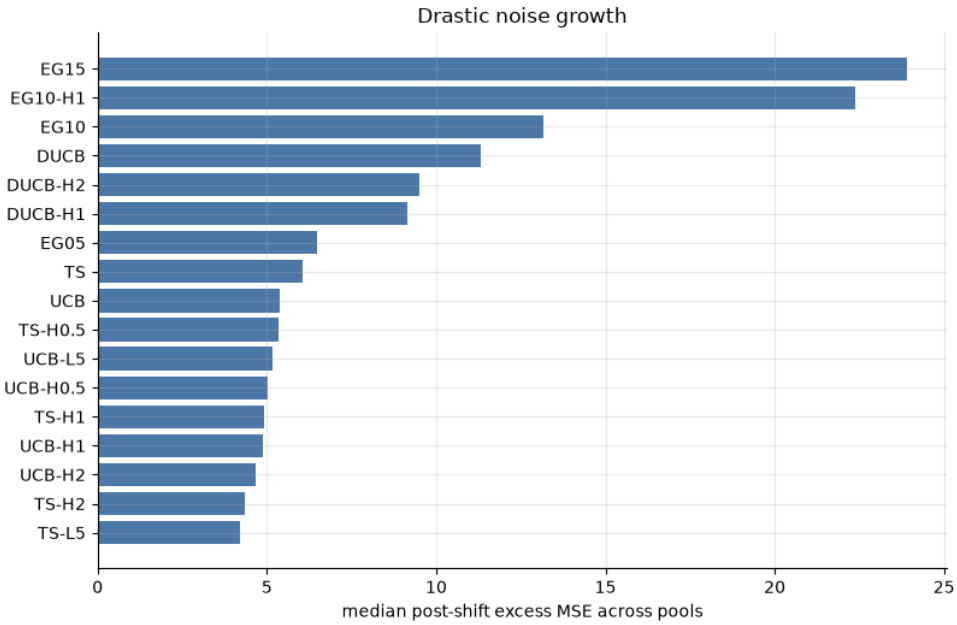


Figure 34: Median post-shift excess across the six pools, drastic noise growth. Lower is better.

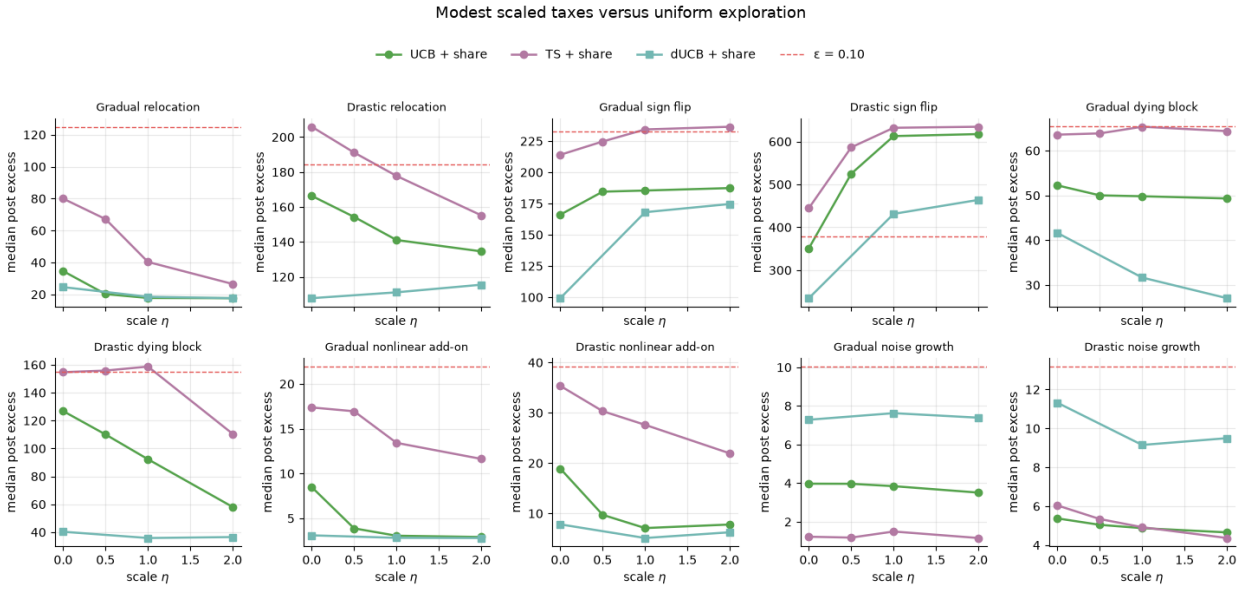


Figure 35: Post-shift excess against η, median across pools.

A.4 Policy horse-race

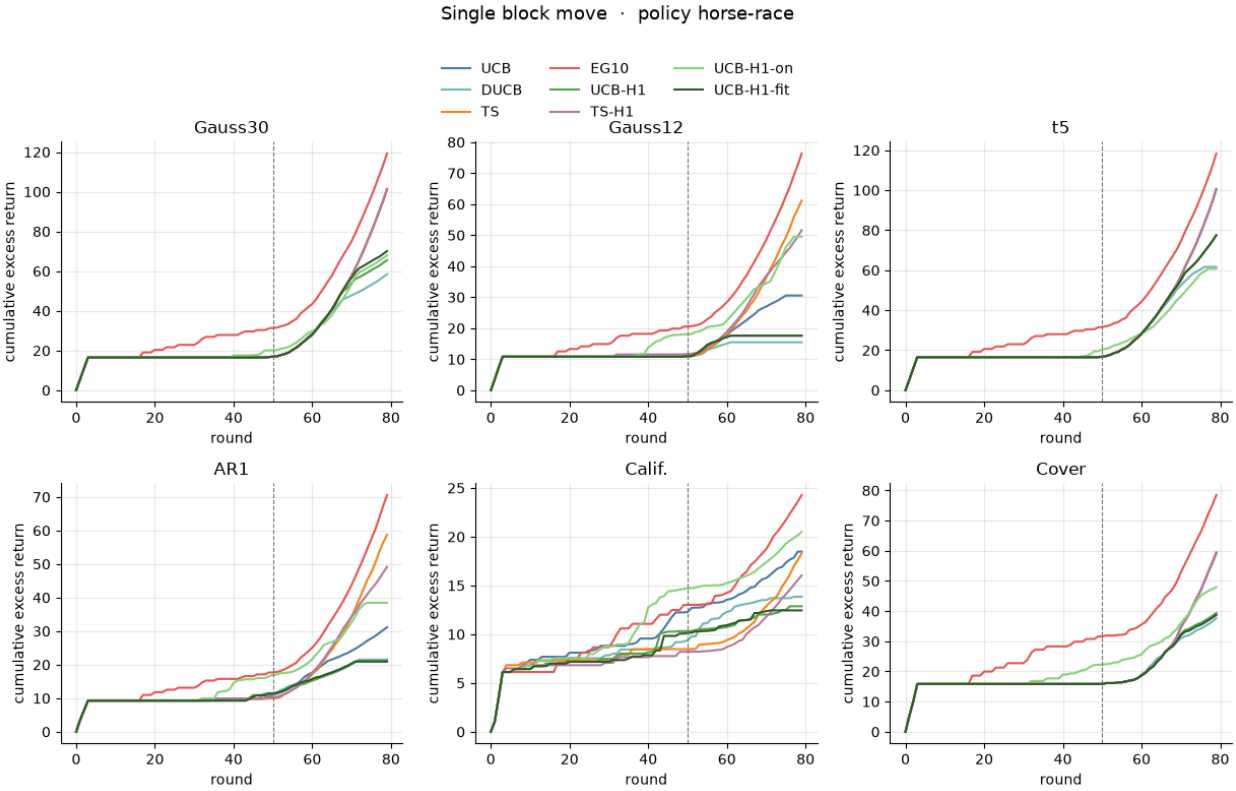


Figure 36: Cumulative excess return, single block move.

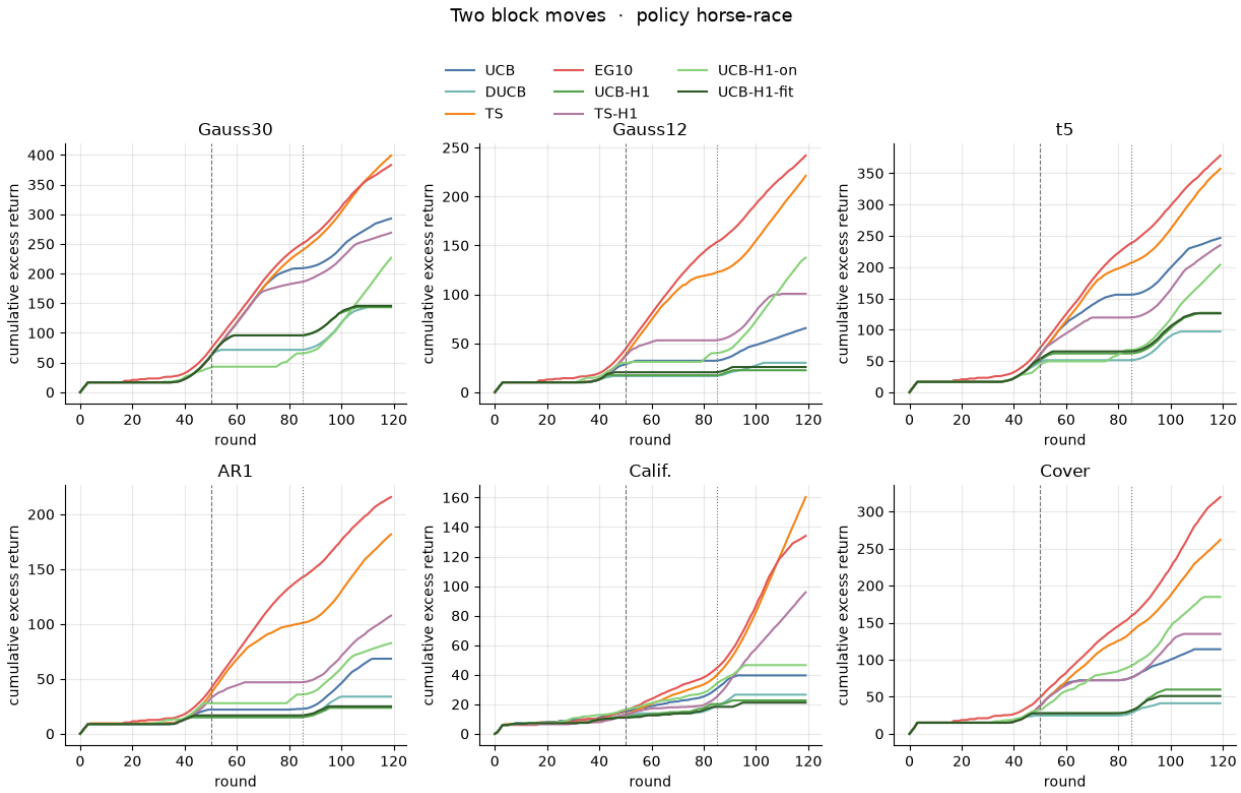


Figure 37: Cumulative excess return, two block moves. The second dashed line is the second phase.

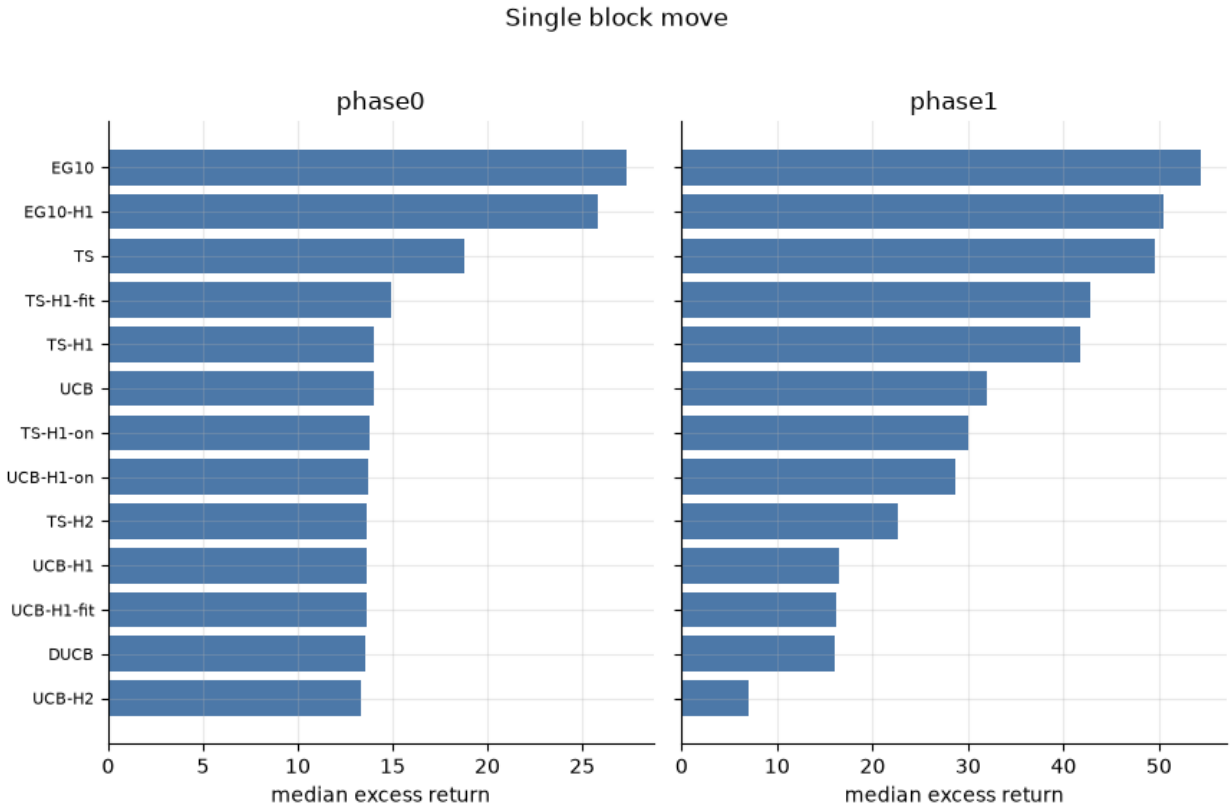


Figure 38: Median excess return, single block move.

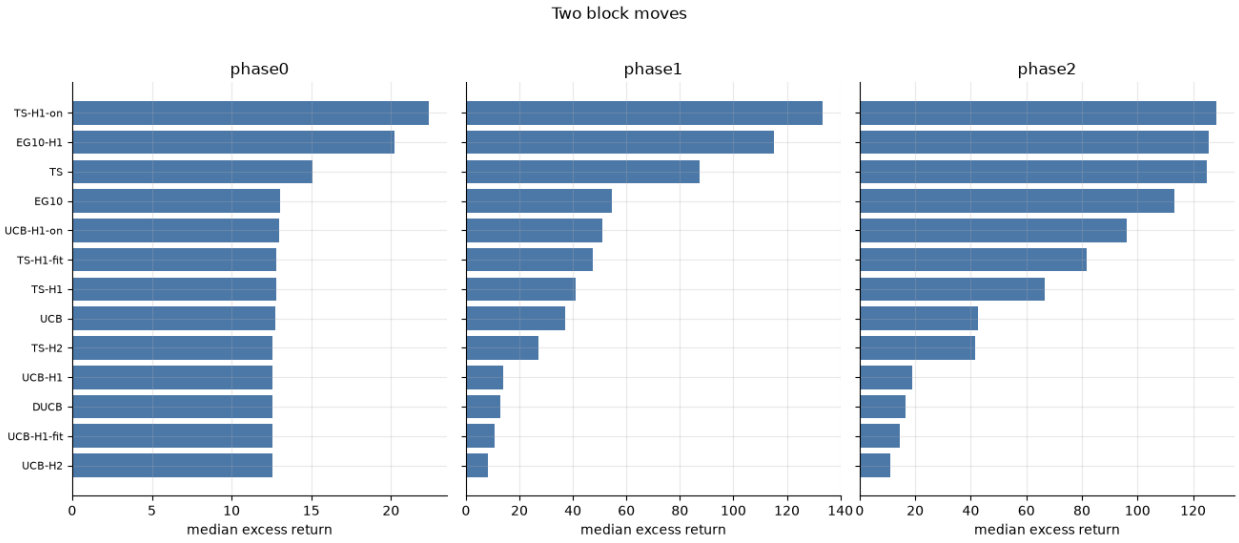


Figure 39: Median excess return, two block moves.

B Tables

The tables are all here. Post-shift excess, the ramp, the rounds before the ramp, hit rates, PO-risk shares, and the reward scale are one table per path, including the jump paths. Policy excess and hit rate are one table per phase. The jump-path rows are kept so the ramp numbers can be read next to them. They are not used in the claim.

Table 7: Median across covariate pools of the mean post-shift excess MSE. Each pool contributes the average over four seeds. Lower is better. The last column is the lowest median among all rules, not only the columns shown.

	UCB	dUCB	$\text{dUCB}_{\eta=1}$	$\text{dUCB}_{\eta=2}$	TS	$\varepsilon.10$	$\text{UCB}_{\eta=1}$	$\text{UCB}_{\eta=2}$	$\text{TS}_{\eta=2}$	$\text{UCB}_{\lambda=5}$	best
Gradual relocation	34.58	24.57	18.52	17.64	79.95	124.98	17.78	17.64	26.52	17.64	$\text{UCB}_{\eta=2}$
Drastic relocation	166.35	107.75	111.10	115.47	205.94	184.12	141.07	134.55	155.20	132.39	dUCB
Gradual sign flip	166.08	99.25	168.07	174.65	214.02	232.78	185.41	187.38	236.48	188.67	dUCB
Drastic sign flip	350.98	234.64	431.29	463.92	444.62	377.94	612.73	617.39	634.52	535.59	dUCB
Gradual dying block	52.25	41.63	31.66	27.06	63.58	65.48	49.82	49.33	64.36	50.47	$\text{dUCB}_{\eta=2}$
Drastic dying block	126.79	40.31	35.81	36.50	154.69	155.22	92.28	58.04	110.44	89.80	$\text{dUCB}_{\eta=1}$
Gradual nonlinear add-on	8.48	3.10	2.82	2.80	17.37	21.93	3.06	2.91	11.63	3.15	$\text{dUCB}_{\eta=2}$
Drastic nonlinear add-on	18.92	7.83	5.12	6.26	35.36	39.19	7.10	7.80	21.94	7.63	$\text{dUCB}_{\eta=1}$
Gradual noise growth	3.97	7.28	7.62	7.39	1.23	10.04	3.84	3.51	1.16	3.94	$\text{TS}_{\lambda=5}$
Drastic noise growth	5.37	11.32	9.15	9.50	6.04	13.18	4.87	4.65	4.36	5.16	$\text{TS}_{\lambda=5}$

η multiplies the reference-window standard deviation of arm rewards. $\lambda = 5$ is an absolute penalty on the share, not scaled.

Table 8: Post-shift excess MSE: Gradual relocation. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	50.61 (8.51)	22.87 (0.43)	46.29 (4.72)	18.22 (1.10)	4.23 (0.63)	73.10 (20.68)
dUCB	32.71 (6.18)	16.43 (1.31)	34.85 (3.66)	13.86 (0.66)	1.65 (1.50)	42.22 (0.64)
TS	116.40 (30.74)	69.17 (25.32)	90.73 (25.06)	55.57 (6.56)	15.08 (4.61)	125.18 (31.74)
$\varepsilon.05$	148.46 (64.87)	61.53 (20.98)	139.65 (66.59)	44.88 (17.42)	13.44 (4.51)	164.54 (85.35)
$\varepsilon.10$	187.11 (32.16)	79.78 (9.26)	170.17 (33.87)	57.42 (9.38)	18.75 (1.63)	216.59 (51.67)
$\varepsilon.15$	130.94 (28.90)	49.64 (9.53)	115.62 (21.78)	38.70 (7.38)	12.31 (3.73)	135.13 (39.55)
$\mathrm{UCB}_{\eta=.5}$	26.97 (2.90)	15.43 (1.65)	24.98 (1.09)	15.01 (0.40)	3.04 (1.27)	37.92 (12.61)
$\mathrm{UCB}_{\eta=1}$	25.12 (1.51)	10.31 (0.13)	24.98 (1.09)	10.59 (2.12)	3.03 (1.46)	32.00 (1.05)
$\mathrm{UCB}_{\eta=2}$	25.12 (1.51)	10.31 (0.13)	24.98 (1.09)	8.80 (0.52)	1.49 (0.58)	31.17 (1.16)
$\mathrm{TS}_{\eta=.5}$	76.00 (28.02)	64.06 (15.24)	70.23 (27.75)	47.12 (7.38)	12.91 (3.86)	72.64 (20.20)
$\mathrm{TS}_{\eta=1}$	54.18 (27.21)	37.63 (18.19)	43.02 (12.73)	33.93 (4.65)	9.57 (5.08)	79.34 (49.36)
$\mathrm{TS}_{\eta=2}$	37.17 (7.16)	20.13 (7.30)	32.91 (6.79)	19.01 (7.32)	7.76 (2.66)	35.15 (3.04)
$\mathrm{UCB}_{\lambda=5}$	25.12 (1.51)	10.31 (0.13)	24.98 (1.09)	8.16 (0.42)	0.23 (0.34)	37.92 (12.61)
$\mathrm{TS}_{\lambda=5}$	59.23 (20.17)	18.62 (5.12)	40.29 (9.48)	15.45 (0.65)	3.67 (1.41)	75.60 (29.30)
$\varepsilon.10_{\eta=1}$	103.61 (15.10)	45.62 (12.98)	99.39 (17.96)	33.64 (6.23)	16.29 (5.91)	107.87 (18.76)
$\mathrm{dUCB}_{\eta=1}$	25.12 (1.51)	12.06 (2.04)	24.98 (1.09)	11.53 (1.68)	0.26 (0.18)	31.17 (1.16)
$\mathrm{dUCB}_{\eta=2}$	25.12 (1.51)	10.31 (0.13)	24.98 (1.09)	8.34 (0.59)	0.22 (0.35)	31.17 (1.16)

Sum of per-round gaps to the best scorecard once the mixing weight reaches 1/2.

Table 9: Post-shift excess MSE: Drastic relocation. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	249.54 (8.01)	94.33 (1.83)	238.36 (6.20)	75.01 (2.01)	10.03 (2.94)	314.64 (31.71)
dUCB	161.70 (6.07)	64.27 (1.86)	151.22 (4.07)	52.96 (1.23)	6.82 (6.65)	217.02 (28.16)
TS	306.12 (7.78)	120.61 (4.82)	291.26 (4.01)	95.62 (2.21)	28.00 (1.22)	406.47 (71.10)
$\varepsilon.05$	255.62 (5.81)	104.08 (6.30)	240.51 (13.15)	82.54 (1.76)	26.35 (3.85)	376.86 (81.63)
$\varepsilon.10$	266.81 (10.88)	108.43 (2.79)	259.80 (9.02)	85.33 (1.94)	28.45 (5.92)	352.51 (72.97)
$\varepsilon.15$	266.81 (13.16)	107.10 (5.41)	260.44 (12.15)	85.54 (4.57)	25.58 (2.89)	372.09 (64.51)
$\mathrm{UCB}_{\eta=.5}$	228.59 (5.70)	90.53 (1.37)	218.02 (6.87)	72.06 (2.91)	7.93 (1.98)	309.14 (43.30)
$\mathrm{UCB}_{\eta=1}$	209.62 (5.69)	83.45 (1.63)	198.69 (8.74)	68.67 (2.31)	5.57 (1.35)	305.33 (55.07)
$\mathrm{UCB}_{\eta=2}$	206.07 (5.29)	72.48 (13.35)	196.62 (9.99)	58.78 (7.33)	6.71 (0.88)	286.96 (49.54)
$\mathrm{TS}_{\eta=.5}$	280.12 (3.92)	112.18 (2.62)	270.12 (9.02)	91.33 (2.74)	25.73 (1.65)	362.32 (42.53)
$\mathrm{TS}_{\eta=1}$	268.78 (7.57)	104.92 (6.31)	250.64 (8.10)	83.88 (4.08)	22.89 (5.84)	341.22 (51.00)
$\mathrm{TS}_{\eta=2}$	224.97 (5.42)	93.75 (4.36)	216.64 (10.35)	76.60 (3.77)	20.60 (2.77)	307.29 (55.09)
$\mathrm{UCB}_{\lambda=5}$	208.95 (3.72)	71.08 (16.60)	193.70 (4.21)	50.20 (9.99)	7.84 (1.49)	316.96 (65.68)
$\mathrm{TS}_{\lambda=5}$	266.79 (5.96)	87.36 (3.17)	252.40 (15.42)	63.73 (8.05)	15.37 (6.29)	366.06 (38.16)
$\varepsilon.10_{\eta=1}$	231.06 (14.03)	94.03 (6.65)	224.32 (19.04)	74.73 (3.60)	24.65 (1.99)	342.49 (64.93)
$\mathrm{dUCB}_{\eta=1}$	166.69 (4.55)	63.45 (1.22)	158.75 (5.35)	51.09 (1.08)	4.12 (2.41)	216.30 (24.50)
$\mathrm{dUCB}_{\eta=2}$	174.14 (6.29)	66.89 (1.54)	164.04 (9.31)	52.56 (2.20)	3.97 (0.34)	232.18 (36.17)

Sum of per-round gaps to the best scorecard once the mixing weight reaches 1/2.

Table 10: Post-shift excess MSE: Gradual sign flip. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	272.27 (14.77)	63.51 (10.45)	268.65 (20.49)	39.79 (7.51)	28.04 (2.94)	349.63 (7.18)
dUCB	148.69 (15.65)	49.80 (7.16)	155.78 (18.24)	35.19 (2.40)	22.40 (1.03)	202.93 (17.80)
TS	334.90 (14.05)	97.15 (17.33)	330.89 (10.39)	89.30 (17.28)	62.83 (14.16)	423.23 (12.58)
$\varepsilon.05$	279.04 (98.84)	122.11 (34.25)	278.05 (102.37)	82.59 (25.00)	57.83 (23.55)	353.18 (100.96)
$\varepsilon.10$	340.18 (29.58)	138.03 (13.48)	327.53 (23.94)	101.38 (6.21)	67.93 (11.21)	430.32 (51.67)
$\varepsilon.15$	246.18 (42.88)	107.22 (15.57)	255.93 (31.79)	81.21 (12.25)	54.02 (13.24)	329.84 (72.09)
$\mathrm{UCB}_{\eta=.5}$	285.19 (14.86)	94.69 (16.10)	274.35 (13.55)	57.24 (2.04)	37.27 (0.78)	370.06 (18.01)
$\mathrm{UCB}_{\eta=1}$	287.78 (13.55)	102.44 (10.03)	268.38 (13.15)	66.76 (2.50)	44.65 (2.04)	367.79 (20.64)
$\mathrm{UCB}_{\eta=2}$	281.92 (19.04)	104.33 (6.75)	270.42 (15.66)	67.63 (2.36)	46.35 (1.96)	364.21 (16.63)
$\mathrm{TS}_{\eta=.5}$	333.78 (15.45)	125.59 (19.74)	323.77 (18.63)	81.76 (9.99)	70.13 (13.69)	427.58 (25.12)
$\mathrm{TS}_{\eta=1}$	337.07 (22.83)	135.44 (11.26)	333.18 (25.39)	94.70 (5.86)	67.34 (7.41)	425.00 (24.94)
$\mathrm{TS}_{\eta=2}$	349.34 (16.17)	137.13 (5.55)	335.84 (16.88)	104.72 (4.80)	67.54 (3.12)	438.70 (17.82)
$\mathrm{UCB}_{\lambda=5}$	284.25 (14.93)	101.14 (10.13)	276.20 (12.27)	68.02 (1.51)	46.54 (2.04)	361.97 (12.23)
$\mathrm{TS}_{\lambda=5}$	342.93 (13.39)	134.76 (8.77)	335.03 (10.33)	100.24 (5.86)	69.07 (3.23)	424.18 (25.07)
$\varepsilon.10_{\eta=1}$	333.63 (33.37)	136.04 (14.06)	325.50 (42.42)	100.79 (11.11)	71.66 (7.19)	431.92 (33.55)
$\mathrm{dUCB}_{\eta=1}$	253.35 (10.32)	87.75 (10.14)	248.40 (14.26)	55.12 (4.22)	36.02 (1.96)	329.53 (5.36)
$\mathrm{dUCB}_{\eta=2}$	252.82 (9.36)	99.89 (5.49)	249.41 (13.03)	66.71 (5.14)	43.87 (3.71)	328.25 (6.18)

Sum of per-round gaps to the best scorecard once the mixing weight reaches 1/2.

Table 11: Post-shift excess MSE: Drastic sign flip. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	507.85 (14.60)	203.27 (3.97)	498.69 (16.25)	147.99 (5.04)	105.09 (3.81)	647.66 (12.24)
dUCB	331.12 (5.22)	139.24 (2.43)	330.04 (5.91)	102.65 (3.62)	75.33 (1.10)	417.74 (11.44)
TS	646.12 (15.71)	259.37 (10.48)	629.87 (13.88)	191.04 (3.44)	141.80 (4.87)	795.40 (33.88)
$\varepsilon.05$	520.53 (5.66)	215.25 (6.89)	512.99 (31.75)	161.26 (10.29)	115.45 (4.37)	652.30 (12.29)
$\varepsilon.10$	533.72 (1.26)	225.03 (7.20)	530.85 (7.06)	162.73 (8.02)	119.34 (5.13)	669.45 (32.59)
$\varepsilon.15$	536.30 (17.01)	218.68 (8.04)	531.15 (36.12)	166.43 (10.15)	119.39 (4.53)	678.69 (17.29)
$\mathrm{UCB}_{\eta=.5}$	774.85 (28.16)	283.24 (9.80)	766.67 (50.32)	195.71 (3.84)	134.62 (2.09)	1011.00 (52.31)
$\mathrm{UCB}_{\eta=1}$	916.23 (11.19)	325.52 (8.15)	899.93 (31.79)	228.49 (3.87)	155.90 (3.31)	1172.01 (21.49)
$\mathrm{UCB}_{\eta=2}$	920.32 (13.18)	332.46 (5.51)	902.33 (26.14)	239.85 (4.25)	169.27 (4.32)	1181.34 (23.95)
$\mathrm{TS}_{\eta=.5}$	838.97 (35.07)	334.04 (13.65)	857.20 (37.11)	235.38 (7.91)	163.47 (5.20)	1059.57 (14.66)
$\mathrm{TS}_{\eta=1}$	916.51 (28.52)	362.61 (6.56)	901.57 (48.98)	265.17 (3.19)	193.77 (3.55)	1179.96 (20.10)
$\mathrm{TS}_{\eta=2}$	912.12 (15.54)	367.67 (13.60)	901.37 (16.18)	268.20 (7.15)	198.13 (8.84)	1171.51 (27.73)
$\mathrm{UCB}_{\lambda=5}$	765.75 (17.47)	327.33 (5.89)	743.84 (17.71)	239.57 (4.72)	168.99 (4.34)	918.73 (24.42)
$\mathrm{TS}_{\lambda=5}$	836.90 (31.88)	364.22 (4.25)	829.87 (36.54)	265.65 (2.13)	195.81 (5.37)	1026.04 (18.49)
$\varepsilon.10_{\eta=1}$	911.76 (42.67)	364.63 (20.63)	883.02 (50.01)	267.38 (14.50)	192.86 (7.93)	1151.46 (27.01)
$\mathrm{dUCB}_{\eta=1}$	655.70 (0.90)	228.52 (12.28)	634.07 (32.12)	159.66 (7.66)	108.49 (3.46)	788.12 (55.73)
$\mathrm{dUCB}_{\eta=2}$	670.25 (9.12)	270.46 (3.17)	657.37 (16.06)	197.05 (3.37)	128.48 (26.53)	860.77 (13.86)

Sum of per-round gaps to the best scorecard once the mixing weight reaches 1/2.

Table 12: Post-shift excess MSE: Gradual dying block. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	100.19 (2.92)	1.95 (1.70)	96.70 (0.57)	7.80 (9.47)	2.80 (1.39)	145.04 (39.83)
dUCB	82.27 (10.64)	4.55 (2.68)	78.70 (21.80)	0.70 (0.96)	0.43 (0.37)	121.14 (70.19)
TS	100.19 (2.92)	30.45 (8.46)	96.70 (0.57)	30.28 (1.72)	11.61 (0.95)	156.07 (18.01)
$\varepsilon.05$	102.55 (5.01)	38.98 (2.47)	98.28 (7.28)	30.83 (3.13)	12.11 (2.01)	161.70 (15.49)
$\varepsilon.10$	97.51 (8.25)	36.84 (2.71)	94.13 (4.53)	28.93 (1.41)	11.81 (1.21)	153.59 (9.70)
$\varepsilon.15$	109.73 (8.63)	42.03 (3.11)	106.10 (11.36)	32.75 (3.40)	14.85 (3.45)	164.01 (20.00)
$\mathrm{UCB}_{\eta=.5}$	100.19 (2.92)	3.16 (3.25)	96.70 (0.57)	3.34 (2.94)	2.25 (0.71)	141.53 (46.83)
$\mathrm{UCB}_{\eta=1}$	100.19 (2.92)	2.84 (1.75)	96.70 (0.57)	2.94 (4.03)	0.72 (0.35)	139.07 (51.73)
$\mathrm{UCB}_{\eta=2}$	100.19 (2.92)	3.60 (1.95)	95.06 (3.65)	2.53 (3.37)	0.73 (0.61)	132.97 (63.90)
$\mathrm{TS}_{\eta=.5}$	100.19 (2.92)	31.01 (8.60)	96.70 (0.57)	28.34 (3.95)	11.61 (0.95)	159.13 (12.12)
$\mathrm{TS}_{\eta=1}$	100.19 (2.92)	33.98 (8.30)	96.70 (0.57)	27.64 (4.83)	11.79 (0.84)	150.60 (28.79)
$\mathrm{TS}_{\eta=2}$	100.19 (2.92)	32.03 (10.55)	96.70 (0.57)	25.96 (4.27)	10.67 (2.10)	148.50 (32.95)
$\mathrm{UCB}_{\lambda=5}$	100.19 (2.92)	4.24 (2.28)	96.70 (0.57)	2.59 (3.33)	0.99 (0.84)	141.53 (46.83)
$\mathrm{TS}_{\lambda=5}$	100.19 (2.92)	30.20 (11.02)	96.70 (0.57)	26.97 (6.52)	11.35 (1.08)	159.13 (12.12)
$\varepsilon.10_{\eta=1}$	105.89 (2.11)	40.61 (1.68)	102.14 (4.15)	31.57 (1.69)	13.43 (2.61)	161.84 (13.73)
$\mathrm{dUCB}_{\eta=1}$	60.07 (9.11)	4.18 (1.65)	59.14 (15.82)	0.40 (0.20)	0.25 (0.15)	109.06 (67.14)
$\mathrm{dUCB}_{\eta=2}$	53.42 (1.17)	4.99 (1.85)	49.14 (10.26)	1.39 (1.04)	0.35 (0.15)	100.40 (71.89)

Sum of per-round gaps to the best scorecard once the mixing weight reaches 1/2.

Table 13: Post-shift excess MSE: Drastic dying block. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	260.38 (8.77)	14.39 (5.79)	230.55 (37.96)	23.02 (5.00)	7.85 (3.66)	350.17 (166.91)
dUCB	73.71 (10.02)	10.61 (3.27)	70.02 (27.50)	8.19 (6.73)	1.54 (1.13)	190.27 (164.61)
TS	230.11 (61.06)	72.54 (32.53)	253.31 (5.46)	79.27 (4.07)	31.82 (2.81)	325.72 (132.51)
$\varepsilon.05$	250.02 (22.39)	98.82 (3.87)	252.01 (9.17)	77.43 (3.78)	31.74 (3.52)	409.33 (41.98)
$\varepsilon.10$	221.90 (38.75)	88.54 (15.41)	223.25 (38.27)	68.79 (11.88)	28.80 (6.57)	327.10 (49.96)
$\varepsilon.15$	254.41 (23.74)	99.34 (6.26)	246.06 (28.21)	75.23 (8.46)	32.88 (4.77)	368.30 (82.72)
$\mathrm{UCB}_{\eta=.5}$	216.93 (26.52)	19.92 (2.39)	200.14 (51.67)	18.62 (3.15)	4.67 (1.59)	308.90 (170.65)
$\mathrm{UCB}_{\eta=1}$	174.07 (34.70)	20.70 (2.07)	163.86 (54.75)	9.87 (7.17)	4.04 (2.52)	280.93 (179.90)
$\mathrm{UCB}_{\eta=2}$	103.44 (31.15)	20.02 (1.02)	96.06 (32.73)	12.30 (3.52)	3.58 (1.99)	265.37 (175.46)
$\mathrm{TS}_{\eta=.5}$	262.48 (6.15)	57.33 (16.03)	241.52 (24.35)	70.18 (18.17)	31.82 (2.81)	354.97 (148.53)
$\mathrm{TS}_{\eta=1}$	256.88 (8.36)	49.38 (17.32)	248.44 (10.71)	68.98 (19.34)	26.54 (10.54)	303.59 (155.57)
$\mathrm{TS}_{\eta=2}$	202.80 (56.22)	35.10 (9.51)	180.97 (75.18)	39.90 (29.18)	19.66 (6.69)	274.86 (166.23)
$\mathrm{UCB}_{\lambda=5}$	170.52 (32.22)	20.02 (1.02)	159.58 (56.10)	15.21 (0.42)	6.79 (0.84)	308.90 (170.65)
$\mathrm{TS}_{\lambda=5}$	240.04 (45.91)	34.14 (10.15)	226.69 (41.78)	27.78 (12.95)	10.08 (4.38)	312.93 (168.78)
$\varepsilon.10_{\eta=1}$	232.65 (55.69)	93.37 (11.62)	225.34 (42.81)	69.04 (16.13)	31.97 (5.18)	313.40 (70.18)
$\mathrm{dUCB}_{\eta=1}$	54.33 (1.91)	17.60 (3.70)	54.03 (4.91)	5.24 (2.60)	1.90 (0.98)	157.25 (112.74)
$\mathrm{dUCB}_{\eta=2}$	52.47 (2.00)	20.70 (2.07)	52.31 (2.16)	9.70 (3.99)	2.46 (0.60)	132.65 (106.33)

Sum of per-round gaps to the best scorecard once the mixing weight reaches 1/2.

Table 14: Post-shift excess MSE: Gradual nonlinear add-on. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	2.34 (0.28)	2.27 (0.48)	13.14 (11.69)	3.82 (1.18)	1745.25 (2817.19)	228.43 (35.78)
dUCB	0.28 (0.13)	0.16 (0.15)	5.92 (9.64)	0.10 (0.03)	1695.80 (2855.67)	110.64 (110.75)
TS	10.41 (0.69)	11.18 (0.70)	23.55 (2.92)	10.82 (1.21)	290.75 (200.52)	226.84 (35.44)
$\varepsilon.05$	16.37 (8.45)	12.73 (3.45)	27.71 (13.57)	12.27 (3.83)	1669.34 (2873.04)	234.10 (48.58)
$\varepsilon.10$	16.08 (12.24)	12.23 (4.79)	27.78 (12.81)	11.71 (3.45)	1719.57 (2825.33)	225.68 (38.72)
$\varepsilon.15$	38.94 (16.69)	20.82 (5.76)	51.30 (15.82)	18.07 (5.37)	225.21 (158.71)	258.15 (50.00)
$\mathrm{UCB}_{\eta=.5}$	0.92 (0.27)	1.05 (0.11)	6.21 (9.48)	1.54 (0.95)	198.69 (191.21)	208.56 (19.91)
$\mathrm{UCB}_{\eta=1}$	0.47 (0.09)	0.53 (0.14)	5.58 (9.88)	0.41 (0.17)	200.73 (194.17)	161.24 (70.04)
$\mathrm{UCB}_{\eta=2}$	0.43 (0.18)	0.37 (0.10)	5.40 (10.00)	0.32 (0.08)	180.34 (209.10)	94.76 (83.06)
$\mathrm{TS}_{\eta=.5}$	10.41 (0.69)	10.59 (0.64)	23.13 (3.86)	10.75 (1.22)	192.72 (191.70)	226.84 (35.44)
$\mathrm{TS}_{\eta=1}$	10.41 (0.69)	10.59 (0.64)	16.24 (3.55)	10.37 (0.55)	182.45 (205.13)	211.62 (18.82)
$\mathrm{TS}_{\eta=2}$	9.52 (1.92)	9.77 (2.22)	12.50 (7.64)	10.75 (1.22)	170.26 (213.06)	140.07 (97.89)
$\mathrm{UCB}_{\lambda=5}$	0.64 (0.28)	0.19 (0.12)	5.67 (9.81)	0.29 (0.01)	1732.79 (2836.28)	208.56 (19.91)
$\mathrm{TS}_{\lambda=5}$	10.41 (0.69)	10.59 (0.64)	17.23 (3.21)	10.79 (0.93)	266.52 (236.04)	226.84 (35.44)
$\varepsilon.10_{\eta=1}$	28.93 (17.23)	16.77 (4.71)	40.74 (14.47)	16.28 (5.58)	192.49 (191.80)	239.38 (28.46)
$\mathrm{dUCB}_{\eta=1}$	0.27 (0.06)	0.13 (0.08)	5.38 (10.01)	0.19 (0.10)	183.81 (205.39)	88.84 (102.20)
$\mathrm{dUCB}_{\eta=2}$	0.30 (0.11)	0.13 (0.07)	5.30 (10.05)	0.25 (0.09)	176.81 (213.17)	56.02 (48.53)

Sum of per-round gaps to the best scorecard once the mixing weight reaches 1/2.

Table 15: Post-shift excess MSE: Drastic nonlinear add-on. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	4.69 (0.52)	5.35 (1.85)	31.74 (25.27)	6.09 (1.31)	6881.65 (8936.77)	359.45 (170.25)
dUCB	1.75 (0.26)	1.70 (0.81)	13.91 (25.84)	1.34 (0.16)	5289.18 (9742.23)	128.53 (66.51)
TS	24.64 (5.43)	28.26 (1.07)	42.46 (10.26)	24.82 (4.64)	5576.16 (9369.74)	442.39 (128.64)
$\varepsilon.05$	33.15 (8.38)	29.37 (3.17)	54.93 (20.19)	26.92 (5.45)	5102.50 (9676.15)	535.73 (93.55)
$\varepsilon.10$	29.50 (14.98)	26.34 (7.39)	48.88 (10.97)	23.82 (6.56)	6709.47 (9080.82)	439.94 (78.74)
$\varepsilon.15$	54.11 (19.54)	36.76 (7.14)	76.64 (24.35)	32.85 (6.54)	5300.45 (9558.91)	484.08 (161.92)
$\mathrm{UCB}_{\eta=.5}$	1.53 (0.19)	2.04 (0.33)	16.51 (24.41)	2.93 (1.26)	4995.11 (9759.41)	286.82 (217.37)
$\mathrm{UCB}_{\eta=1}$	2.00 (1.07)	2.33 (0.69)	11.27 (17.35)	2.92 (1.51)	4995.05 (9761.10)	250.04 (243.91)
$\mathrm{UCB}_{\eta=2}$	3.38 (0.92)	3.39 (0.78)	12.21 (5.53)	2.99 (0.44)	4993.54 (9757.37)	215.55 (175.16)
$\mathrm{TS}_{\eta=.5}$	26.99 (0.47)	28.29 (2.45)	32.37 (16.01)	27.96 (0.79)	4995.24 (9750.92)	360.84 (189.26)
$\mathrm{TS}_{\eta=1}$	23.99 (6.72)	27.65 (1.30)	27.58 (16.99)	23.46 (6.79)	4993.10 (9755.40)	293.36 (222.07)
$\mathrm{TS}_{\eta=2}$	13.51 (8.27)	14.15 (3.21)	24.50 (19.20)	19.38 (5.17)	4991.48 (9754.37)	262.87 (237.63)
$\mathrm{UCB}_{\lambda=5}$	2.02 (0.85)	4.74 (0.30)	10.52 (16.79)	3.48 (0.38)	5269.94 (9578.09)	281.62 (221.39)
$\mathrm{TS}_{\lambda=5}$	24.85 (5.01)	11.87 (5.06)	25.45 (18.86)	11.40 (1.89)	5333.57 (9524.38)	303.08 (221.35)
$\varepsilon.10_{\eta=1}$	51.90 (17.10)	33.34 (6.62)	63.37 (23.17)	32.36 (6.48)	5053.59 (9716.62)	503.13 (143.42)
$\mathrm{dUCB}_{\eta=1}$	1.50 (0.31)	1.91 (0.70)	8.33 (12.25)	1.56 (1.13)	4992.45 (9757.10)	123.65 (60.02)
$\mathrm{dUCB}_{\eta=2}$	3.57 (0.81)	3.38 (1.34)	8.95 (7.15)	2.21 (0.32)	4994.98 (9756.30)	131.83 (34.99)

Sum of per-round gaps to the best scorecard once the mixing weight reaches 1/2.

Table 16: Post-shift excess MSE: Gradual noise growth. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	3.00 (0.54)	5.02 (1.26)	3.09 (1.35)	4.85 (0.78)	7.81 (0.63)	2.86 (1.18)
dUCB	6.22 (0.92)	8.34 (0.74)	5.44 (0.79)	11.12 (3.45)	11.76 (0.55)	6.20 (0.63)
TS	0.36 (0.73)	3.41 (1.77)	0.35 (0.71)	1.90 (0.27)	3.69 (1.85)	0.55 (1.10)
$\varepsilon.05$	7.86 (5.96)	3.04 (2.28)	6.61 (6.34)	2.90 (2.44)	2.26 (1.15)	10.04 (8.62)
$\varepsilon.10$	14.46 (12.59)	6.87 (3.92)	13.20 (11.69)	4.96 (3.91)	3.47 (1.13)	18.51 (15.75)
$\varepsilon.15$	31.47 (12.47)	12.53 (4.46)	29.34 (10.63)	9.69 (3.86)	4.90 (0.89)	39.38 (13.40)
$\mathrm{UCB}_{\eta=.5}$	3.15 (0.53)	5.56 (0.76)	2.96 (1.04)	4.78 (0.28)	7.33 (0.89)	2.68 (1.34)
$\mathrm{UCB}_{\eta=1}$	3.18 (0.22)	4.31 (1.92)	2.48 (1.09)	4.86 (0.58)	7.33 (0.72)	3.38 (1.44)
$\mathrm{UCB}_{\eta=2}$	2.79 (0.09)	4.95 (1.66)	1.81 (0.54)	4.09 (1.46)	7.84 (1.50)	2.93 (1.44)
$\mathrm{TS}_{\eta=.5}$	0.00 (0.00)	2.36 (0.67)	0.00 (0.00)	2.49 (1.14)	2.99 (1.07)	0.00 (0.00)
$\mathrm{TS}_{\eta=1}$	0.00 (0.00)	3.63 (2.05)	0.00 (0.00)	3.30 (1.84)	2.99 (1.73)	0.00 (0.00)
$\mathrm{TS}_{\eta=2}$	0.00 (0.00)	2.15 (0.49)	0.25 (0.50)	2.07 (0.84)	3.05 (1.83)	0.00 (0.00)
$\mathrm{UCB}_{\lambda=5}$	3.18 (0.22)	5.06 (1.66)	2.48 (1.09)	4.60 (0.23)	6.42 (0.31)	3.28 (1.57)
$\mathrm{TS}_{\lambda=5}$	0.00 (0.00)	2.06 (0.52)	0.36 (0.72)	1.39 (0.54)	1.90 (0.71)	0.42 (0.84)
$\varepsilon.10_{\eta=1}$	23.16 (19.45)	8.40 (6.70)	19.95 (17.99)	8.73 (7.14)	3.53 (2.38)	38.65 (39.72)
$\mathrm{dUCB}_{\eta=1}$	5.92 (0.60)	9.03 (1.14)	6.22 (0.90)	9.23 (3.35)	11.33 (0.49)	5.73 (0.94)
$\mathrm{dUCB}_{\eta=2}$	5.62 (1.24)	8.49 (1.13)	6.29 (0.47)	8.72 (2.66)	10.73 (1.04)	4.88 (0.91)

Sum of per-round gaps to the best scorecard once the mixing weight reaches 1/2.

Table 17: Post-shift excess MSE: Drastic noise growth. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	3.48 (0.54)	7.09 (1.18)	3.63 (1.00)	10.31 (3.18)	10.57 (0.90)	3.66 (0.23)
dUCB	7.94 (2.27)	15.65 (1.69)	6.72 (3.06)	14.69 (2.71)	15.06 (0.46)	6.12 (0.66)
TS	3.97 (0.77)	7.72 (2.49)	3.87 (1.58)	7.02 (1.79)	10.55 (1.35)	5.06 (0.95)
$\varepsilon.05$	10.28 (2.82)	3.90 (1.78)	8.85 (4.13)	3.45 (2.21)	4.11 (0.83)	12.78 (5.51)
$\varepsilon.10$	19.05 (10.53)	8.38 (3.30)	17.97 (9.67)	6.14 (3.05)	5.43 (0.64)	24.78 (12.44)
$\varepsilon.15$	35.66 (16.35)	14.40 (6.34)	33.39 (14.27)	11.59 (4.79)	7.71 (2.18)	44.01 (18.56)
$\mathrm{UCB}_{\eta=.5}$	3.45 (0.49)	5.94 (0.47)	3.54 (0.87)	9.15 (4.33)	10.01 (0.33)	4.15 (0.65)
$\mathrm{UCB}_{\eta=1}$	3.38 (0.55)	6.33 (0.68)	3.30 (0.49)	8.23 (2.75)	9.62 (0.37)	3.41 (0.83)
$\mathrm{UCB}_{\eta=2}$	3.13 (0.16)	6.17 (0.75)	3.13 (0.66)	7.20 (1.80)	8.83 (1.11)	2.64 (0.57)
$\mathrm{TS}_{\eta=.5}$	3.27 (1.09)	6.62 (2.22)	3.34 (0.81)	7.21 (3.15)	9.56 (1.66)	4.05 (0.46)
$\mathrm{TS}_{\eta=1}$	4.45 (0.84)	7.51 (3.33)	4.53 (1.12)	5.31 (1.59)	7.94 (1.25)	4.26 (0.81)
$\mathrm{TS}_{\eta=2}$	3.87 (0.41)	4.62 (1.36)	4.10 (1.18)	4.68 (2.10)	8.22 (1.05)	3.89 (0.76)
$\mathrm{UCB}_{\lambda=5}$	3.19 (0.20)	6.18 (0.70)	3.30 (0.49)	6.79 (2.45)	8.27 (0.81)	4.15 (0.65)
$\mathrm{TS}_{\lambda=5}$	3.78 (1.30)	3.65 (1.54)	3.79 (1.00)	4.62 (2.05)	8.50 (1.70)	5.20 (1.12)
$\varepsilon.10_{\eta=1}$	35.39 (19.28)	12.98 (6.61)	31.75 (17.23)	12.43 (7.39)	6.17 (2.43)	53.75 (42.59)
$\mathrm{dUCB}_{\eta=1}$	6.70 (0.32)	11.60 (3.76)	5.84 (0.51)	13.35 (1.86)	13.56 (1.11)	5.81 (0.72)
$\mathrm{dUCB}_{\eta=2}$	6.89 (0.56)	12.10 (3.32)	5.90 (0.73)	12.25 (2.17)	13.27 (0.65)	5.85 (0.66)

Sum of per-round gaps to the best scorecard once the mixing weight reaches 1/2.

Table 18: Ramp excess MSE: Gradual relocation. Rounds with mixing weight in $(0, 1/2)$.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	24.03 (0.74)	2.16 (0.51)	16.27 (10.05)	3.32 (0.57)	3.44 (0.17)	24.58 (7.75)
dUCB	16.55 (3.94)	0.90 (0.78)	10.18 (6.00)	1.29 (0.71)	1.11 (0.61)	15.72 (9.72)
TS	24.03 (0.74)	9.43 (0.32)	23.48 (0.85)	7.97 (0.95)	2.25 (0.11)	28.42 (0.36)
$\varepsilon.05$	30.26 (5.36)	12.23 (2.18)	29.63 (4.44)	9.23 (1.72)	2.94 (0.47)	36.53 (6.31)
$\varepsilon.10$	34.07 (6.10)	13.56 (2.34)	34.07 (7.79)	10.57 (2.18)	3.67 (1.04)	41.62 (8.92)
$\varepsilon.15$	41.75 (10.06)	16.84 (5.26)	39.31 (9.36)	13.15 (3.19)	4.20 (1.44)	50.18 (13.80)
$\text{UCB}_{\eta=.5}$	20.26 (3.36)	0.97 (0.36)	13.83 (7.69)	2.02 (0.55)	2.74 (0.69)	22.41 (4.67)
$\text{UCB}_{\eta=1}$	17.92 (0.95)	0.72 (0.23)	13.62 (7.21)	0.94 (0.37)	2.29 (0.43)	23.21 (3.85)
$\text{UCB}_{\eta=2}$	17.92 (0.95)	0.97 (0.13)	13.62 (7.21)	0.76 (0.29)	1.47 (0.43)	21.19 (0.37)
$\text{TS}_{\eta=.5}$	24.03 (0.74)	9.43 (0.32)	23.48 (0.85)	7.41 (0.26)	2.71 (0.90)	25.15 (6.61)
$\text{TS}_{\eta=1}$	21.82 (4.71)	8.73 (1.29)	21.14 (4.67)	7.34 (0.38)	2.83 (0.70)	28.42 (0.36)
$\text{TS}_{\eta=2}$	24.03 (0.74)	8.58 (1.97)	21.29 (3.78)	7.47 (0.54)	2.62 (0.76)	25.13 (6.73)
$\text{UCB}_{\lambda=5}$	17.92 (0.95)	1.04 (0.08)	13.04 (6.98)	0.79 (0.05)	0.34 (0.14)	22.41 (4.67)
$\text{TS}_{\lambda=5}$	24.03 (0.74)	9.02 (0.66)	22.77 (2.20)	7.34 (0.38)	2.27 (0.34)	27.73 (1.60)
$\varepsilon.10_{\eta=1}$	33.45 (3.05)	13.32 (1.71)	33.44 (1.75)	10.28 (1.33)	4.04 (0.54)	40.10 (4.22)
$\text{dUCB}_{\eta=1}$	15.84 (2.77)	0.90 (0.33)	10.18 (6.00)	0.46 (0.34)	1.59 (0.70)	14.37 (5.22)
$\text{dUCB}_{\eta=2}$	16.58 (1.73)	0.96 (0.25)	10.18 (6.00)	0.79 (0.05)	1.15 (0.74)	12.96 (5.56)

The ramp is where a historical mean is already stale and the fresh arm is not yet dominant.

Table 19: Ramp excess MSE: Gradual sign flip. Rounds with mixing weight in $(0, 1/2)$.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	16.76 (2.60)	6.37 (1.98)	16.71 (1.71)	5.42 (0.76)	3.99 (1.62)	20.53 (2.63)
dUCB	13.02 (3.19)	4.49 (0.45)	13.17 (1.77)	4.29 (1.03)	4.10 (0.78)	14.04 (2.42)
TS	55.63 (5.75)	23.63 (0.71)	58.30 (2.37)	16.24 (1.40)	12.97 (0.52)	63.73 (10.80)
$\varepsilon.05$	70.56 (9.51)	29.39 (4.29)	72.75 (11.38)	20.98 (2.99)	16.15 (2.51)	90.73 (12.42)
$\varepsilon.10$	68.13 (9.79)	27.68 (4.77)	71.93 (10.24)	19.96 (2.67)	14.68 (2.18)	85.44 (14.75)
$\varepsilon.15$	85.02 (12.21)	35.00 (6.03)	85.88 (14.38)	25.81 (4.03)	19.62 (2.90)	109.85 (19.10)
$\mathrm{UCB}_{\eta=.5}$	15.97 (3.76)	7.45 (0.95)	15.84 (2.53)	4.89 (0.98)	3.82 (0.48)	23.49 (1.12)
$\mathrm{UCB}_{\eta=1}$	16.41 (3.82)	6.70 (0.69)	16.34 (4.00)	5.31 (0.55)	3.51 (0.56)	22.45 (0.70)
$\mathrm{UCB}_{\eta=2}$	14.27 (1.97)	6.37 (0.48)	16.25 (4.09)	5.09 (0.41)	4.16 (0.71)	23.21 (3.36)
$\mathrm{TS}_{\eta=.5}$	46.26 (6.33)	21.10 (3.43)	45.70 (8.40)	14.19 (2.03)	12.02 (1.81)	63.19 (8.06)
$\mathrm{TS}_{\eta=1}$	51.43 (10.74)	21.14 (3.64)	53.83 (9.00)	16.17 (1.64)	11.87 (1.61)	60.44 (7.27)
$\mathrm{TS}_{\eta=2}$	56.45 (3.55)	21.77 (1.78)	54.64 (6.20)	16.84 (0.60)	10.98 (1.13)	64.67 (8.42)
$\mathrm{UCB}_{\lambda=5}$	16.94 (2.97)	7.32 (0.59)	15.84 (2.53)	5.30 (0.34)	4.19 (0.78)	22.20 (2.98)
$\mathrm{TS}_{\lambda=5}$	53.18 (2.44)	21.80 (1.39)	56.54 (3.52)	16.23 (1.08)	10.00 (2.60)	64.70 (5.36)
$\varepsilon.10_{\eta=1}$	60.74 (20.25)	24.60 (8.05)	62.27 (22.48)	17.68 (6.28)	13.67 (4.57)	78.70 (26.95)
$\mathrm{dUCB}_{\eta=1}$	10.82 (0.89)	4.84 (0.71)	13.82 (1.62)	3.01 (0.95)	2.52 (0.62)	14.32 (1.82)
$\mathrm{dUCB}_{\eta=2}$	10.97 (1.73)	4.01 (0.26)	12.18 (1.63)	3.38 (0.35)	2.38 (0.69)	14.10 (3.44)

The ramp is where a historical mean is already stale and the fresh arm is not yet dominant.

Table 20: Ramp excess MSE: Gradual dying block. Rounds with mixing weight in $(0, 1/2)$.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.01 (0.02)	0.61 (1.20)	0.00 (0.00)	1.70 (1.50)	2.30 (0.62)	0.07 (0.13)
dUCB	0.01 (0.02)	0.01 (0.01)	0.00 (0.00)	1.74 (1.74)	1.27 (0.82)	0.07 (0.13)
TS	0.01 (0.02)	0.01 (0.01)	0.00 (0.00)	0.32 (0.63)	0.75 (0.85)	0.07 (0.13)
$\varepsilon.05$	4.72 (3.47)	2.09 (1.71)	4.49 (3.55)	1.56 (1.15)	0.67 (0.69)	8.04 (5.42)
$\varepsilon.10$	11.69 (10.04)	4.82 (4.09)	11.80 (10.21)	3.65 (3.17)	2.53 (2.36)	17.15 (15.15)
$\varepsilon.15$	16.24 (10.02)	6.89 (4.68)	15.05 (10.48)	4.82 (3.39)	2.92 (2.16)	23.46 (16.05)
$\text{UCB}_{\eta=.5}$	0.01 (0.02)	0.01 (0.01)	0.00 (0.00)	1.35 (0.40)	1.43 (0.78)	0.07 (0.13)
$\text{UCB}_{\eta=1}$	0.01 (0.02)	0.01 (0.01)	0.00 (0.00)	0.65 (0.47)	1.63 (1.10)	0.07 (0.13)
$\text{UCB}_{\eta=2}$	0.01 (0.02)	0.01 (0.01)	0.00 (0.00)	0.31 (0.29)	0.94 (0.88)	0.07 (0.13)
$\text{TS}_{\eta=.5}$	0.01 (0.02)	0.66 (1.30)	0.00 (0.00)	0.85 (0.98)	0.71 (0.78)	0.07 (0.13)
$\text{TS}_{\eta=1}$	0.01 (0.02)	0.06 (0.11)	0.00 (0.00)	0.50 (1.00)	0.48 (0.65)	0.07 (0.13)
$\text{TS}_{\eta=2}$	0.01 (0.02)	0.08 (0.16)	0.00 (0.00)	0.68 (0.99)	0.75 (0.65)	0.07 (0.13)
$\text{UCB}_{\lambda=5}$	0.01 (0.02)	0.01 (0.01)	0.00 (0.00)	0.05 (0.06)	0.30 (0.28)	0.07 (0.13)
$\text{TS}_{\lambda=5}$	0.01 (0.02)	0.01 (0.01)	0.00 (0.00)	0.00 (0.00)	0.04 (0.03)	0.07 (0.13)
$\varepsilon.10_{\eta=1}$	17.07 (10.59)	6.52 (3.90)	17.48 (10.64)	4.81 (2.85)	3.79 (2.68)	23.07 (15.92)
$\text{dUCB}_{\eta=1}$	0.01 (0.02)	0.01 (0.01)	0.00 (0.00)	1.34 (1.07)	1.20 (0.59)	0.07 (0.13)
$\text{dUCB}_{\eta=2}$	0.01 (0.02)	0.01 (0.01)	0.00 (0.00)	0.16 (0.27)	0.41 (0.15)	0.07 (0.13)

The ramp is where a historical mean is already stale and the fresh arm is not yet dominant.

Table 21: Ramp excess MSE: Gradual nonlinear add-on. Rounds with mixing weight in $(0, 1/2)$.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	1.45 (0.12)	2.30 (1.36)	1.74 (1.41)	1.93 (0.81)	43.69 (71.87)	0.00 (0.00)
dUCB	0.72 (0.21)	0.40 (0.45)	0.80 (0.52)	1.25 (1.16)	20.88 (11.22)	0.00 (0.00)
TS	0.03 (0.02)	0.05 (0.04)	0.17 (0.22)	0.01 (0.01)	115.44 (150.41)	0.00 (0.00)
$\varepsilon.05$	1.78 (2.73)	1.15 (1.44)	2.09 (2.96)	0.77 (0.91)	8.05 (9.38)	7.36 (8.50)
$\varepsilon.10$	9.94 (10.98)	4.18 (4.34)	10.13 (10.69)	3.44 (3.68)	108.43 (206.84)	19.27 (15.25)
$\varepsilon.15$	10.51 (9.84)	4.61 (3.56)	9.93 (8.28)	3.62 (2.97)	1309.60 (2599.51)	29.94 (27.60)
$\mathrm{UCB}_{\eta=.5}$	1.49 (0.26)	1.42 (0.24)	1.09 (0.73)	1.85 (0.94)	7.00 (12.75)	0.00 (0.00)
$\mathrm{UCB}_{\eta=1}$	1.59 (0.23)	1.55 (0.34)	0.95 (0.64)	2.14 (0.94)	6.73 (13.20)	0.00 (0.00)
$\mathrm{UCB}_{\eta=2}$	1.15 (0.18)	1.12 (0.31)	0.49 (0.41)	1.29 (0.18)	6.32 (12.44)	0.00 (0.00)
$\mathrm{TS}_{\eta=.5}$	0.20 (0.33)	0.79 (1.16)	0.05 (0.10)	0.98 (0.79)	6.69 (11.86)	0.00 (0.00)
$\mathrm{TS}_{\eta=1}$	0.21 (0.37)	0.21 (0.31)	0.18 (0.15)	0.60 (0.92)	6.75 (12.50)	0.00 (0.00)
$\mathrm{TS}_{\eta=2}$	0.11 (0.16)	0.10 (0.11)	0.17 (0.22)	0.13 (0.24)	5.38 (10.57)	0.00 (0.00)
$\mathrm{UCB}_{\lambda=5}$	1.54 (0.30)	0.98 (0.32)	0.98 (0.62)	1.02 (0.20)	9.89 (10.92)	0.00 (0.00)
$\mathrm{TS}_{\lambda=5}$	0.03 (0.02)	0.05 (0.04)	0.05 (0.10)	0.01 (0.01)	7.70 (9.73)	0.00 (0.00)
$\varepsilon.10_{\eta=1}$	19.63 (16.53)	7.48 (5.89)	18.71 (16.38)	6.49 (5.08)	8.90 (11.85)	30.31 (18.79)
$\mathrm{dUCB}_{\eta=1}$	0.40 (0.27)	0.65 (0.36)	0.62 (0.35)	0.66 (0.15)	7.09 (12.64)	0.00 (0.00)
$\mathrm{dUCB}_{\eta=2}$	0.17 (0.23)	0.38 (0.32)	0.30 (0.09)	0.42 (0.10)	6.55 (12.89)	0.00 (0.00)

The ramp is where a historical mean is already stale and the fresh arm is not yet dominant.

Table 22: Ramp excess MSE: Gradual noise growth. Rounds with mixing weight in $(0, 1/2)$.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	1.56 (0.11)	4.22 (1.47)	1.86 (0.79)	5.42 (0.54)	6.08 (0.65)	1.97 (0.05)
dUCB	1.85 (0.81)	4.12 (1.71)	2.53 (0.84)	4.92 (0.54)	5.48 (0.62)	2.26 (0.22)
TS	0.00 (0.00)	0.18 (0.36)	0.00 (0.00)	0.53 (1.06)	0.40 (0.46)	0.44 (0.88)
$\varepsilon.05$	6.38 (5.14)	2.92 (2.50)	6.74 (4.93)	1.92 (1.43)	1.70 (1.21)	8.22 (5.96)
$\varepsilon.10$	14.59 (7.28)	6.13 (2.71)	15.32 (7.14)	4.94 (2.16)	2.60 (0.47)	19.50 (8.91)
$\varepsilon.15$	19.46 (10.27)	7.63 (3.93)	18.19 (8.80)	5.79 (2.93)	4.01 (1.70)	23.27 (10.61)
$\mathrm{UCB}_{\eta=.5}$	1.56 (0.11)	3.60 (1.34)	1.86 (0.79)	4.75 (0.39)	5.87 (0.81)	1.97 (0.05)
$\mathrm{UCB}_{\eta=1}$	1.59 (0.06)	3.53 (1.16)	1.93 (0.74)	4.67 (0.46)	5.44 (0.43)	2.05 (0.14)
$\mathrm{UCB}_{\eta=2}$	1.59 (0.07)	3.01 (0.08)	2.10 (0.68)	4.34 (1.07)	4.88 (0.73)	2.15 (0.22)
$\mathrm{TS}_{\eta=.5}$	0.33 (0.66)	0.67 (1.34)	0.73 (0.87)	0.00 (0.00)	1.44 (1.33)	1.56 (1.04)
$\mathrm{TS}_{\eta=1}$	0.00 (0.00)	0.00 (0.00)	0.47 (0.94)	0.15 (0.29)	1.01 (0.89)	1.04 (1.21)
$\mathrm{TS}_{\eta=2}$	0.86 (0.99)	0.00 (0.00)	0.90 (1.05)	0.16 (0.32)	1.21 (0.97)	1.42 (0.97)
$\mathrm{UCB}_{\lambda=5}$	1.59 (0.06)	2.93 (0.20)	1.93 (0.74)	2.80 (0.25)	3.05 (0.63)	2.00 (0.05)
$\mathrm{TS}_{\lambda=5}$	0.76 (0.89)	0.00 (0.00)	0.83 (0.97)	0.42 (0.56)	0.59 (0.26)	1.00 (1.16)
$\varepsilon.10_{\eta=1}$	25.99 (13.96)	9.97 (5.24)	25.26 (13.67)	8.37 (4.80)	3.44 (1.73)	32.85 (16.65)
$\mathrm{dUCB}_{\eta=1}$	1.83 (0.81)	3.58 (1.48)	2.12 (0.81)	4.84 (0.48)	5.46 (0.91)	2.26 (0.22)
$\mathrm{dUCB}_{\eta=2}$	2.10 (0.72)	3.06 (0.33)	2.12 (0.80)	4.25 (0.89)	5.33 (0.54)	2.68 (0.86)

The ramp is where a historical mean is already stale and the fresh arm is not yet dominant.

Table 23: Excess MSE before the shift starts: Gradual relocation. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	23.54 (1.47)	10.42 (1.06)	21.82 (0.54)	8.95 (0.37)	7.06 (0.03)	27.33 (6.33)
dUCB	23.54 (1.47)	10.29 (1.03)	21.82 (0.54)	8.44 (0.56)	6.67 (0.84)	27.33 (6.33)
TS	23.54 (1.47)	10.79 (2.56)	21.82 (0.54)	8.25 (1.12)	6.28 (0.71)	27.33 (6.33)
$\varepsilon.05$	25.21 (4.21)	9.94 (0.98)	23.48 (3.19)	7.56 (0.65)	3.37 (0.68)	28.89 (4.54)
$\varepsilon.10$	30.03 (5.39)	11.73 (2.26)	27.99 (4.88)	9.03 (1.50)	3.87 (0.95)	35.60 (12.84)
$\varepsilon.15$	31.77 (8.47)	12.39 (2.57)	29.81 (8.15)	9.71 (2.28)	4.29 (1.18)	36.09 (8.44)
$\mathrm{UCB}_{\eta=.5}$	23.54 (1.47)	10.42 (1.06)	21.82 (0.54)	8.43 (0.54)	6.82 (0.59)	27.33 (6.33)
$\mathrm{UCB}_{\eta=1}$	23.54 (1.47)	10.42 (1.06)	21.82 (0.54)	8.43 (0.54)	6.74 (0.71)	27.33 (6.33)
$\mathrm{UCB}_{\eta=2}$	23.54 (1.47)	10.42 (1.06)	21.82 (0.54)	8.43 (0.54)	6.62 (0.67)	27.33 (6.33)
$\mathrm{TS}_{\eta=.5}$	23.54 (1.47)	11.10 (0.77)	21.82 (0.54)	8.66 (0.75)	5.12 (1.63)	27.33 (6.33)
$\mathrm{TS}_{\eta=1}$	23.54 (1.47)	10.89 (1.45)	21.82 (0.54)	8.70 (0.96)	4.97 (1.36)	27.33 (6.33)
$\mathrm{TS}_{\eta=2}$	23.54 (1.47)	10.78 (1.16)	21.82 (0.54)	8.41 (0.47)	5.43 (1.33)	27.33 (6.33)
$\mathrm{UCB}_{\lambda=5}$	23.54 (1.47)	10.42 (1.06)	21.82 (0.54)	8.43 (0.54)	6.51 (0.56)	27.33 (6.33)
$\mathrm{TS}_{\lambda=5}$	23.54 (1.47)	10.13 (1.15)	21.82 (0.54)	8.15 (0.27)	5.45 (0.76)	27.33 (6.33)
$\varepsilon.10_{\eta=1}$	33.90 (10.22)	13.53 (4.36)	31.91 (10.57)	10.51 (3.15)	4.80 (1.87)	38.16 (10.57)
$\mathrm{dUCB}_{\eta=1}$	23.54 (1.47)	9.97 (1.14)	21.82 (0.54)	8.20 (0.36)	5.64 (0.59)	27.33 (6.33)
$\mathrm{dUCB}_{\eta=2}$	23.54 (1.47)	9.97 (1.14)	21.82 (0.54)	8.20 (0.36)	5.45 (0.76)	27.33 (6.33)

Rounds with mixing weight exactly zero. The share is still refreshed from round 16, so a tax can already be active on a gradual path's later reference window.

Table 24: Excess MSE before the shift starts: Drastic relocation. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	23.54 (1.47)	11.65 (1.09)	21.82 (0.54)	12.03 (0.08)	10.18 (0.60)	27.33 (6.33)
dUCB	23.54 (1.47)	10.61 (0.61)	21.82 (0.54)	9.00 (0.34)	7.65 (0.30)	27.33 (6.33)
TS	23.54 (1.47)	10.79 (2.56)	21.82 (0.54)	9.04 (1.09)	6.28 (0.71)	27.33 (6.33)
$\varepsilon.05$	34.98 (9.12)	14.36 (3.30)	32.90 (7.18)	10.67 (2.44)	4.43 (1.11)	41.95 (8.67)
$\varepsilon.10$	43.94 (14.76)	17.52 (6.04)	42.25 (14.96)	13.69 (4.65)	5.75 (2.14)	54.54 (25.08)
$\varepsilon.15$	58.75 (4.45)	23.25 (3.97)	53.22 (4.42)	18.10 (1.87)	7.13 (1.37)	67.69 (12.07)
$\mathrm{UCB}_{\eta=.5}$	23.54 (1.47)	10.74 (0.65)	21.82 (0.54)	11.05 (1.30)	9.66 (0.59)	27.33 (6.33)
$\mathrm{UCB}_{\eta=1}$	23.54 (1.47)	10.69 (0.69)	21.82 (0.54)	9.25 (0.39)	8.33 (0.30)	27.33 (6.33)
$\mathrm{UCB}_{\eta=2}$	23.54 (1.47)	10.42 (1.06)	21.82 (0.54)	8.43 (0.54)	8.13 (0.20)	27.33 (6.33)
$\mathrm{TS}_{\eta=.5}$	23.54 (1.47)	11.10 (0.77)	21.82 (0.54)	8.66 (0.75)	5.76 (0.73)	27.33 (6.33)
$\mathrm{TS}_{\eta=1}$	23.54 (1.47)	11.16 (1.94)	21.82 (0.54)	8.70 (0.96)	5.36 (0.76)	27.33 (6.33)
$\mathrm{TS}_{\eta=2}$	23.54 (1.47)	10.78 (1.16)	21.82 (0.54)	8.67 (0.96)	5.82 (0.57)	27.33 (6.33)
$\mathrm{UCB}_{\lambda=5}$	23.54 (1.47)	10.42 (1.06)	21.82 (0.54)	8.43 (0.54)	6.88 (0.50)	27.33 (6.33)
$\mathrm{TS}_{\lambda=5}$	23.54 (1.47)	10.13 (1.15)	21.82 (0.54)	8.15 (0.27)	5.45 (0.76)	27.33 (6.33)
$\varepsilon.10_{\eta=1}$	48.60 (8.62)	19.57 (2.67)	46.56 (8.78)	15.10 (2.52)	6.72 (1.87)	53.79 (4.05)
$\mathrm{dUCB}_{\eta=1}$	23.54 (1.47)	9.97 (1.14)	21.82 (0.54)	8.20 (0.36)	7.14 (0.70)	27.33 (6.33)
$\mathrm{dUCB}_{\eta=2}$	23.54 (1.47)	9.97 (1.14)	21.82 (0.54)	8.20 (0.36)	6.21 (0.86)	27.33 (6.33)

Rounds with mixing weight exactly zero. The share is still refreshed from round 16, so a tax can already be active on a gradual path's later reference window.

Table 25: Excess MSE before the shift starts: Gradual sign flip. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	35.57 (4.14)	16.32 (2.32)	34.30 (3.53)	12.74 (1.28)	10.52 (0.61)	39.63 (3.61)
dUCB	34.41 (3.48)	15.70 (2.39)	33.09 (3.36)	11.85 (0.79)	9.46 (0.91)	38.86 (3.01)
TS	35.16 (4.25)	15.97 (2.86)	34.24 (3.52)	12.15 (1.42)	8.90 (0.86)	39.28 (3.41)
$\varepsilon.05$	34.42 (4.15)	14.48 (2.22)	33.50 (3.54)	9.86 (0.34)	7.25 (0.77)	39.26 (2.23)
$\varepsilon.10$	44.48 (14.52)	18.05 (5.35)	42.32 (12.71)	12.61 (3.21)	9.26 (2.58)	52.87 (17.25)
$\varepsilon.15$	44.89 (11.68)	19.09 (6.60)	43.72 (14.50)	12.74 (3.19)	9.33 (2.96)	50.37 (10.59)
$\mathrm{UCB}_{\eta=.5}$	35.57 (4.14)	16.31 (2.34)	34.30 (3.53)	12.17 (0.64)	9.71 (0.89)	39.28 (3.41)
$\mathrm{UCB}_{\eta=1}$	35.67 (4.28)	16.31 (2.34)	34.30 (3.53)	12.15 (0.64)	9.71 (0.89)	39.28 (3.41)
$\mathrm{UCB}_{\eta=2}$	35.59 (4.18)	16.37 (2.23)	34.30 (3.53)	12.15 (0.64)	9.65 (0.81)	39.28 (3.41)
$\mathrm{TS}_{\eta=.5}$	34.41 (4.15)	15.26 (2.71)	33.46 (3.52)	10.87 (0.70)	8.67 (0.43)	39.36 (2.02)
$\mathrm{TS}_{\eta=1}$	34.77 (4.01)	14.89 (2.49)	33.75 (3.07)	10.91 (1.31)	8.28 (0.87)	39.34 (3.49)
$\mathrm{TS}_{\eta=2}$	35.29 (4.95)	14.75 (2.09)	33.93 (3.09)	10.75 (0.84)	8.30 (1.45)	39.74 (3.67)
$\mathrm{UCB}_{\lambda=5}$	35.57 (4.14)	16.31 (2.34)	34.30 (3.53)	12.15 (0.64)	9.65 (0.81)	39.63 (3.61)
$\mathrm{TS}_{\lambda=5}$	35.12 (4.50)	14.90 (2.27)	34.31 (3.39)	10.25 (0.33)	8.55 (1.12)	39.65 (3.62)
$\varepsilon.10_{\eta=1}$	52.29 (14.90)	22.11 (8.80)	50.08 (18.65)	15.37 (5.17)	11.20 (4.24)	64.77 (24.48)
$\mathrm{dUCB}_{\eta=1}$	34.41 (3.48)	15.85 (2.57)	33.09 (3.36)	11.33 (0.40)	9.44 (0.89)	38.86 (3.01)
$\mathrm{dUCB}_{\eta=2}$	34.43 (3.46)	15.85 (2.57)	33.09 (3.36)	11.37 (0.43)	9.39 (0.92)	38.86 (3.01)

Rounds with mixing weight exactly zero. The share is still refreshed from round 16, so a tax can already be active on a gradual path's later reference window.

Table 26: Excess MSE before the shift starts: Drastic sign flip. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	35.57 (4.14)	17.69 (2.44)	34.72 (3.52)	14.67 (1.23)	12.64 (0.77)	40.59 (2.95)
dUCB	35.43 (4.02)	16.19 (2.24)	34.20 (3.48)	12.98 (1.00)	10.92 (0.59)	39.30 (3.43)
TS	35.50 (4.11)	15.97 (2.86)	34.24 (3.52)	12.15 (1.42)	9.67 (0.29)	39.76 (3.72)
$\varepsilon.05$	53.49 (16.87)	23.42 (5.29)	53.64 (13.00)	16.11 (4.73)	12.09 (3.64)	65.68 (18.34)
$\varepsilon.10$	66.04 (24.57)	27.71 (11.54)	65.77 (27.14)	19.83 (7.43)	14.31 (5.62)	83.19 (34.58)
$\varepsilon.15$	92.98 (5.99)	38.77 (6.14)	90.46 (7.60)	27.60 (3.17)	21.05 (2.11)	114.48 (19.72)
$\text{UCB}_{\eta=.5}$	35.57 (4.14)	17.57 (2.22)	34.72 (3.52)	13.54 (0.71)	11.17 (0.88)	40.74 (3.03)
$\text{UCB}_{\eta=1}$	35.67 (4.28)	17.63 (2.25)	34.72 (3.52)	13.57 (0.77)	11.14 (0.85)	40.74 (3.03)
$\text{UCB}_{\eta=2}$	35.59 (4.18)	17.68 (2.15)	34.64 (3.50)	13.46 (0.56)	11.20 (1.09)	40.80 (3.07)
$\text{TS}_{\eta=.5}$	34.77 (4.63)	15.26 (2.71)	33.79 (2.86)	10.87 (0.70)	8.67 (0.43)	39.92 (2.57)
$\text{TS}_{\eta=1}$	35.17 (3.91)	15.04 (2.39)	34.20 (3.17)	10.91 (1.31)	8.28 (0.87)	39.81 (2.62)
$\text{TS}_{\eta=2}$	35.71 (4.21)	14.90 (2.29)	34.34 (3.42)	10.75 (0.84)	8.30 (1.45)	40.20 (3.92)
$\text{UCB}_{\lambda=5}$	35.57 (4.14)	17.63 (2.25)	34.72 (3.52)	13.47 (0.57)	11.21 (1.10)	40.59 (2.95)
$\text{TS}_{\lambda=5}$	35.52 (4.37)	14.90 (2.27)	34.76 (3.38)	10.25 (0.33)	8.55 (1.12)	40.11 (2.73)
$\varepsilon.10_{\eta=1}$	73.57 (12.40)	30.44 (1.63)	72.12 (5.69)	21.37 (2.71)	15.87 (2.60)	91.39 (12.01)
$\text{dUCB}_{\eta=1}$	35.41 (4.02)	16.16 (2.22)	34.28 (3.35)	11.97 (0.33)	10.11 (1.11)	39.39 (3.49)
$\text{dUCB}_{\eta=2}$	35.53 (3.98)	16.16 (2.22)	34.26 (3.33)	12.04 (0.41)	10.09 (1.12)	39.43 (3.54)

Rounds with mixing weight exactly zero. The share is still refreshed from round 16, so a tax can already be active on a gradual path's later reference window.

Table 27: Excess MSE before the shift starts: Gradual dying block. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	9.57 (1.50)	8.56 (0.30)	41.31 (10.81)
dUCB	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	7.78 (0.70)	7.59 (0.56)	41.31 (10.81)
TS	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	9.68 (0.68)	7.69 (1.13)	41.31 (10.81)
$\varepsilon.05$	29.02 (5.63)	12.16 (3.30)	29.62 (6.72)	8.19 (2.24)	5.73 (1.57)	46.07 (7.89)
$\varepsilon.10$	34.18 (4.08)	13.49 (2.28)	33.71 (6.53)	9.74 (2.23)	6.87 (1.76)	53.50 (16.12)
$\varepsilon.15$	36.93 (6.20)	15.15 (3.26)	37.48 (8.86)	10.65 (2.65)	7.49 (1.82)	65.33 (25.43)
$\text{UCB}_{\eta=.5}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	8.98 (0.62)	8.53 (0.32)	41.31 (10.81)
$\text{UCB}_{\eta=1}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	8.98 (0.62)	8.03 (0.92)	41.31 (10.81)
$\text{UCB}_{\eta=2}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	8.98 (0.62)	8.03 (0.92)	41.31 (10.81)
$\text{TS}_{\eta=.5}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	9.31 (1.57)	7.57 (1.31)	41.31 (10.81)
$\text{TS}_{\eta=1}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	10.14 (1.62)	6.90 (1.32)	41.31 (10.81)
$\text{TS}_{\eta=2}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	8.33 (0.98)	6.69 (1.55)	41.31 (10.81)
$\text{UCB}_{\lambda=5}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	8.98 (0.62)	8.03 (0.92)	41.31 (10.81)
$\text{TS}_{\lambda=5}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	8.40 (1.18)	6.69 (0.73)	41.31 (10.81)
$\varepsilon.10_{\eta=1}$	37.82 (10.45)	14.59 (3.76)	37.00 (11.44)	10.51 (3.38)	7.67 (2.63)	56.03 (14.92)
$\text{dUCB}_{\eta=1}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	7.35 (0.60)	6.92 (0.83)	41.31 (10.81)
$\text{dUCB}_{\eta=2}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	7.35 (0.60)	6.92 (0.83)	41.31 (10.81)

Rounds with mixing weight exactly zero. The share is still refreshed from round 16, so a tax can already be active on a gradual path's later reference window.

Table 28: Excess MSE before the shift starts: Drastic dying block. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	10.98 (0.79)	10.40 (0.77)	41.31 (10.81)
dUCB	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	9.21 (1.23)	8.33 (0.26)	41.31 (10.81)
TS	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	10.23 (0.73)	8.46 (1.76)	41.31 (10.81)
$\varepsilon.05$	35.11 (8.02)	14.84 (3.93)	35.53 (8.03)	10.10 (2.75)	6.70 (1.88)	56.51 (10.38)
$\varepsilon.10$	48.04 (15.68)	19.29 (6.74)	47.85 (16.69)	14.10 (5.65)	9.79 (3.76)	74.37 (28.08)
$\varepsilon.15$	56.06 (7.40)	22.94 (4.45)	55.02 (9.55)	16.19 (2.97)	10.73 (2.10)	92.52 (26.58)
$\text{UCB}_{\eta=.5}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	10.45 (0.73)	9.86 (1.10)	41.31 (10.81)
$\text{UCB}_{\eta=1}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	9.43 (1.24)	9.13 (0.32)	41.31 (10.81)
$\text{UCB}_{\eta=2}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	8.98 (0.62)	8.63 (0.22)	41.31 (10.81)
$\text{TS}_{\eta=.5}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	9.31 (1.57)	8.31 (1.90)	41.31 (10.81)
$\text{TS}_{\eta=1}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	10.62 (2.36)	7.46 (1.68)	41.31 (10.81)
$\text{TS}_{\eta=2}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	8.33 (0.98)	7.59 (1.77)	41.31 (10.81)
$\text{UCB}_{\lambda=5}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	8.98 (0.62)	8.03 (0.92)	41.31 (10.81)
$\text{TS}_{\lambda=5}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	8.40 (1.18)	6.69 (0.73)	41.31 (10.81)
$\varepsilon.10_{\eta=1}$	52.79 (18.35)	20.32 (6.54)	52.28 (19.95)	14.65 (5.48)	10.37 (4.44)	75.37 (23.87)
$\text{dUCB}_{\eta=1}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	7.82 (0.76)	7.71 (0.60)	41.31 (10.81)
$\text{dUCB}_{\eta=2}$	26.06 (1.74)	10.41 (0.20)	26.36 (0.61)	7.35 (0.60)	6.92 (0.83)	41.31 (10.81)

Rounds with mixing weight exactly zero. The share is still refreshed from round 16, so a tax can already be active on a gradual path's later reference window.

Table 29: Excess MSE before the shift starts: Gradual nonlinear add-on. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	21.11 (0.65)	10.70 (0.74)	22.78 (0.54)	9.85 (1.10)	42.57 (51.44)	33.32 (4.30)
dUCB	20.51 (0.59)	10.48 (0.70)	22.40 (0.85)	8.97 (0.79)	42.07 (51.73)	33.32 (4.30)
TS	19.63 (0.57)	9.49 (0.57)	21.99 (0.50)	8.72 (1.24)	57.83 (50.58)	33.32 (4.30)
$\varepsilon.05$	22.45 (7.03)	9.92 (1.91)	24.56 (6.89)	8.06 (2.24)	33.32 (55.45)	36.55 (4.06)
$\varepsilon.10$	25.30 (7.50)	11.88 (3.05)	28.38 (6.91)	9.30 (2.18)	60.51 (60.88)	46.60 (10.67)
$\varepsilon.15$	28.70 (6.88)	12.94 (3.07)	30.66 (5.68)	10.39 (2.11)	34.24 (56.71)	57.13 (21.15)
$\mathrm{UCB}_{\eta=.5}$	20.81 (0.40)	10.59 (0.50)	22.61 (0.40)	9.23 (1.08)	42.57 (51.44)	33.32 (4.30)
$\mathrm{UCB}_{\eta=1}$	20.34 (0.56)	10.23 (0.62)	22.61 (0.40)	8.84 (0.70)	42.57 (51.44)	33.32 (4.30)
$\mathrm{UCB}_{\eta=2}$	20.34 (0.56)	10.23 (0.62)	22.61 (0.40)	8.84 (0.70)	42.57 (51.44)	33.32 (4.30)
$\mathrm{TS}_{\eta=.5}$	19.61 (0.83)	9.33 (0.90)	21.95 (0.97)	7.97 (0.82)	43.13 (52.41)	33.32 (4.30)
$\mathrm{TS}_{\eta=1}$	19.58 (0.49)	9.32 (0.90)	23.06 (1.47)	8.15 (1.43)	42.59 (50.40)	33.32 (4.30)
$\mathrm{TS}_{\eta=2}$	19.61 (0.47)	9.38 (0.62)	22.14 (1.66)	8.04 (0.48)	40.32 (51.41)	33.32 (4.30)
$\mathrm{UCB}_{\lambda=5}$	20.48 (0.50)	10.23 (0.62)	22.61 (0.40)	8.84 (0.70)	42.57 (51.44)	33.32 (4.30)
$\mathrm{TS}_{\lambda=5}$	19.74 (0.73)	9.45 (0.56)	21.84 (0.93)	7.71 (0.35)	40.07 (52.30)	33.32 (4.30)
$\varepsilon.10_{\eta=1}$	27.08 (8.42)	12.24 (3.16)	29.16 (7.43)	9.67 (2.73)	36.33 (55.60)	57.94 (24.77)
$\mathrm{dUCB}_{\eta=1}$	20.51 (0.59)	9.98 (0.70)	22.04 (0.52)	8.33 (0.20)	41.74 (51.09)	33.32 (4.30)
$\mathrm{dUCB}_{\eta=2}$	20.51 (0.59)	9.98 (0.70)	22.04 (0.52)	8.33 (0.20)	41.74 (51.09)	33.32 (4.30)

Rounds with mixing weight exactly zero. The share is still refreshed from round 16, so a tax can already be active on a gradual path's later reference window.

Table 30: Excess MSE before the shift starts: Drastic nonlinear add-on. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	22.46 (0.77)	12.16 (0.80)	24.91 (2.55)	11.57 (0.94)	44.63 (51.13)	33.33 (4.30)
dUCB	21.32 (0.49)	10.94 (0.77)	22.55 (0.60)	10.19 (1.14)	42.36 (51.50)	33.33 (4.30)
TS	19.63 (0.57)	9.49 (0.57)	22.19 (0.09)	8.72 (1.24)	174.46 (249.12)	33.33 (4.30)
$\varepsilon.05$	24.34 (6.70)	11.22 (2.31)	26.86 (6.95)	8.96 (2.34)	34.05 (54.93)	47.44 (14.38)
$\varepsilon.10$	35.48 (13.77)	16.15 (5.85)	38.68 (13.41)	12.97 (4.66)	227.20 (366.19)	67.09 (24.60)
$\varepsilon.15$	38.08 (12.28)	17.27 (4.11)	39.68 (8.38)	13.99 (2.86)	1483.01 (2946.89)	88.42 (35.39)
$\mathrm{UCB}_{\eta=.5}$	21.72 (0.39)	11.66 (0.57)	23.22 (1.12)	10.60 (0.91)	42.57 (51.44)	33.33 (4.30)
$\mathrm{UCB}_{\eta=1}$	21.42 (0.56)	11.28 (0.60)	22.61 (0.40)	10.09 (0.63)	42.57 (51.44)	33.33 (4.30)
$\mathrm{UCB}_{\eta=2}$	20.77 (0.47)	10.70 (0.63)	22.61 (0.40)	9.53 (0.76)	42.57 (51.44)	33.33 (4.30)
$\mathrm{TS}_{\eta=.5}$	19.79 (0.49)	9.48 (0.74)	21.95 (0.97)	8.17 (0.69)	43.13 (52.41)	33.33 (4.30)
$\mathrm{TS}_{\eta=1}$	19.78 (0.58)	9.51 (0.60)	23.28 (1.24)	8.26 (1.34)	42.59 (50.40)	33.33 (4.30)
$\mathrm{TS}_{\eta=2}$	19.80 (0.64)	9.52 (0.66)	22.45 (1.14)	8.18 (0.74)	40.32 (51.41)	33.33 (4.30)
$\mathrm{UCB}_{\lambda=5}$	21.43 (0.62)	10.85 (0.81)	22.61 (0.40)	8.96 (0.58)	42.57 (51.44)	33.33 (4.30)
$\mathrm{TS}_{\lambda=5}$	19.74 (0.73)	9.45 (0.56)	21.84 (0.93)	7.71 (0.35)	41.37 (51.86)	33.33 (4.30)
$\varepsilon.10_{\eta=1}$	37.66 (20.67)	16.47 (6.95)	39.27 (18.57)	13.65 (6.27)	38.85 (54.92)	76.43 (30.07)
$\mathrm{dUCB}_{\eta=1}$	20.51 (0.59)	10.44 (0.42)	22.04 (0.52)	8.72 (0.61)	41.74 (51.09)	33.33 (4.30)
$\mathrm{dUCB}_{\eta=2}$	20.51 (0.59)	10.23 (0.89)	22.04 (0.52)	8.33 (0.20)	41.74 (51.09)	33.33 (4.30)

Rounds with mixing weight exactly zero. The share is still refreshed from round 16, so a tax can already be active on a gradual path's later reference window.

Table 31: Excess MSE before the shift starts: Gradual noise growth. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	21.98 (0.65)	10.59 (0.91)	22.33 (1.27)	9.06 (0.82)	7.31 (0.46)	27.46 (6.64)
dUCB	21.72 (1.22)	10.01 (0.90)	21.86 (0.87)	8.28 (0.64)	7.11 (0.23)	27.03 (5.95)
TS	21.74 (1.26)	9.88 (1.13)	22.37 (1.32)	8.59 (1.13)	6.04 (0.40)	27.59 (5.18)
$\varepsilon.05$	23.84 (7.13)	9.81 (2.11)	24.51 (7.08)	7.74 (1.95)	3.11 (0.72)	29.84 (6.97)
$\varepsilon.10$	25.97 (4.63)	11.03 (3.37)	27.03 (6.92)	8.45 (1.71)	3.54 (0.58)	33.89 (12.96)
$\varepsilon.15$	29.12 (8.61)	12.45 (3.83)	29.78 (9.10)	9.54 (2.27)	3.81 (1.09)	45.32 (24.12)
$\text{UCB}_{\eta=.5}$	21.98 (0.65)	10.59 (0.91)	22.33 (1.27)	8.65 (0.67)	7.24 (0.34)	27.46 (6.64)
$\text{UCB}_{\eta=1}$	21.98 (0.65)	10.59 (0.91)	22.33 (1.27)	8.65 (0.67)	7.29 (0.42)	27.49 (6.70)
$\text{UCB}_{\eta=2}$	21.98 (0.65)	10.59 (0.91)	22.33 (1.27)	8.65 (0.67)	7.29 (0.42)	27.49 (6.70)
$\text{TS}_{\eta=.5}$	21.64 (0.58)	9.50 (1.03)	22.06 (1.12)	8.71 (0.55)	5.05 (1.48)	27.27 (7.38)
$\text{TS}_{\eta=1}$	21.99 (0.56)	9.38 (0.89)	22.29 (1.00)	8.11 (0.88)	5.05 (1.31)	27.69 (5.81)
$\text{TS}_{\eta=2}$	21.25 (1.02)	9.33 (0.98)	21.95 (1.79)	8.14 (0.83)	4.76 (1.34)	27.22 (5.86)
$\text{UCB}_{\lambda=5}$	21.98 (0.65)	10.59 (0.91)	22.33 (1.27)	8.65 (0.67)	7.22 (0.30)	27.46 (6.64)
$\text{TS}_{\lambda=5}$	21.14 (1.01)	9.97 (0.91)	22.01 (1.74)	8.08 (1.37)	5.47 (0.70)	27.48 (6.11)
$\varepsilon.10_{\eta=1}$	27.83 (9.43)	11.75 (3.83)	28.28 (9.47)	9.13 (2.90)	4.11 (1.49)	37.35 (14.49)
$\text{dUCB}_{\eta=1}$	21.72 (1.22)	10.01 (0.90)	21.86 (0.87)	8.28 (0.64)	6.43 (0.74)	27.03 (5.95)
$\text{dUCB}_{\eta=2}$	21.72 (1.22)	10.01 (0.90)	21.91 (0.97)	8.28 (0.64)	6.43 (0.74)	27.03 (5.95)

Rounds with mixing weight exactly zero. The share is still refreshed from round 16, so a tax can already be active on a gradual path's later reference window.

Table 32: Excess MSE before the shift starts: Drastic noise growth. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	21.98 (0.65)	12.59 (0.91)	23.08 (0.47)	11.97 (0.92)	10.54 (0.79)	28.47 (6.15)
dUCB	22.04 (0.66)	10.62 (0.88)	22.25 (1.16)	9.88 (0.69)	7.87 (0.26)	27.53 (6.80)
TS	21.74 (1.26)	10.06 (1.03)	22.37 (1.32)	9.14 (0.62)	6.04 (0.40)	28.03 (5.95)
$\varepsilon.05$	28.59 (8.60)	12.05 (2.89)	29.55 (7.71)	9.31 (2.36)	4.31 (1.24)	36.43 (6.23)
$\varepsilon.10$	37.65 (11.80)	15.90 (5.50)	39.02 (13.05)	12.39 (3.35)	5.36 (1.96)	49.19 (19.61)
$\varepsilon.15$	45.24 (8.04)	18.51 (3.27)	44.44 (6.37)	14.28 (1.91)	6.97 (0.68)	63.78 (23.36)
$\text{UCB}_{\eta=.5}$	21.98 (0.65)	11.99 (0.92)	23.08 (0.47)	11.03 (0.73)	9.40 (0.51)	28.47 (6.15)
$\text{UCB}_{\eta=1}$	21.98 (0.65)	11.87 (0.92)	23.08 (0.47)	10.06 (0.75)	9.27 (0.41)	28.50 (6.21)
$\text{UCB}_{\eta=2}$	21.98 (0.65)	11.87 (0.92)	23.08 (0.47)	10.08 (0.75)	8.82 (0.45)	28.56 (6.14)
$\text{TS}_{\eta=.5}$	21.97 (0.41)	9.50 (1.03)	22.83 (1.15)	8.71 (0.55)	5.32 (1.06)	28.29 (7.19)
$\text{TS}_{\eta=1}$	21.99 (0.56)	9.38 (0.89)	22.29 (1.00)	8.11 (0.88)	5.36 (0.85)	28.74 (6.21)
$\text{TS}_{\eta=2}$	22.10 (0.60)	9.33 (0.98)	22.84 (0.89)	8.14 (0.83)	5.49 (0.77)	28.17 (5.54)
$\text{UCB}_{\lambda=5}$	21.98 (0.65)	11.87 (0.92)	23.08 (0.47)	10.09 (0.74)	8.76 (0.33)	28.47 (6.15)
$\text{TS}_{\lambda=5}$	21.51 (0.56)	9.97 (0.91)	22.39 (1.86)	8.08 (1.37)	5.47 (0.70)	27.97 (6.88)
$\varepsilon.10_{\eta=1}$	42.88 (17.83)	17.42 (6.33)	42.68 (16.79)	13.81 (5.91)	6.18 (1.56)	55.27 (21.33)
$\text{dUCB}_{\eta=1}$	22.04 (0.66)	10.65 (0.90)	22.25 (1.16)	8.96 (0.67)	7.82 (0.25)	27.55 (6.84)
$\text{dUCB}_{\eta=2}$	22.04 (0.66)	10.65 (0.90)	22.30 (1.21)	8.95 (0.70)	7.12 (0.76)	27.55 (6.84)

Rounds with mixing weight exactly zero. The share is still refreshed from round 16, so a tax can already be active on a gradual path's later reference window.

Table 33: Post-shift hit rate: Gradual relocation. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.38 (0.09)	0.44 (0.02)	0.42 (0.06)	0.42 (0.02)	0.52 (0.10)	0.38 (0.13)
dUCB	0.51 (0.02)	0.49 (0.02)	0.47 (0.00)	0.48 (0.02)	0.75 (0.17)	0.47 (0.06)
TS	0.07 (0.07)	0.07 (0.13)	0.13 (0.09)	0.01 (0.02)	0.03 (0.05)	0.14 (0.13)
$\varepsilon.05$	0.05 (0.04)	0.04 (0.04)	0.07 (0.07)	0.06 (0.06)	0.03 (0.03)	0.12 (0.12)
$\varepsilon.10$	0.03 (0.03)	0.02 (0.03)	0.03 (0.03)	0.03 (0.03)	0.03 (0.03)	0.03 (0.03)
$\varepsilon.15$	0.06 (0.06)	0.05 (0.04)	0.07 (0.07)	0.06 (0.06)	0.05 (0.03)	0.12 (0.13)
$\text{UCB}_{\eta=.5}$	0.49 (0.02)	0.50 (0.03)	0.47 (0.00)	0.49 (0.03)	0.67 (0.14)	0.45 (0.10)
$\text{UCB}_{\eta=1}$	0.51 (0.02)	0.50 (0.03)	0.47 (0.00)	0.47 (0.03)	0.59 (0.07)	0.49 (0.03)
$\text{UCB}_{\eta=2}$	0.51 (0.02)	0.50 (0.03)	0.47 (0.00)	0.50 (0.03)	0.73 (0.03)	0.50 (0.03)
$\text{TS}_{\eta=.5}$	0.17 (0.12)	0.00 (0.00)	0.16 (0.12)	0.00 (0.00)	0.01 (0.02)	0.31 (0.11)
$\text{TS}_{\eta=1}$	0.28 (0.18)	0.12 (0.21)	0.31 (0.11)	0.01 (0.02)	0.13 (0.17)	0.27 (0.18)
$\text{TS}_{\eta=2}$	0.40 (0.05)	0.30 (0.14)	0.39 (0.07)	0.23 (0.15)	0.14 (0.17)	0.47 (0.05)
$\text{UCB}_{\lambda=5}$	0.51 (0.02)	0.50 (0.03)	0.47 (0.00)	0.50 (0.03)	0.92 (0.08)	0.45 (0.10)
$\text{TS}_{\lambda=5}$	0.24 (0.15)	0.32 (0.13)	0.33 (0.07)	0.31 (0.03)	0.40 (0.12)	0.28 (0.15)
$\varepsilon.10_{\eta=1}$	0.03 (0.03)	0.03 (0.03)	0.01 (0.02)	0.02 (0.02)	0.02 (0.02)	0.13 (0.07)
$\text{dUCB}_{\eta=1}$	0.51 (0.02)	0.50 (0.03)	0.47 (0.00)	0.50 (0.03)	0.87 (0.08)	0.50 (0.03)
$\text{dUCB}_{\eta=2}$	0.51 (0.02)	0.50 (0.03)	0.47 (0.00)	0.50 (0.03)	0.92 (0.08)	0.50 (0.03)

Fraction of post-shift rounds on the single best scorecard. A blend can have low excess and a low hit rate.

Table 34: Post-shift hit rate: Drastic relocation. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.63 (0.15)	0.00 (0.00)
dUCB	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.62 (0.44)	0.00 (0.00)
TS	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
$\varepsilon.05$	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)
$\varepsilon.10$	0.04 (0.01)	0.04 (0.01)	0.04 (0.01)	0.04 (0.01)	0.04 (0.01)	0.12 (0.15)
$\varepsilon.15$	0.04 (0.03)	0.04 (0.03)	0.04 (0.03)	0.04 (0.03)	0.04 (0.03)	0.04 (0.03)
$\text{UCB}_{\eta=.5}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.71 (0.06)	0.00 (0.00)
$\text{UCB}_{\eta=1}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.79 (0.03)	0.00 (0.00)
$\text{UCB}_{\eta=2}$	0.00 (0.00)	0.10 (0.20)	0.00 (0.00)	0.07 (0.15)	0.79 (0.01)	0.00 (0.00)
$\text{TS}_{\eta=.5}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
$\text{TS}_{\eta=1}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.12 (0.25)	0.00 (0.00)
$\text{TS}_{\eta=2}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.12 (0.24)	0.00 (0.00)
$\text{UCB}_{\lambda=5}$	0.00 (0.00)	0.14 (0.29)	0.00 (0.00)	0.30 (0.26)	0.80 (0.00)	0.00 (0.00)
$\text{TS}_{\lambda=5}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.07 (0.15)	0.39 (0.45)	0.00 (0.00)
$\varepsilon.10_{\eta=1}$	0.03 (0.04)	0.03 (0.04)	0.03 (0.04)	0.03 (0.04)	0.03 (0.04)	0.03 (0.04)
$\text{dUCB}_{\eta=1}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.84 (0.08)	0.00 (0.00)
$\text{dUCB}_{\eta=2}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.84 (0.05)	0.00 (0.00)

Fraction of post-shift rounds on the single best scorecard. A blend can have low excess and a low hit rate.

Table 35: Post-shift hit rate: Gradual sign flip. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.00 (0.00)	0.18 (0.18)	0.00 (0.00)	0.31 (0.18)	0.35 (0.06)	0.00 (0.00)
dUCB	0.00 (0.00)	0.17 (0.12)	0.00 (0.00)	0.26 (0.11)	0.37 (0.09)	0.00 (0.00)
TS	0.00 (0.00)	0.06 (0.06)	0.00 (0.00)	0.03 (0.05)	0.03 (0.07)	0.00 (0.00)
$\varepsilon.05$	0.03 (0.03)	0.03 (0.03)	0.03 (0.03)	0.04 (0.04)	0.08 (0.08)	0.03 (0.03)
$\varepsilon.10$	0.03 (0.03)	0.03 (0.03)	0.03 (0.03)	0.03 (0.03)	0.03 (0.03)	0.03 (0.03)
$\varepsilon.15$	0.05 (0.04)	0.03 (0.03)	0.03 (0.03)	0.03 (0.03)	0.07 (0.09)	0.06 (0.06)
$\mathrm{UCB}_{\eta=.5}$	0.00 (0.00)	0.12 (0.19)	0.00 (0.00)	0.38 (0.07)	0.41 (0.03)	0.00 (0.00)
$\mathrm{UCB}_{\eta=1}$	0.00 (0.00)	0.08 (0.13)	0.00 (0.00)	0.36 (0.10)	0.43 (0.03)	0.00 (0.00)
$\mathrm{UCB}_{\eta=2}$	0.00 (0.00)	0.07 (0.11)	0.00 (0.00)	0.31 (0.09)	0.42 (0.06)	0.00 (0.00)
$\mathrm{TS}_{\eta=.5}$	0.00 (0.00)	0.09 (0.11)	0.00 (0.00)	0.14 (0.10)	0.04 (0.08)	0.00 (0.00)
$\mathrm{TS}_{\eta=1}$	0.00 (0.00)	0.01 (0.02)	0.00 (0.00)	0.10 (0.08)	0.08 (0.10)	0.00 (0.00)
$\mathrm{TS}_{\eta=2}$	0.00 (0.00)	0.03 (0.05)	0.00 (0.00)	0.03 (0.05)	0.08 (0.10)	0.00 (0.00)
$\mathrm{UCB}_{\lambda=5}$	0.00 (0.00)	0.08 (0.13)	0.00 (0.00)	0.33 (0.08)	0.42 (0.06)	0.00 (0.00)
$\mathrm{TS}_{\lambda=5}$	0.00 (0.00)	0.03 (0.07)	0.00 (0.00)	0.08 (0.10)	0.03 (0.07)	0.00 (0.00)
$\varepsilon.10_{\eta=1}$	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)	0.02 (0.02)
$\mathrm{dUCB}_{\eta=1}$	0.00 (0.00)	0.09 (0.10)	0.00 (0.00)	0.33 (0.14)	0.36 (0.11)	0.00 (0.00)
$\mathrm{dUCB}_{\eta=2}$	0.00 (0.00)	0.06 (0.08)	0.00 (0.00)	0.25 (0.13)	0.32 (0.12)	0.00 (0.00)

Fraction of post-shift rounds on the single best scorecard. A blend can have low excess and a low hit rate.

Table 36: Post-shift hit rate: Drastic sign flip. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
dUCB	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
TS	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
$\varepsilon.05$	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)
$\varepsilon.10$	0.04 (0.01)	0.04 (0.01)	0.04 (0.01)	0.04 (0.01)	0.04 (0.01)	0.04 (0.01)
$\varepsilon.15$	0.04 (0.03)	0.04 (0.03)	0.04 (0.03)	0.04 (0.03)	0.04 (0.03)	0.04 (0.03)
$\mathrm{UCB}_{\eta=.5}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
$\mathrm{UCB}_{\eta=1}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
$\mathrm{UCB}_{\eta=2}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
$\mathrm{TS}_{\eta=.5}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
$\mathrm{TS}_{\eta=1}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
$\mathrm{TS}_{\eta=2}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
$\mathrm{UCB}_{\lambda=5}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
$\mathrm{TS}_{\lambda=5}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
$\varepsilon.10_{\eta=1}$	0.03 (0.04)	0.03 (0.04)	0.03 (0.04)	0.03 (0.04)	0.03 (0.04)	0.03 (0.04)
$\mathrm{dUCB}_{\eta=1}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.01 (0.03)	0.00 (0.00)
$\mathrm{dUCB}_{\eta=2}$	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.11 (0.21)	0.00 (0.00)

Fraction of post-shift rounds on the single best scorecard. A blend can have low excess and a low hit rate.

Table 37: Post-shift hit rate: Gradual dying block. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.01 (0.02)	0.82 (0.10)	0.00 (0.00)	0.62 (0.31)	0.76 (0.02)	0.05 (0.10)
dUCB	0.12 (0.08)	0.71 (0.07)	0.12 (0.11)	0.90 (0.09)	0.89 (0.05)	0.18 (0.28)
TS	0.01 (0.02)	0.17 (0.17)	0.00 (0.00)	0.01 (0.02)	0.03 (0.03)	0.01 (0.02)
$\varepsilon.05$	0.03 (0.03)	0.03 (0.03)	0.02 (0.02)	0.03 (0.03)	0.05 (0.02)	0.02 (0.02)
$\varepsilon.10$	0.05 (0.02)	0.06 (0.02)	0.04 (0.02)	0.05 (0.02)	0.07 (0.03)	0.04 (0.02)
$\varepsilon.15$	0.04 (0.03)	0.05 (0.04)	0.03 (0.04)	0.04 (0.05)	0.05 (0.06)	0.03 (0.04)
$\text{UCB}_{\eta=.5}$	0.01 (0.02)	0.78 (0.10)	0.00 (0.00)	0.78 (0.16)	0.77 (0.07)	0.07 (0.13)
$\text{UCB}_{\eta=1}$	0.01 (0.02)	0.76 (0.07)	0.00 (0.00)	0.79 (0.17)	0.84 (0.06)	0.08 (0.17)
$\text{UCB}_{\eta=2}$	0.01 (0.02)	0.73 (0.08)	0.01 (0.02)	0.78 (0.17)	0.84 (0.06)	0.12 (0.23)
$\text{TS}_{\eta=.5}$	0.01 (0.02)	0.14 (0.17)	0.00 (0.00)	0.06 (0.12)	0.03 (0.03)	0.00 (0.00)
$\text{TS}_{\eta=1}$	0.01 (0.02)	0.10 (0.18)	0.00 (0.00)	0.06 (0.10)	0.03 (0.03)	0.03 (0.05)
$\text{TS}_{\eta=2}$	0.01 (0.02)	0.14 (0.18)	0.00 (0.00)	0.10 (0.08)	0.08 (0.08)	0.03 (0.07)
$\text{UCB}_{\lambda=5}$	0.01 (0.02)	0.70 (0.10)	0.00 (0.00)	0.78 (0.16)	0.81 (0.07)	0.07 (0.13)
$\text{TS}_{\lambda=5}$	0.01 (0.02)	0.14 (0.18)	0.00 (0.00)	0.08 (0.10)	0.05 (0.02)	0.00 (0.00)
$\varepsilon.10_{\eta=1}$	0.03 (0.04)	0.04 (0.03)	0.03 (0.03)	0.03 (0.05)	0.06 (0.03)	0.03 (0.03)
$\text{dUCB}_{\eta=1}$	0.23 (0.07)	0.70 (0.07)	0.22 (0.10)	0.94 (0.02)	0.93 (0.03)	0.22 (0.27)
$\text{dUCB}_{\eta=2}$	0.27 (0.02)	0.67 (0.07)	0.29 (0.07)	0.86 (0.03)	0.91 (0.05)	0.27 (0.31)

Fraction of post-shift rounds on the single best scorecard. A blend can have low excess and a low hit rate.

Table 38: Post-shift hit rate: Drastic dying block. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.02 (0.02)	0.86 (0.04)	0.10 (0.14)	0.72 (0.06)	0.78 (0.12)	0.19 (0.35)
dUCB	0.73 (0.04)	0.90 (0.02)	0.74 (0.09)	0.90 (0.08)	0.96 (0.04)	0.56 (0.40)
TS	0.12 (0.23)	0.30 (0.32)	0.01 (0.01)	0.00 (0.00)	0.00 (0.00)	0.23 (0.27)
$\varepsilon.05$	0.06 (0.08)	0.03 (0.03)	0.02 (0.01)	0.02 (0.01)	0.02 (0.01)	0.03 (0.03)
$\varepsilon.10$	0.15 (0.15)	0.12 (0.15)	0.12 (0.15)	0.12 (0.15)	0.12 (0.15)	0.21 (0.20)
$\varepsilon.15$	0.06 (0.07)	0.04 (0.04)	0.06 (0.08)	0.06 (0.08)	0.04 (0.03)	0.11 (0.18)
$\text{UCB}_{\eta=.5}$	0.17 (0.12)	0.81 (0.01)	0.22 (0.21)	0.77 (0.04)	0.88 (0.02)	0.30 (0.38)
$\text{UCB}_{\eta=1}$	0.33 (0.15)	0.80 (0.00)	0.36 (0.22)	0.88 (0.09)	0.91 (0.03)	0.35 (0.41)
$\text{UCB}_{\eta=2}$	0.61 (0.13)	0.80 (0.00)	0.62 (0.13)	0.84 (0.04)	0.89 (0.05)	0.39 (0.41)
$\text{TS}_{\eta=.5}$	0.00 (0.00)	0.44 (0.17)	0.06 (0.11)	0.12 (0.25)	0.00 (0.00)	0.17 (0.30)
$\text{TS}_{\eta=1}$	0.02 (0.04)	0.52 (0.18)	0.03 (0.04)	0.12 (0.25)	0.17 (0.34)	0.29 (0.34)
$\text{TS}_{\eta=2}$	0.22 (0.22)	0.66 (0.09)	0.29 (0.30)	0.51 (0.35)	0.42 (0.28)	0.36 (0.38)
$\text{UCB}_{\lambda=5}$	0.34 (0.14)	0.80 (0.00)	0.38 (0.22)	0.80 (0.00)	0.81 (0.02)	0.30 (0.38)
$\text{TS}_{\lambda=5}$	0.09 (0.17)	0.66 (0.10)	0.11 (0.17)	0.66 (0.16)	0.70 (0.17)	0.29 (0.38)
$\varepsilon.10_{\eta=1}$	0.12 (0.23)	0.09 (0.14)	0.13 (0.19)	0.12 (0.23)	0.06 (0.10)	0.22 (0.20)
$\text{dUCB}_{\eta=1}$	0.79 (0.01)	0.83 (0.02)	0.79 (0.01)	0.94 (0.02)	0.94 (0.02)	0.62 (0.26)
$\text{dUCB}_{\eta=2}$	0.80 (0.00)	0.80 (0.00)	0.80 (0.00)	0.89 (0.04)	0.93 (0.03)	0.67 (0.26)

Fraction of post-shift rounds on the single best scorecard. A blend can have low excess and a low hit rate.

Table 39: Post-shift hit rate: Gradual nonlinear add-on. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.71 (0.05)	0.72 (0.02)	0.34 (0.38)	0.68 (0.02)	0.03 (0.05)	0.05 (0.03)
dUCB	0.89 (0.02)	0.93 (0.06)	0.62 (0.40)	0.94 (0.03)	0.26 (0.30)	0.42 (0.28)
TS	0.04 (0.02)	0.05 (0.06)	0.03 (0.03)	0.04 (0.02)	0.15 (0.16)	0.05 (0.03)
$\varepsilon.05$	0.06 (0.03)	0.07 (0.06)	0.07 (0.07)	0.06 (0.02)	0.22 (0.35)	0.07 (0.05)
$\varepsilon.10$	0.08 (0.02)	0.08 (0.04)	0.06 (0.02)	0.08 (0.03)	0.16 (0.19)	0.09 (0.03)
$\varepsilon.15$	0.07 (0.06)	0.08 (0.10)	0.05 (0.06)	0.07 (0.05)	0.25 (0.27)	0.08 (0.03)
$\text{UCB}_{\eta=.5}$	0.84 (0.03)	0.80 (0.04)	0.62 (0.39)	0.81 (0.02)	0.41 (0.34)	0.08 (0.06)
$\text{UCB}_{\eta=1}$	0.88 (0.03)	0.88 (0.02)	0.67 (0.43)	0.86 (0.05)	0.41 (0.34)	0.21 (0.18)
$\text{UCB}_{\eta=2}$	0.87 (0.00)	0.89 (0.06)	0.72 (0.46)	0.88 (0.02)	0.52 (0.35)	0.42 (0.26)
$\text{TS}_{\eta=.5}$	0.04 (0.02)	0.05 (0.06)	0.03 (0.02)	0.04 (0.02)	0.45 (0.32)	0.05 (0.03)
$\text{TS}_{\eta=1}$	0.04 (0.02)	0.05 (0.06)	0.21 (0.13)	0.04 (0.02)	0.51 (0.36)	0.07 (0.05)
$\text{TS}_{\eta=2}$	0.09 (0.10)	0.11 (0.10)	0.31 (0.25)	0.04 (0.02)	0.58 (0.41)	0.28 (0.29)
$\text{UCB}_{\lambda=5}$	0.84 (0.05)	0.91 (0.06)	0.65 (0.41)	0.90 (0.03)	0.10 (0.16)	0.08 (0.06)
$\text{TS}_{\lambda=5}$	0.04 (0.02)	0.05 (0.06)	0.16 (0.10)	0.04 (0.02)	0.26 (0.29)	0.05 (0.03)
$\varepsilon.10_{\eta=1}$	0.12 (0.16)	0.07 (0.07)	0.04 (0.04)	0.07 (0.03)	0.49 (0.31)	0.11 (0.06)
$\text{dUCB}_{\eta=1}$	0.88 (0.03)	0.92 (0.04)	0.70 (0.45)	0.91 (0.04)	0.50 (0.33)	0.46 (0.31)
$\text{dUCB}_{\eta=2}$	0.91 (0.04)	0.92 (0.03)	0.72 (0.46)	0.91 (0.03)	0.56 (0.38)	0.56 (0.26)

Fraction of post-shift rounds on the single best scorecard. A blend can have low excess and a low hit rate.

Table 40: Post-shift hit rate: Drastic nonlinear add-on. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.82 (0.01)	0.83 (0.01)	0.45 (0.43)	0.82 (0.02)	0.31 (0.37)	0.38 (0.22)
dUCB	0.93 (0.01)	0.93 (0.04)	0.73 (0.49)	0.95 (0.00)	0.48 (0.34)	0.82 (0.05)
TS	0.10 (0.20)	0.00 (0.00)	0.25 (0.07)	0.08 (0.16)	0.16 (0.16)	0.26 (0.17)
$\varepsilon.05$	0.02 (0.01)	0.02 (0.01)	0.14 (0.17)	0.08 (0.12)	0.55 (0.37)	0.12 (0.12)
$\varepsilon.10$	0.14 (0.19)	0.12 (0.15)	0.20 (0.10)	0.17 (0.16)	0.34 (0.37)	0.25 (0.09)
$\varepsilon.15$	0.11 (0.18)	0.06 (0.07)	0.17 (0.17)	0.07 (0.09)	0.47 (0.34)	0.23 (0.30)
$\text{UCB}_{\eta=.5}$	0.94 (0.01)	0.92 (0.01)	0.68 (0.46)	0.91 (0.01)	0.59 (0.40)	0.54 (0.33)
$\text{UCB}_{\eta=1}$	0.93 (0.04)	0.91 (0.03)	0.78 (0.31)	0.91 (0.02)	0.60 (0.40)	0.60 (0.37)
$\text{UCB}_{\eta=2}$	0.88 (0.03)	0.87 (0.04)	0.76 (0.12)	0.89 (0.01)	0.60 (0.40)	0.65 (0.27)
$\text{TS}_{\eta=.5}$	0.03 (0.04)	0.00 (0.00)	0.41 (0.30)	0.00 (0.00)	0.58 (0.39)	0.39 (0.24)
$\text{TS}_{\eta=1}$	0.12 (0.24)	0.00 (0.00)	0.48 (0.33)	0.14 (0.23)	0.60 (0.40)	0.52 (0.32)
$\text{TS}_{\eta=2}$	0.51 (0.30)	0.49 (0.11)	0.53 (0.36)	0.29 (0.21)	0.60 (0.40)	0.58 (0.36)
$\text{UCB}_{\lambda=5}$	0.93 (0.04)	0.82 (0.02)	0.79 (0.30)	0.87 (0.01)	0.36 (0.37)	0.55 (0.34)
$\text{TS}_{\lambda=5}$	0.09 (0.17)	0.57 (0.19)	0.54 (0.37)	0.58 (0.10)	0.39 (0.30)	0.49 (0.32)
$\varepsilon.10_{\eta=1}$	0.14 (0.23)	0.12 (0.20)	0.34 (0.25)	0.11 (0.14)	0.53 (0.34)	0.24 (0.23)
$\text{dUCB}_{\eta=1}$	0.94 (0.01)	0.92 (0.02)	0.84 (0.23)	0.94 (0.05)	0.60 (0.40)	0.81 (0.06)
$\text{dUCB}_{\eta=2}$	0.86 (0.03)	0.87 (0.06)	0.81 (0.15)	0.92 (0.01)	0.60 (0.40)	0.80 (0.05)

Fraction of post-shift rounds on the single best scorecard. A blend can have low excess and a low hit rate.

Table 41: Post-shift hit rate: Gradual noise growth. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.93 (0.00)	0.82 (0.02)	0.93 (0.03)	0.76 (0.02)	0.64 (0.02)	0.96 (0.02)
dUCB	0.88 (0.02)	0.67 (0.00)	0.88 (0.02)	0.59 (0.02)	0.50 (0.00)	0.90 (0.00)
TS	0.99 (0.02)	0.88 (0.02)	0.99 (0.02)	0.87 (0.00)	0.82 (0.06)	0.99 (0.02)
$\varepsilon.05$	0.96 (0.03)	0.93 (0.03)	0.96 (0.03)	0.93 (0.03)	0.91 (0.03)	0.96 (0.03)
$\varepsilon.10$	0.94 (0.03)	0.89 (0.04)	0.94 (0.03)	0.89 (0.04)	0.88 (0.06)	0.94 (0.03)
$\varepsilon.15$	0.88 (0.02)	0.84 (0.03)	0.88 (0.02)	0.84 (0.03)	0.82 (0.03)	0.88 (0.02)
$\text{UCB}_{\eta=.5}$	0.93 (0.00)	0.82 (0.02)	0.94 (0.02)	0.77 (0.00)	0.66 (0.02)	0.96 (0.02)
$\text{UCB}_{\eta=1}$	0.93 (0.00)	0.83 (0.03)	0.95 (0.02)	0.77 (0.00)	0.65 (0.02)	0.95 (0.02)
$\text{UCB}_{\eta=2}$	0.93 (0.00)	0.83 (0.03)	0.96 (0.02)	0.77 (0.03)	0.66 (0.02)	0.96 (0.02)
$\text{TS}_{\eta=.5}$	1.00 (0.00)	0.91 (0.02)	1.00 (0.00)	0.86 (0.05)	0.84 (0.03)	1.00 (0.00)
$\text{TS}_{\eta=1}$	1.00 (0.00)	0.87 (0.04)	1.00 (0.00)	0.84 (0.05)	0.82 (0.05)	1.00 (0.00)
$\text{TS}_{\eta=2}$	1.00 (0.00)	0.90 (0.03)	0.99 (0.02)	0.87 (0.03)	0.84 (0.06)	1.00 (0.00)
$\text{UCB}_{\lambda=5}$	0.93 (0.00)	0.82 (0.03)	0.95 (0.02)	0.77 (0.03)	0.68 (0.03)	0.95 (0.02)
$\text{TS}_{\lambda=5}$	1.00 (0.00)	0.90 (0.03)	0.99 (0.02)	0.88 (0.03)	0.88 (0.02)	0.99 (0.02)
$\varepsilon.10_{\eta=1}$	0.92 (0.06)	0.88 (0.06)	0.92 (0.06)	0.86 (0.06)	0.87 (0.06)	0.92 (0.06)
$\text{dUCB}_{\eta=1}$	0.88 (0.02)	0.67 (0.03)	0.87 (0.00)	0.62 (0.02)	0.50 (0.00)	0.91 (0.02)
$\text{dUCB}_{\eta=2}$	0.88 (0.03)	0.68 (0.02)	0.87 (0.00)	0.61 (0.02)	0.52 (0.02)	0.92 (0.02)

Fraction of post-shift rounds on the single best scorecard. A blend can have low excess and a low hit rate.

Table 42: Post-shift hit rate: Drastic noise growth. Mean over four seeds, seed standard deviation in parentheses.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.95 (0.00)	0.82 (0.02)	0.94 (0.01)	0.75 (0.02)	0.66 (0.01)	0.95 (0.00)
dUCB	0.90 (0.00)	0.69 (0.01)	0.91 (0.01)	0.62 (0.05)	0.55 (0.04)	0.93 (0.00)
TS	0.93 (0.01)	0.85 (0.02)	0.94 (0.01)	0.82 (0.01)	0.72 (0.00)	0.94 (0.01)
$\varepsilon.05$	0.95 (0.00)	0.93 (0.02)	0.95 (0.00)	0.94 (0.01)	0.89 (0.01)	0.95 (0.00)
$\varepsilon.10$	0.92 (0.01)	0.89 (0.03)	0.92 (0.01)	0.90 (0.04)	0.86 (0.05)	0.92 (0.01)
$\varepsilon.15$	0.88 (0.03)	0.86 (0.04)	0.88 (0.04)	0.86 (0.03)	0.83 (0.02)	0.89 (0.03)
$\text{UCB}_{\eta=.5}$	0.95 (0.00)	0.84 (0.01)	0.94 (0.01)	0.75 (0.03)	0.65 (0.00)	0.95 (0.00)
$\text{UCB}_{\eta=1}$	0.95 (0.00)	0.82 (0.02)	0.94 (0.01)	0.77 (0.01)	0.64 (0.01)	0.96 (0.01)
$\text{UCB}_{\eta=2}$	0.95 (0.00)	0.82 (0.02)	0.94 (0.01)	0.76 (0.01)	0.67 (0.02)	0.96 (0.01)
$\text{TS}_{\eta=.5}$	0.94 (0.01)	0.86 (0.03)	0.94 (0.01)	0.84 (0.02)	0.76 (0.02)	0.95 (0.00)
$\text{TS}_{\eta=1}$	0.93 (0.01)	0.85 (0.04)	0.93 (0.01)	0.84 (0.03)	0.76 (0.01)	0.95 (0.00)
$\text{TS}_{\eta=2}$	0.94 (0.01)	0.87 (0.02)	0.93 (0.01)	0.84 (0.04)	0.78 (0.04)	0.95 (0.00)
$\text{UCB}_{\lambda=5}$	0.95 (0.00)	0.83 (0.02)	0.94 (0.01)	0.78 (0.02)	0.69 (0.01)	0.95 (0.00)
$\text{TS}_{\lambda=5}$	0.94 (0.01)	0.88 (0.02)	0.94 (0.01)	0.86 (0.02)	0.76 (0.02)	0.94 (0.01)
$\varepsilon.10_{\eta=1}$	0.89 (0.05)	0.87 (0.06)	0.89 (0.05)	0.88 (0.05)	0.84 (0.02)	0.91 (0.06)
$\text{dUCB}_{\eta=1}$	0.91 (0.01)	0.73 (0.04)	0.92 (0.01)	0.65 (0.04)	0.55 (0.00)	0.93 (0.00)
$\text{dUCB}_{\eta=2}$	0.91 (0.01)	0.73 (0.02)	0.91 (0.01)	0.64 (0.05)	0.55 (0.02)	0.93 (0.00)

Fraction of post-shift rounds on the single best scorecard. A blend can have low excess and a low hit rate.

Table 43: Post-shift PO-risk shares: Gradual relocation. Arms are stale, fresh, blend, zero. Mean of the refresh-time shares on post-shift rounds, then mean and seed sd across four seeds.

	stale	fresh	blend	zero
Gaussian $d = 30$	0.350 (0.009)	0.112 (0.005)	0.086 (0.003)	0.452 (0.006)
Gaussian $d = 12$	0.340 (0.004)	0.134 (0.005)	0.113 (0.003)	0.413 (0.007)
Student-t_5	0.353 (0.008)	0.119 (0.005)	0.082 (0.003)	0.446 (0.004)
AR(1) Gaussian	0.339 (0.006)	0.141 (0.005)	0.130 (0.004)	0.390 (0.006)
California housing	0.247 (0.014)	0.176 (0.009)	0.140 (0.009)	0.437 (0.017)
Covertype	0.343 (0.031)	0.086 (0.007)	0.077 (0.010)	0.494 (0.019)

The share is the residual pseudo-outcome risk of that score, divided by the sum across the four scores. It does not depend on which arm was served.

Table 44: Post-shift PO-risk shares: Drastic relocation. Arms are stale, fresh, blend, zero. Mean of the refresh-time shares on post-shift rounds, then mean and seed sd across four seeds.

	stale	fresh	blend	zero
Gaussian $d = 30$	0.317 (0.004)	0.089 (0.003)	0.202 (0.002)	0.392 (0.002)
Gaussian $d = 12$	0.308 (0.003)	0.104 (0.003)	0.212 (0.004)	0.376 (0.003)
Student-t_5	0.319 (0.004)	0.090 (0.002)	0.198 (0.003)	0.392 (0.005)
AR(1) Gaussian	0.302 (0.005)	0.107 (0.002)	0.219 (0.003)	0.372 (0.005)
California housing	0.274 (0.018)	0.148 (0.005)	0.181 (0.014)	0.397 (0.009)
Covertype	0.312 (0.018)	0.086 (0.010)	0.201 (0.026)	0.402 (0.003)

The share is the residual pseudo-outcome risk of that score, divided by the sum across the four scores. It does not depend on which arm was served.

Table 45: Post-shift PO-risk shares: Gradual sign flip. Arms are positive, negative, zero, half. Mean of the refresh-time shares on post-shift rounds, then mean and seed sd across four seeds.

	positive	negative	zero	half
Gaussian $d = 30$	0.150 (0.003)	0.147 (0.004)	0.555 (0.011)	0.149 (0.006)
Gaussian $d = 12$	0.187 (0.003)	0.184 (0.003)	0.446 (0.004)	0.183 (0.005)
Student-t_5	0.151 (0.003)	0.150 (0.002)	0.550 (0.005)	0.150 (0.002)
AR(1) Gaussian	0.195 (0.001)	0.191 (0.002)	0.423 (0.007)	0.191 (0.005)
California housing	0.208 (0.002)	0.204 (0.003)	0.384 (0.007)	0.203 (0.006)
Covertype	0.142 (0.002)	0.137 (0.005)	0.581 (0.009)	0.139 (0.004)

The share is the residual pseudo-outcome risk of that score, divided by the sum across the four scores. It does not depend on which arm was served.

Table 46: Post-shift PO-risk shares: Drastic sign flip. Arms are positive, negative, zero, half. Mean of the refresh-time shares on post-shift rounds, then mean and seed sd across four seeds.

	positive	negative	zero	half
Gaussian $d = 30$	0.034 (0.001)	0.034 (0.001)	0.898 (0.002)	0.034 (0.001)
Gaussian $d = 12$	0.072 (0.002)	0.071 (0.001)	0.787 (0.005)	0.070 (0.002)
Student-t_5	0.035 (0.001)	0.035 (0.001)	0.895 (0.002)	0.035 (0.001)
AR(1) Gaussian	0.087 (0.002)	0.087 (0.002)	0.740 (0.005)	0.086 (0.002)
California housing	0.106 (0.001)	0.105 (0.001)	0.685 (0.004)	0.103 (0.002)
Covertype	0.029 (0.001)	0.029 (0.001)	0.914 (0.004)	0.028 (0.001)

The share is the residual pseudo-outcome risk of that score, divided by the sum across the four scores. It does not depend on which arm was served.

Table 47: Post-shift PO-risk shares: Gradual dying block. Arms are both, stable, dying, zero. Mean of the refresh-time shares on post-shift rounds, then mean and seed sd across four seeds.

	both	stable	dying	zero
Gaussian $d = 30$	0.107 (0.004)	0.056 (0.001)	0.393 (0.002)	0.444 (0.003)
Gaussian $d = 12$	0.118 (0.002)	0.071 (0.002)	0.381 (0.004)	0.429 (0.002)
Student-t_5	0.108 (0.001)	0.057 (0.004)	0.393 (0.008)	0.442 (0.005)
AR(1) Gaussian	0.134 (0.003)	0.083 (0.001)	0.381 (0.007)	0.402 (0.009)
California housing	0.093 (0.001)	0.079 (0.005)	0.337 (0.013)	0.492 (0.017)
Covertype	0.096 (0.012)	0.058 (0.003)	0.396 (0.022)	0.450 (0.014)

The share is the residual pseudo-outcome risk of that score, divided by the sum across the four scores. It does not depend on which arm was served.

Table 48: Post-shift PO-risk shares: Drastic dying block. Arms are both, stable, dying, zero. Mean of the refresh-time shares on post-shift rounds, then mean and seed sd across four seeds.

	both	stable	dying	zero
Gaussian $d = 30$	0.164 (0.003)	0.064 (0.004)	0.362 (0.003)	0.409 (0.005)
Gaussian $d = 12$	0.165 (0.002)	0.077 (0.002)	0.354 (0.007)	0.404 (0.004)
Student-t_5	0.161 (0.004)	0.062 (0.001)	0.364 (0.009)	0.413 (0.007)
AR(1) Gaussian	0.182 (0.004)	0.086 (0.001)	0.346 (0.004)	0.386 (0.002)
California housing	0.121 (0.007)	0.081 (0.000)	0.338 (0.009)	0.460 (0.005)
Covertype	0.157 (0.016)	0.066 (0.010)	0.367 (0.028)	0.410 (0.013)

The share is the residual pseudo-outcome risk of that score, divided by the sum across the four scores. It does not depend on which arm was served.

Table 49: Post-shift PO-risk shares: Gradual nonlinear add-on. Arms are linear, linear_plus, zero, other_block. Mean of the refresh-time shares on post-shift rounds, then mean and seed sd across four seeds.

	linear	linear_plus	zero	other_block
Gaussian $d = 30$	0.035 (0.001)	0.020 (0.001)	0.469 (0.005)	0.476 (0.004)
Gaussian $d = 12$	0.074 (0.003)	0.043 (0.002)	0.437 (0.008)	0.446 (0.007)
Student-t_5	0.062 (0.012)	0.026 (0.003)	0.454 (0.021)	0.458 (0.009)
AR(1) Gaussian	0.097 (0.002)	0.057 (0.001)	0.425 (0.008)	0.420 (0.009)
California housing	0.134 (0.050)	0.179 (0.170)	0.310 (0.098)	0.378 (0.110)
Covertype	0.180 (0.038)	0.029 (0.004)	0.395 (0.017)	0.396 (0.041)

The share is the residual pseudo-outcome risk of that score, divided by the sum across the four scores. It does not depend on which arm was served.

Table 50: Post-shift PO-risk shares: Drastic nonlinear add-on. Arms are linear, linear_plus, zero, other_block. Mean of the refresh-time shares on post-shift rounds, then mean and seed sd across four seeds.

	linear	linear_plus	zero	other_block
Gaussian $d = 30$	0.044 (0.001)	0.023 (0.001)	0.466 (0.002)	0.466 (0.002)
Gaussian $d = 12$	0.089 (0.003)	0.048 (0.002)	0.430 (0.010)	0.433 (0.008)
Student-t_5	0.080 (0.006)	0.026 (0.001)	0.450 (0.017)	0.445 (0.010)
AR(1) Gaussian	0.115 (0.002)	0.061 (0.001)	0.413 (0.004)	0.411 (0.004)
California housing	0.165 (0.062)	0.100 (0.074)	0.304 (0.074)	0.431 (0.150)
Covertype	0.197 (0.036)	0.056 (0.008)	0.374 (0.013)	0.373 (0.034)

The share is the residual pseudo-outcome risk of that score, divided by the sum across the four scores. It does not depend on which arm was served.

Table 51: Post-shift PO-risk shares: Gradual noise growth. Arms are linear, zero, half, irrelevant. Mean of the refresh-time shares on post-shift rounds, then mean and seed sd across four seeds.

	linear	zero	half	irrelevant
Gaussian $d = 30$	0.102 (0.002)	0.395 (0.009)	0.102 (0.005)	0.401 (0.006)
Gaussian $d = 12$	0.160 (0.003)	0.338 (0.005)	0.159 (0.007)	0.342 (0.009)
Student-t_5	0.105 (0.004)	0.393 (0.010)	0.105 (0.007)	0.397 (0.011)
AR(1) Gaussian	0.179 (0.004)	0.322 (0.006)	0.178 (0.009)	0.321 (0.006)
California housing	0.194 (0.003)	0.316 (0.009)	0.194 (0.010)	0.295 (0.009)
Covertype	0.091 (0.006)	0.401 (0.029)	0.091 (0.004)	0.416 (0.038)

The share is the residual pseudo-outcome risk of that score, divided by the sum across the four scores. It does not depend on which arm was served.

Table 52: Post-shift PO-risk shares: Drastic noise growth. Arms are linear, zero, half, irrelevant. Mean of the refresh-time shares on post-shift rounds, then mean and seed sd across four seeds.

	linear	zero	half	irrelevant
Gaussian $d = 30$	0.114 (0.002)	0.384 (0.007)	0.114 (0.003)	0.387 (0.006)
Gaussian $d = 12$	0.159 (0.002)	0.342 (0.002)	0.158 (0.003)	0.342 (0.001)
Student-t_5	0.116 (0.003)	0.383 (0.008)	0.115 (0.006)	0.386 (0.009)
AR(1) Gaussian	0.173 (0.001)	0.326 (0.004)	0.172 (0.005)	0.329 (0.003)
California housing	0.182 (0.004)	0.326 (0.004)	0.182 (0.010)	0.309 (0.014)
Covertype	0.104 (0.004)	0.397 (0.017)	0.103 (0.002)	0.396 (0.021)

The share is the residual pseudo-outcome risk of that score, divided by the sum across the four scores. It does not depend on which arm was served.

Table 53: Reward scale σ on the reference window, and the resulting penalty multiplier $\eta\sigma$ at $\eta = 1$.

	mean σ	mean $\eta\sigma$	min $\eta\sigma$	max $\eta\sigma$
Gaussian $d = 30$	6.07	6.07	4.83	10.36
Gaussian $d = 12$	2.50	2.50	1.91	4.36
Student-t_5	6.02	6.02	4.74	10.60
AR(1) Gaussian	1.87	1.87	1.26	3.14
California housing	266.11	266.11	0.65	1325.66
Covertype	8.72	8.72	6.49	12.72

σ is the standard deviation of the four arm rewards on the first six batches. The index subtracts $\eta\sigma w_k$.

Table 54: Policy horse-race excess return: Single block move, Phase 0, before the first block has majority weight. Sum of per-round gaps to the best policy. Mean over four seeds.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	16.74 (0.71)	10.87 (0.53)	16.58 (0.47)	10.68 (1.90)	12.24 (0.54)	15.84 (0.26)
dUCB	16.74 (0.71)	10.87 (0.53)	16.58 (0.47)	11.32 (1.16)	9.25 (0.98)	15.84 (0.26)
TS	16.74 (0.71)	10.87 (0.53)	16.58 (0.47)	9.94 (1.66)	8.48 (2.01)	15.84 (0.26)
$\varepsilon.05$	26.17 (5.75)	16.77 (3.91)	25.55 (4.93)	14.60 (3.89)	9.88 (1.49)	25.61 (5.13)
$\varepsilon.10$	31.20 (3.58)	20.49 (3.21)	31.46 (3.53)	17.68 (2.96)	13.01 (1.70)	31.59 (4.00)
$\varepsilon.15$	43.07 (8.71)	27.26 (5.59)	41.81 (8.17)	23.38 (4.32)	17.53 (3.91)	42.08 (8.10)
$\text{UCB}_{\eta=1}$	16.74 (0.71)	10.87 (0.53)	16.58 (0.47)	11.46 (1.06)	10.32 (0.37)	15.84 (0.26)
$\text{UCB}_{\eta=2}$	16.74 (0.71)	12.14 (0.92)	16.58 (0.47)	11.07 (1.09)	9.63 (0.36)	15.84 (0.26)
$\text{TS}_{\eta=1}$	16.74 (0.71)	11.60 (1.89)	16.58 (0.47)	10.29 (1.74)	8.20 (1.23)	15.84 (0.26)
$\text{TS}_{\eta=2}$	16.74 (0.71)	11.45 (1.24)	16.58 (0.47)	10.62 (0.58)	8.22 (1.68)	15.84 (0.26)
$\varepsilon.10_{\eta=1}$	34.18 (6.79)	21.64 (5.14)	33.83 (6.80)	19.12 (5.25)	13.75 (2.68)	33.11 (7.71)
$\text{UCB}_{\eta=1}$, on-policy	20.11 (2.22)	17.99 (1.87)	19.54 (0.64)	16.60 (1.34)	14.64 (0.45)	22.23 (3.37)
$\text{TS}_{\eta=1}$, on-policy	18.38 (1.43)	12.38 (1.20)	17.41 (1.54)	11.32 (0.88)	10.30 (2.13)	18.16 (1.49)
$\text{UCB}_{\eta=1}$, fitted	16.74 (0.71)	10.87 (0.53)	16.58 (0.47)	11.42 (1.03)	10.07 (0.27)	15.84 (0.26)
$\text{TS}_{\eta=1}$, fitted	16.74 (0.71)	11.70 (2.08)	16.58 (0.47)	10.16 (2.21)	9.12 (1.88)	15.84 (0.26)

Oracle rows score every policy on every round. On-policy rows only retain the served policy. Fitted rows imagine every policy from a least-squares reward fit on the served transitions.

Table 55: Policy horse-race hit rate: Single block move, Phase 0, before the first block has majority weight.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.91 (0.01)	0.92 (0.01)	0.93 (0.01)	0.91 (0.02)	0.84 (0.02)	0.94 (0.00)
dUCB	0.91 (0.01)	0.92 (0.01)	0.93 (0.01)	0.90 (0.02)	0.86 (0.02)	0.94 (0.00)
TS	0.91 (0.01)	0.92 (0.01)	0.93 (0.01)	0.92 (0.02)	0.91 (0.02)	0.94 (0.00)
$\varepsilon.05$	0.88 (0.03)	0.89 (0.01)	0.90 (0.03)	0.89 (0.03)	0.91 (0.02)	0.91 (0.02)
$\varepsilon.10$	0.84 (0.02)	0.86 (0.03)	0.86 (0.02)	0.86 (0.03)	0.87 (0.03)	0.87 (0.03)
$\varepsilon.15$	0.79 (0.05)	0.80 (0.06)	0.80 (0.04)	0.80 (0.04)	0.81 (0.05)	0.81 (0.05)
$\text{UCB}_{\eta=1}$	0.91 (0.01)	0.92 (0.01)	0.93 (0.01)	0.90 (0.02)	0.86 (0.00)	0.94 (0.00)
$\text{UCB}_{\eta=2}$	0.91 (0.01)	0.91 (0.02)	0.93 (0.01)	0.91 (0.01)	0.87 (0.01)	0.94 (0.00)
$\text{TS}_{\eta=1}$	0.91 (0.01)	0.92 (0.02)	0.93 (0.01)	0.91 (0.02)	0.91 (0.01)	0.94 (0.00)
$\text{TS}_{\eta=2}$	0.91 (0.01)	0.92 (0.02)	0.93 (0.01)	0.92 (0.01)	0.91 (0.01)	0.94 (0.00)
$\varepsilon.10_{\eta=1}$	0.83 (0.03)	0.84 (0.04)	0.84 (0.03)	0.84 (0.03)	0.85 (0.03)	0.85 (0.03)
$\text{UCB}_{\eta=1}$, on-policy	0.89 (0.01)	0.80 (0.04)	0.90 (0.04)	0.83 (0.02)	0.76 (0.03)	0.87 (0.04)
$\text{TS}_{\eta=1}$, on-policy	0.91 (0.01)	0.92 (0.01)	0.93 (0.01)	0.91 (0.01)	0.88 (0.03)	0.92 (0.01)
$\text{UCB}_{\eta=1}$, fitted	0.91 (0.01)	0.92 (0.01)	0.93 (0.01)	0.91 (0.01)	0.86 (0.00)	0.94 (0.00)
$\text{TS}_{\eta=1}$, fitted	0.91 (0.01)	0.92 (0.02)	0.93 (0.01)	0.92 (0.02)	0.90 (0.02)	0.94 (0.00)

Hit rate is the fraction of rounds whose served policy has the highest mean return on that round.

Table 56: Policy horse-race excess return: Single block move, Phase 1, first moved block has majority weight. Sum of per-round gaps to the best policy. Mean over four seeds.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	84.48 (2.22)	19.68 (16.27)	84.02 (2.46)	20.47 (22.58)	6.23 (0.76)	43.60 (2.17)
dUCB	41.68 (29.86)	4.61 (4.95)	45.10 (19.43)	10.20 (15.53)	4.59 (0.53)	21.89 (17.63)
TS	84.55 (2.36)	50.34 (7.94)	84.02 (2.46)	48.82 (6.60)	9.81 (1.04)	43.36 (2.27)
$\varepsilon.05$	82.64 (3.70)	52.64 (1.86)	81.90 (3.35)	50.18 (1.07)	8.85 (0.80)	42.17 (2.31)
$\varepsilon.10$	88.05 (3.98)	55.84 (0.87)	86.89 (0.94)	52.88 (1.05)	11.26 (1.67)	46.88 (3.69)
$\varepsilon.15$	85.46 (2.06)	54.19 (3.00)	84.71 (5.63)	51.72 (2.84)	11.38 (1.98)	46.00 (4.38)
$\text{UCB}_{\eta=1}$	48.81 (24.24)	6.76 (2.99)	60.94 (26.25)	9.42 (13.63)	2.55 (1.37)	23.53 (15.16)
$\text{UCB}_{\eta=2}$	27.29 (17.15)	3.96 (2.28)	23.91 (8.79)	5.64 (5.46)	0.88 (0.62)	8.49 (3.68)
$\text{TS}_{\eta=1}$	84.73 (2.27)	40.01 (17.37)	84.04 (2.48)	38.85 (16.99)	7.84 (2.12)	43.36 (2.27)
$\text{TS}_{\eta=2}$	59.59 (29.66)	23.07 (15.12)	57.26 (23.92)	20.87 (10.52)	4.14 (1.69)	22.20 (9.57)
$\varepsilon.10_{\eta=1}$	80.18 (8.05)	51.10 (5.19)	80.29 (7.80)	49.68 (5.81)	9.03 (3.11)	44.38 (2.06)
$\text{UCB}_{\eta=1}$, on-policy	47.86 (9.39)	31.54 (3.43)	41.22 (11.26)	21.85 (4.79)	5.86 (0.69)	25.73 (2.11)
$\text{TS}_{\eta=1}$, on-policy	54.89 (21.47)	30.07 (5.31)	64.00 (20.94)	30.02 (6.21)	6.12 (1.00)	29.09 (11.58)
$\text{UCB}_{\eta=1}$, fitted	53.35 (21.61)	6.71 (3.19)	60.94 (26.25)	9.58 (12.72)	2.38 (1.56)	22.91 (14.95)
$\text{TS}_{\eta=1}$, fitted	84.87 (2.79)	41.67 (17.13)	83.92 (2.47)	40.32 (16.08)	6.69 (1.37)	43.82 (1.63)

Oracle rows score every policy on every round. On-policy rows only retain the served policy. Fitted rows imagine every policy from a least-squares reward fit on the served transitions.

Table 57: Policy horse-race hit rate: Single block move, Phase 1, first moved block has majority weight.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.03 (0.00)	0.47 (0.23)	0.03 (0.02)	0.49 (0.32)	0.40 (0.03)	0.20 (0.07)
dUCB	0.38 (0.24)	0.80 (0.13)	0.31 (0.19)	0.72 (0.31)	0.65 (0.02)	0.51 (0.21)
TS	0.03 (0.00)	0.07 (0.09)	0.03 (0.02)	0.07 (0.07)	0.22 (0.02)	0.20 (0.07)
$\varepsilon.05$	0.06 (0.02)	0.05 (0.03)	0.05 (0.02)	0.05 (0.02)	0.24 (0.03)	0.23 (0.07)
$\varepsilon.10$	0.03 (0.00)	0.03 (0.02)	0.03 (0.02)	0.03 (0.02)	0.22 (0.03)	0.20 (0.07)
$\varepsilon.15$	0.07 (0.03)	0.06 (0.03)	0.06 (0.02)	0.06 (0.02)	0.22 (0.03)	0.22 (0.05)
$\text{UCB}_{\eta=1}$	0.29 (0.18)	0.73 (0.07)	0.19 (0.21)	0.69 (0.26)	0.72 (0.11)	0.46 (0.18)
$\text{UCB}_{\eta=2}$	0.51 (0.16)	0.79 (0.06)	0.52 (0.12)	0.78 (0.14)	0.79 (0.12)	0.67 (0.10)
$\text{TS}_{\eta=1}$	0.03 (0.00)	0.18 (0.18)	0.03 (0.02)	0.18 (0.18)	0.30 (0.11)	0.20 (0.07)
$\text{TS}_{\eta=2}$	0.22 (0.21)	0.39 (0.21)	0.21 (0.19)	0.42 (0.16)	0.51 (0.12)	0.46 (0.13)
$\varepsilon.10_{\eta=1}$	0.08 (0.08)	0.07 (0.06)	0.07 (0.07)	0.08 (0.09)	0.31 (0.15)	0.20 (0.07)
$\text{UCB}_{\eta=1}$, on-policy	0.37 (0.10)	0.39 (0.05)	0.44 (0.06)	0.50 (0.05)	0.43 (0.05)	0.43 (0.03)
$\text{TS}_{\eta=1}$, on-policy	0.29 (0.20)	0.38 (0.04)	0.22 (0.21)	0.32 (0.14)	0.37 (0.03)	0.41 (0.18)
$\text{UCB}_{\eta=1}$, fitted	0.25 (0.16)	0.73 (0.08)	0.19 (0.21)	0.67 (0.25)	0.72 (0.10)	0.47 (0.19)
$\text{TS}_{\eta=1}$, fitted	0.03 (0.00)	0.17 (0.20)	0.03 (0.02)	0.17 (0.17)	0.35 (0.08)	0.20 (0.07)

Hit rate is the fraction of rounds whose served policy has the highest mean return on that round.

Table 58: Policy horse-race excess return: Two block moves, Phase 0, before the first block has majority weight. Sum of per-round gaps to the best policy. Mean over four seeds.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	16.69 (0.65)	10.17 (0.25)	16.56 (1.28)	8.87 (0.36)	9.66 (0.97)	14.93 (0.65)
dUCB	16.69 (0.65)	10.17 (0.25)	16.56 (1.28)	8.87 (0.36)	8.40 (0.40)	14.93 (0.65)
TS	16.69 (0.65)	10.66 (1.13)	16.56 (1.28)	9.92 (1.24)	8.05 (1.38)	14.93 (0.65)
$\varepsilon.05$	25.63 (5.72)	15.81 (3.91)	25.29 (5.43)	13.70 (2.45)	9.48 (1.60)	23.42 (3.80)
$\varepsilon.10$	26.01 (3.09)	16.18 (2.56)	25.69 (3.33)	14.04 (1.45)	9.83 (1.54)	24.28 (2.79)
$\varepsilon.15$	33.14 (5.34)	20.70 (3.35)	33.23 (5.44)	18.18 (3.50)	13.29 (1.73)	32.93 (5.72)
$\text{UCB}_{\eta=1}$	16.69 (0.65)	10.17 (0.25)	16.56 (1.28)	8.87 (0.36)	7.57 (0.38)	14.93 (0.65)
$\text{UCB}_{\eta=2}$	16.69 (0.65)	10.17 (0.25)	16.56 (1.28)	8.87 (0.36)	7.63 (0.99)	14.93 (0.65)
$\text{TS}_{\eta=1}$	16.69 (0.65)	10.68 (0.94)	16.56 (1.28)	9.32 (0.95)	7.71 (0.76)	14.93 (0.65)
$\text{TS}_{\eta=2}$	16.69 (0.65)	10.54 (0.58)	16.56 (1.28)	9.56 (0.82)	7.59 (1.17)	14.93 (0.65)
$\varepsilon.10_{\eta=1}$	28.26 (8.33)	17.58 (5.14)	27.88 (9.34)	14.74 (4.30)	10.53 (2.77)	27.20 (9.99)
$\text{UCB}_{\eta=1}$, on-policy	17.69 (1.36)	13.18 (0.94)	16.98 (0.77)	12.68 (0.98)	11.75 (1.94)	16.96 (1.47)
$\text{TS}_{\eta=1}$, on-policy	16.69 (0.65)	10.17 (0.25)	16.56 (1.28)	10.60 (0.68)	7.95 (0.88)	15.43 (1.57)
$\text{UCB}_{\eta=1}$, fitted	16.69 (0.65)	10.17 (0.25)	16.56 (1.28)	8.87 (0.36)	7.41 (0.41)	14.93 (0.65)
$\text{TS}_{\eta=1}$, fitted	16.69 (0.65)	11.02 (1.59)	16.56 (1.28)	9.63 (1.53)	8.98 (1.50)	14.93 (0.65)

Oracle rows score every policy on every round. On-policy rows only retain the served policy. Fitted rows imagine every policy from a least-squares reward fit on the served transitions.

Table 59: Policy horse-race hit rate: Two block moves, Phase 0, before the first block has majority weight.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.91 (0.01)	0.91 (0.01)	0.91 (0.01)	0.91 (0.00)	0.82 (0.03)	0.91 (0.00)
dUCB	0.91 (0.01)	0.91 (0.01)	0.91 (0.01)	0.91 (0.00)	0.84 (0.01)	0.91 (0.00)
TS	0.91 (0.01)	0.90 (0.03)	0.91 (0.01)	0.90 (0.02)	0.87 (0.02)	0.91 (0.00)
$\varepsilon.05$	0.86 (0.02)	0.86 (0.04)	0.86 (0.02)	0.86 (0.03)	0.86 (0.03)	0.86 (0.03)
$\varepsilon.10$	0.84 (0.02)	0.84 (0.03)	0.84 (0.02)	0.85 (0.01)	0.85 (0.01)	0.85 (0.01)
$\varepsilon.15$	0.80 (0.04)	0.80 (0.02)	0.80 (0.04)	0.81 (0.04)	0.81 (0.04)	0.81 (0.04)
$\text{UCB}_{\eta=1}$	0.91 (0.01)	0.91 (0.01)	0.91 (0.01)	0.91 (0.00)	0.86 (0.01)	0.91 (0.00)
$\text{UCB}_{\eta=2}$	0.91 (0.01)	0.91 (0.01)	0.91 (0.01)	0.91 (0.00)	0.87 (0.03)	0.91 (0.00)
$\text{TS}_{\eta=1}$	0.91 (0.01)	0.90 (0.02)	0.91 (0.01)	0.91 (0.01)	0.87 (0.02)	0.91 (0.00)
$\text{TS}_{\eta=2}$	0.91 (0.01)	0.90 (0.02)	0.91 (0.01)	0.90 (0.02)	0.88 (0.01)	0.91 (0.00)
$\varepsilon.10_{\eta=1}$	0.84 (0.04)	0.84 (0.05)	0.84 (0.05)	0.84 (0.05)	0.84 (0.05)	0.84 (0.05)
$\text{UCB}_{\eta=1}$, on-policy	0.90 (0.03)	0.84 (0.02)	0.90 (0.03)	0.83 (0.02)	0.76 (0.04)	0.89 (0.02)
$\text{TS}_{\eta=1}$, on-policy	0.91 (0.01)	0.91 (0.01)	0.91 (0.01)	0.89 (0.00)	0.87 (0.02)	0.91 (0.01)
$\text{UCB}_{\eta=1}$, fitted	0.91 (0.01)	0.91 (0.01)	0.91 (0.01)	0.91 (0.00)	0.87 (0.02)	0.91 (0.00)
$\text{TS}_{\eta=1}$, fitted	0.91 (0.01)	0.90 (0.02)	0.91 (0.01)	0.91 (0.01)	0.86 (0.01)	0.91 (0.00)

Hit rate is the fraction of rounds whose served policy has the highest mean return on that round.

Table 60: Policy horse-race excess return: Two block moves, Phase 1, first moved block has majority weight. Sum of per-round gaps to the best policy. Mean over four seeds.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	192.20 (21.55)	22.02 (14.05)	139.31 (55.69)	13.72 (7.93)	19.64 (1.04)	59.72 (12.35)
dUCB	54.80 (10.68)	6.44 (5.20)	34.69 (16.92)	6.32 (2.71)	9.60 (2.75)	12.06 (4.74)
TS	219.19 (4.16)	110.86 (22.33)	188.57 (24.76)	90.48 (33.93)	29.95 (2.63)	119.40 (9.01)
$\varepsilon.05$	187.27 (40.24)	111.81 (24.98)	173.85 (37.65)	107.73 (19.75)	29.81 (1.08)	124.82 (2.70)
$\varepsilon.10$	221.52 (5.36)	135.37 (1.73)	210.35 (3.23)	126.72 (2.13)	33.20 (2.26)	131.54 (2.32)
$\varepsilon.15$	140.77 (56.42)	90.11 (30.92)	140.53 (45.90)	83.85 (29.48)	31.62 (1.31)	124.49 (7.29)
$\text{UCB}_{\eta=1}$	79.10 (17.08)	7.21 (3.42)	45.41 (22.28)	6.19 (2.65)	12.52 (2.16)	15.26 (3.56)
$\text{UCB}_{\eta=2}$	30.11 (0.98)	6.16 (2.54)	21.34 (8.58)	5.33 (3.07)	8.13 (2.69)	8.28 (3.01)
$\text{TS}_{\eta=1}$	168.67 (34.92)	42.36 (20.79)	102.82 (53.22)	37.60 (22.42)	15.86 (1.84)	59.82 (18.43)
$\text{TS}_{\eta=2}$	87.76 (23.71)	20.98 (1.51)	56.16 (36.72)	25.23 (14.65)	10.14 (2.22)	29.03 (12.81)
$\varepsilon.10_{\eta=1}$	136.00 (58.82)	80.64 (42.14)	135.03 (64.83)	73.11 (32.59)	19.98 (4.69)	93.99 (39.58)
$\text{UCB}_{\eta=1}$, on-policy	47.77 (3.48)	26.73 (2.79)	50.56 (8.04)	23.08 (0.73)	20.82 (1.80)	73.23 (1.73)
$\text{TS}_{\eta=1}$, on-policy	108.10 (50.42)	36.30 (10.77)	81.28 (41.59)	33.71 (4.78)	21.41 (0.85)	72.55 (15.74)
$\text{UCB}_{\eta=1}$, fitted	79.20 (16.64)	10.23 (3.81)	48.70 (20.33)	7.87 (3.24)	10.85 (2.83)	15.11 (3.41)
$\text{TS}_{\eta=1}$, fitted	196.43 (25.98)	34.23 (27.28)	99.56 (59.13)	20.60 (6.15)	15.12 (2.25)	60.83 (43.35)

Oracle rows score every policy on every round. On-policy rows only retain the served policy. Fitted rows imagine every policy from a least-squares reward fit on the served transitions.

Table 61: Policy horse-race hit rate: Two block moves, Phase 1, first moved block has majority weight.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.14 (0.11)	0.73 (0.11)	0.33 (0.21)	0.80 (0.06)	0.42 (0.02)	0.48 (0.09)
dUCB	0.66 (0.03)	0.86 (0.07)	0.73 (0.07)	0.89 (0.03)	0.71 (0.07)	0.78 (0.06)
TS	0.01 (0.01)	0.20 (0.17)	0.11 (0.13)	0.29 (0.22)	0.11 (0.08)	0.12 (0.04)
$\varepsilon.05$	0.16 (0.15)	0.18 (0.20)	0.18 (0.18)	0.17 (0.16)	0.10 (0.02)	0.10 (0.01)
$\varepsilon.10$	0.01 (0.01)	0.01 (0.02)	0.01 (0.01)	0.01 (0.02)	0.09 (0.04)	0.08 (0.02)
$\varepsilon.15$	0.34 (0.22)	0.33 (0.23)	0.32 (0.22)	0.33 (0.23)	0.14 (0.04)	0.12 (0.05)
$\text{UCB}_{\eta=1}$	0.57 (0.05)	0.85 (0.06)	0.69 (0.09)	0.88 (0.03)	0.66 (0.06)	0.75 (0.04)
$\text{UCB}_{\eta=2}$	0.75 (0.01)	0.87 (0.05)	0.79 (0.05)	0.89 (0.03)	0.76 (0.09)	0.83 (0.03)
$\text{TS}_{\eta=1}$	0.24 (0.15)	0.61 (0.12)	0.47 (0.19)	0.64 (0.14)	0.49 (0.08)	0.48 (0.11)
$\text{TS}_{\eta=2}$	0.54 (0.09)	0.74 (0.02)	0.65 (0.14)	0.71 (0.10)	0.64 (0.12)	0.68 (0.09)
$\varepsilon.10_{\eta=1}$	0.36 (0.21)	0.39 (0.27)	0.35 (0.26)	0.41 (0.20)	0.45 (0.13)	0.30 (0.27)
$\text{UCB}_{\eta=1}$, on-policy	0.69 (0.01)	0.72 (0.03)	0.68 (0.06)	0.74 (0.00)	0.40 (0.04)	0.38 (0.06)
$\text{TS}_{\eta=1}$, on-policy	0.43 (0.19)	0.64 (0.07)	0.52 (0.15)	0.65 (0.03)	0.34 (0.02)	0.41 (0.10)
$\text{UCB}_{\eta=1}$, fitted	0.57 (0.05)	0.82 (0.04)	0.67 (0.08)	0.85 (0.03)	0.70 (0.07)	0.76 (0.03)
$\text{TS}_{\eta=1}$, fitted	0.12 (0.12)	0.66 (0.16)	0.48 (0.23)	0.73 (0.06)	0.49 (0.06)	0.49 (0.27)

Hit rate is the fraction of rounds whose served policy has the highest mean return on that round.

Table 62: Policy horse-race excess return: Two block moves, Phase 2, second moved block has majority weight. Sum of per-round gaps to the best policy. Mean over four seeds.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	83.90 (67.36)	33.33 (42.83)	90.83 (46.68)	45.80 (26.11)	10.24 (8.14)	39.48 (50.79)
dUCB	72.21 (23.33)	13.49 (19.04)	45.83 (17.59)	18.67 (11.73)	8.50 (9.16)	14.17 (14.62)
TS	163.10 (27.82)	99.59 (5.88)	151.95 (26.87)	81.33 (15.95)	122.30 (2.27)	127.38 (84.23)
$\varepsilon.05$	166.66 (12.55)	102.45 (7.70)	161.07 (14.09)	90.61 (5.46)	122.99 (1.97)	191.76 (2.76)
$\varepsilon.10$	135.48 (46.15)	90.33 (28.71)	142.22 (48.66)	74.95 (23.98)	91.02 (25.20)	163.63 (31.33)
$\varepsilon.15$	154.61 (10.70)	96.55 (8.40)	149.71 (11.12)	86.87 (3.03)	102.19 (33.82)	135.55 (53.25)
$\text{UCB}_{\eta=1}$	47.54 (25.66)	5.08 (1.43)	64.88 (33.84)	8.63 (7.08)	2.50 (4.99)	29.47 (15.59)
$\text{UCB}_{\eta=2}$	17.85 (7.01)	4.18 (2.37)	19.84 (1.22)	2.28 (2.17)	2.50 (4.99)	18.71 (9.65)
$\text{TS}_{\eta=1}$	83.50 (54.35)	47.57 (9.25)	115.63 (44.79)	60.69 (33.95)	72.24 (55.60)	60.01 (28.94)
$\text{TS}_{\eta=2}$	59.33 (30.24)	32.05 (9.73)	67.82 (18.01)	26.73 (3.86)	13.42 (14.17)	51.37 (11.79)
$\varepsilon.10_{\eta=1}$	152.98 (1.74)	94.62 (2.26)	149.56 (4.07)	86.31 (2.07)	102.00 (34.66)	184.00 (8.45)
$\text{UCB}_{\eta=1}$, on-policy	161.10 (4.64)	97.51 (3.36)	136.16 (42.55)	46.83 (34.16)	13.94 (10.78)	94.51 (54.25)
$\text{TS}_{\eta=1}$, on-policy	162.17 (4.38)	95.30 (10.59)	155.77 (1.99)	95.33 (1.44)	101.19 (37.54)	177.99 (8.46)
$\text{UCB}_{\eta=1}$, fitted	49.81 (24.16)	5.34 (0.92)	60.74 (30.28)	8.26 (5.79)	2.83 (5.66)	20.93 (12.40)
$\text{TS}_{\eta=1}$, fitted	84.12 (51.18)	73.28 (28.96)	138.92 (25.77)	60.58 (35.57)	79.11 (49.40)	157.95 (42.67)

Oracle rows score every policy on every round. On-policy rows only retain the served policy. Fitted rows imagine every policy from a least-squares reward fit on the served transitions.

Table 63: Policy horse-race hit rate: Two block moves, Phase 2, second moved block has majority weight.

	Gaussian $d = 30$	Gaussian $d = 12$	Student-t_5	AR(1) Gaussian	California housing	Covertype
UCB	0.41 (0.43)	0.55 (0.38)	0.32 (0.24)	0.41 (0.23)	0.86 (0.11)	0.70 (0.29)
dUCB	0.43 (0.11)	0.76 (0.21)	0.55 (0.09)	0.66 (0.13)	0.88 (0.12)	0.84 (0.12)
TS	0.07 (0.14)	0.01 (0.01)	0.05 (0.10)	0.09 (0.14)	0.00 (0.00)	0.30 (0.43)
$\varepsilon.05$	0.01 (0.01)	0.00 (0.00)	0.01 (0.01)	0.01 (0.03)	0.00 (0.00)	0.00 (0.00)
$\varepsilon.10$	0.21 (0.23)	0.15 (0.23)	0.15 (0.23)	0.19 (0.21)	0.23 (0.19)	0.13 (0.14)
$\varepsilon.15$	0.05 (0.03)	0.05 (0.04)	0.04 (0.04)	0.06 (0.03)	0.15 (0.24)	0.24 (0.25)
$\text{UCB}_{\eta=1}$	0.56 (0.13)	0.83 (0.04)	0.45 (0.17)	0.78 (0.11)	0.97 (0.06)	0.72 (0.10)
$\text{UCB}_{\eta=2}$	0.73 (0.07)	0.84 (0.05)	0.70 (0.02)	0.89 (0.06)	0.97 (0.06)	0.78 (0.06)
$\text{TS}_{\eta=1}$	0.40 (0.33)	0.41 (0.08)	0.19 (0.21)	0.27 (0.31)	0.36 (0.42)	0.55 (0.16)
$\text{TS}_{\eta=2}$	0.50 (0.16)	0.53 (0.09)	0.44 (0.06)	0.56 (0.03)	0.84 (0.15)	0.59 (0.06)
$\varepsilon.10_{\eta=1}$	0.03 (0.02)	0.02 (0.01)	0.02 (0.03)	0.03 (0.04)	0.14 (0.25)	0.01 (0.02)
$\text{UCB}_{\eta=1}$, on-policy	0.01 (0.02)	0.03 (0.02)	0.11 (0.21)	0.41 (0.33)	0.82 (0.12)	0.41 (0.28)
$\text{TS}_{\eta=1}$, on-policy	0.01 (0.01)	0.06 (0.10)	0.01 (0.01)	0.01 (0.01)	0.14 (0.27)	0.02 (0.04)
$\text{UCB}_{\eta=1}$, fitted	0.54 (0.12)	0.81 (0.02)	0.47 (0.15)	0.78 (0.09)	0.97 (0.06)	0.78 (0.09)
$\text{TS}_{\eta=1}$, fitted	0.40 (0.28)	0.20 (0.23)	0.07 (0.12)	0.27 (0.31)	0.30 (0.35)	0.09 (0.19)

Hit rate is the fraction of rounds whose served policy has the highest mean return on that round.